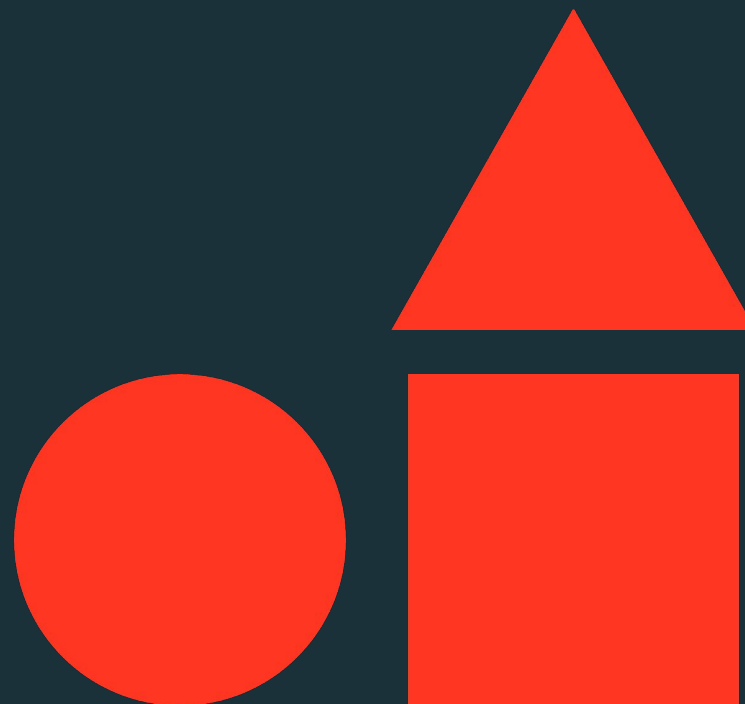


Genie overview

Intelligent analytics
for everyone



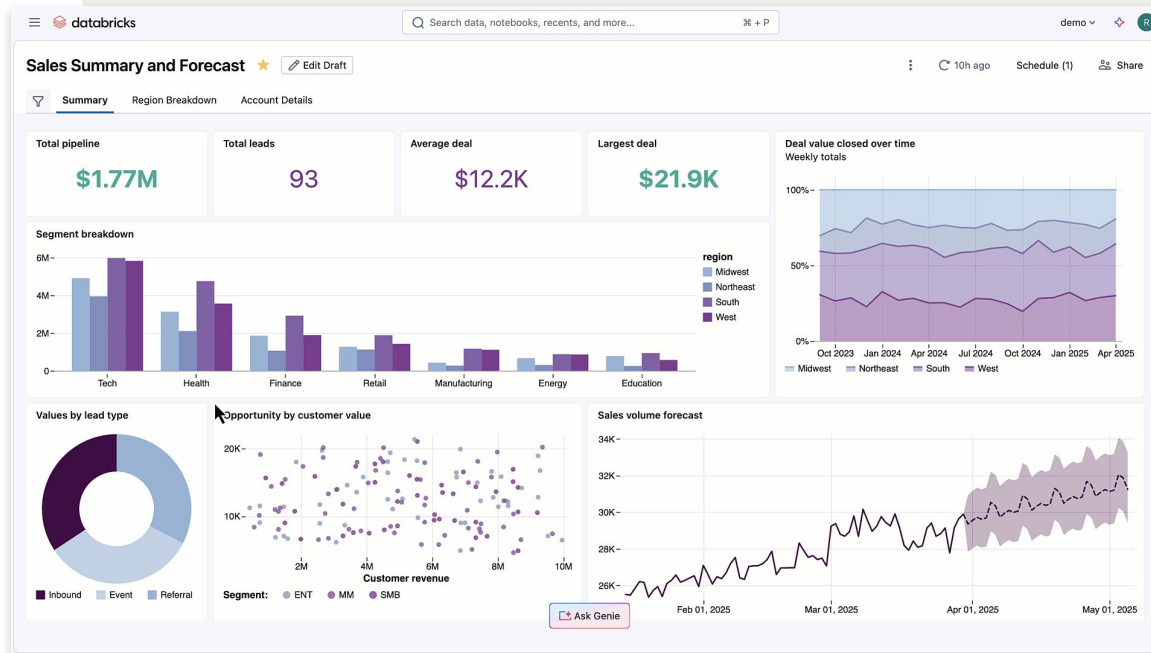
Introducing Databricks Dashboards & Genie

Intelligent analytics for everyone

Governed with Unity Catalog | Powered by Databricks SQL

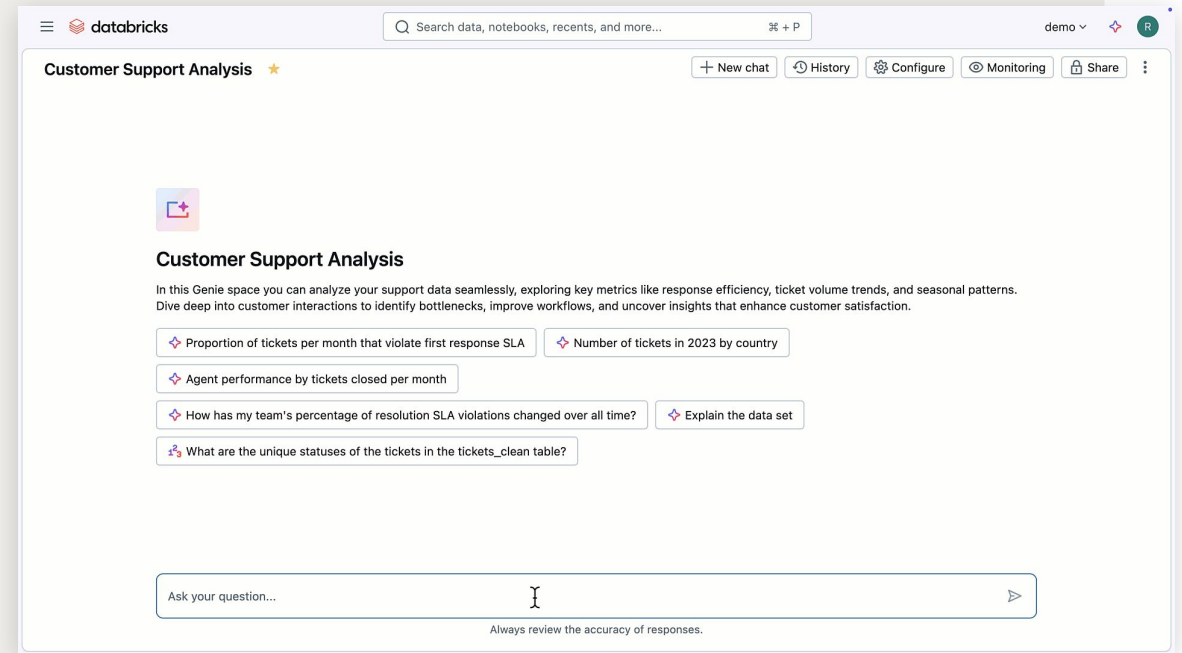
Dashboards

Fast and secure visual insights for your entire organization



Genie

Talk with your data using natural language



Included with Databricks SQL—No extra license fees

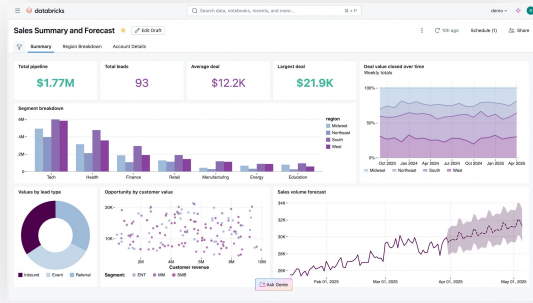


Accessed via Databricks One

The new consumer experience for Dashboards, Genie and Apps

The screenshot shows the Databricks One home interface. At the top, the Databricks logo is on the left, and 'Databricks One Demo' with a user icon is on the right. The main heading is 'What would you like to know?'. Below this is a search bar with the placeholder text 'Search for Dashboards and Genie spaces...'. Under the search bar are two buttons: 'Search' and 'Ask'. Below the search bar are four tabs: 'For you', 'Dashboards', 'Genie spaces', and 'Apps'. Under the 'Dashboards' tab, there are four dashboard cards: 'Dashboard - Customer Support Team Revi...', 'Dashboard - Sales Pipeline Review', 'Dashboard - Portfolio Assistant', and 'Genie space - Customer Support Review'. Each card shows a preview of its content, including various charts and tables.

The first screenshot shows a 'Sales Summary and Forecast' dashboard. It features a top section with key metrics: 'Total pipeline' (\$1.77M), 'Total leads' (93), 'Average deal' (\$12.2K), and 'Largest deal' (\$21.9K). Below this are several charts: 'Segment breakdown' (a bar chart), 'Deal value closed over time' (a line chart), 'Values by lead type' (a donut chart), 'Opportunity by customer value' (a scatter plot), and 'Sales volume forecast' (a line chart). The second screenshot shows a 'Customer Support Analysis' dashboard. It includes a section titled 'Customer Support Analysis' with a description: 'In this Genie space you can analyze your support data seamlessly, exploring key metrics like response efficiency, ticket volume trends, and seasonal patterns. Dive deep into customer interactions to identify bottlenecks, improve workflows, and uncover insights that enhance customer satisfaction.' Below this are several interactive filters and a search bar. The filters include: 'Proportion of tickets per month that violate first response SLA', 'Number of tickets in 2023 by country', 'Agent performance by tickets closed per month', 'How has my team's percentage of resolution SLA violations changed over all time?', and 'What are the unique statuses of the tickets in the tickets_clean table?'. At the bottom, there is a search bar with the placeholder text 'Ask your question...' and a button to 'Ask Genie'.



The 'Sales Genie' interface displays a table of the top 10 sales reps by revenue and deals closed:

name	1.2 total_revenue	2.5 closed_deals
Kathye Osanne	\$96,001.94	8
Alejandro Baldwin	\$72,002.07	6
Morgana Whinney	\$60,000.35	5

Below the table, there is a chat interface with a prompt: "What are the top 10 sales reps by total revenue for the last fiscal year?" and a response: "Which sales rep has the highest revenue?"

Integrated with Unity Catalog Business Semantics

Single source of truth for critical KPIs



Business Semantics

Metric Views

Pre-compute for performance

Measures

Dimensions

Relationship Modeling

Agent metadata

LLM instructions

Synonyms

Reference Queries

Domains

Formatting
(e.g., currency USD)

Certifications

Define and maintain a single set of business semantics inside your data platform

Leverage from Dashboards, Genie and any SQL client

Materialize metrics for best performance and pre-compute to reduce latency

Establish trust and reliability through centralized governance and certification



Genie

Genie

Talk with your data

Conversational analytics
tailored to your business

Governed and secure
answers from AI

Data insights accessible
from anywhere

The screenshot shows the Databricks Genie interface. At the top, there's a navigation bar with the Databricks logo, a search bar, and a '+ P' icon. Below this, the title 'Customer Support Analysis' is followed by a star icon. To the right are buttons for 'New chat', 'History', 'Configure', and 'Monitoring'. The main content area features a Genie icon (a purple square with a white plus sign) and the title 'Customer Support Analysis'. Below the title is a descriptive paragraph: 'In this Genie space you can analyze your support data seamlessly, exploring key metrics like response efficiency, ticket volume trends, and seasonal patterns. Dive deep into customer interactions to identify bottlenecks, improve workflows, and uncover insights that enhance customer satisfaction.' Below this paragraph are five suggested queries, each in a rounded rectangle with a Genie icon: 'Proportion of tickets per month that violate first response SLA', 'Number of tickets in 2023 by country', 'Agent performance by tickets closed per month', 'How has my team's percentage of resolution SLA violations changed over all time?', and 'Explain the data set'. At the bottom, there's a text input field with the placeholder 'Ask your question...' and a Genie icon. Below the input field is a disclaimer: 'Always review the accuracy of responses.'

Customer Support Analysis ★

+ New chat History Configure Monitoring



Customer Support Analysis

In this Genie space you can analyze your support data seamlessly, exploring key metrics like response efficiency, ticket volume trends, and seasonal patterns. Dive deep into customer interactions to identify bottlenecks, improve workflows, and uncover insights that enhance customer satisfaction.

✦ Proportion of tickets per month that violate first response SLA ✦ Number of tickets in 2023 by country

✦ Agent performance by tickets closed per month

✦ How has my team's percentage of resolution SLA violations changed over all time? ✦ Explain the data set

✦ What are the unique statuses of the tickets in the tickets_clean table?

Ask your question... 

Always review the accuracy of responses.

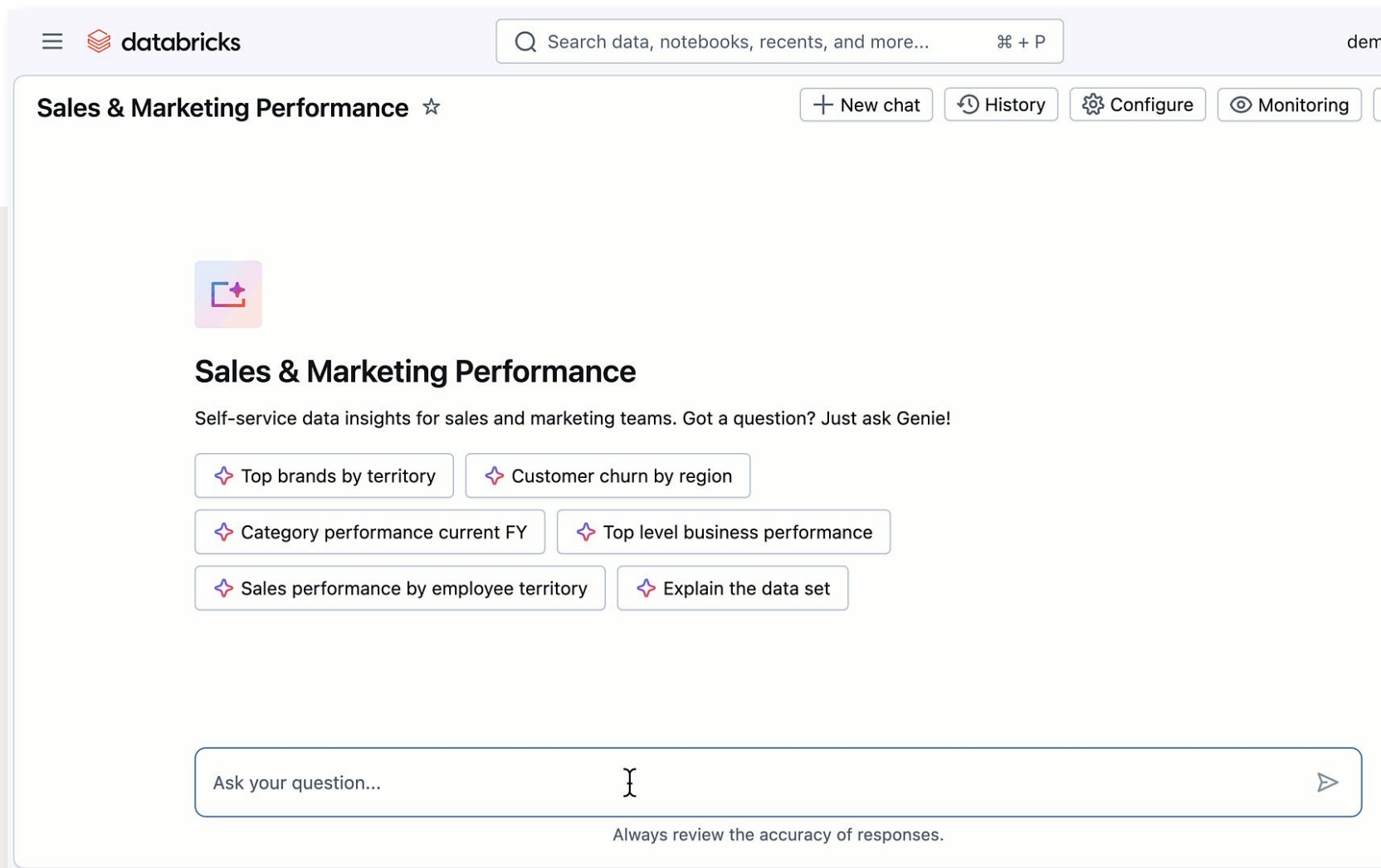


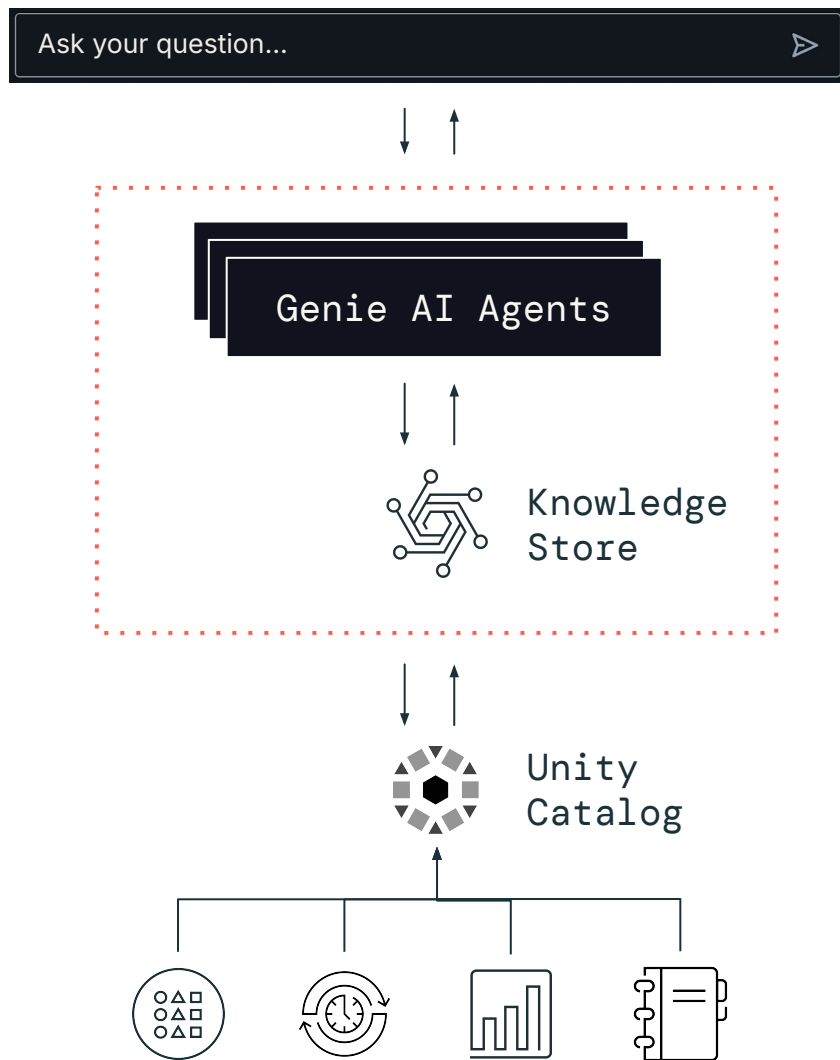
Go beyond your dashboards

Engage in open Q&A **directly**
with your enterprise data

Genie **remembers the**
conversation thread so you
can pick up where you left off

User receives responses in
natural language, data tables,
and visualizations





An agent system

continuously learns your unique data and semantics

Ensemble of specialized AI agents working in concert

Leverages local knowledge store for context

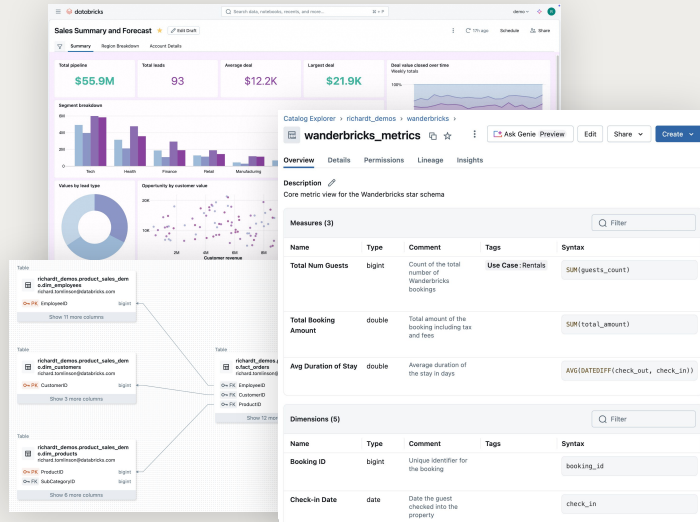
- Unity Catalog Metadata
- Unity Catalog Metrics
- Query history (All Workloads)
- Related assets (Dashboards, SQL Queries, Notebooks)

Seeks clarifications when needed

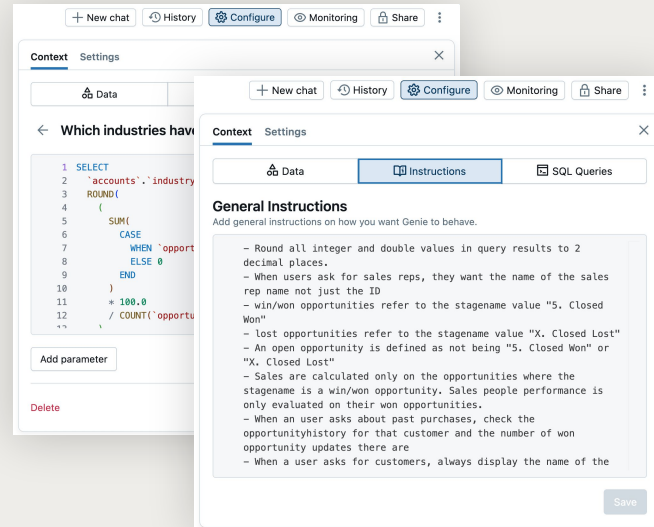
Remembers and learns



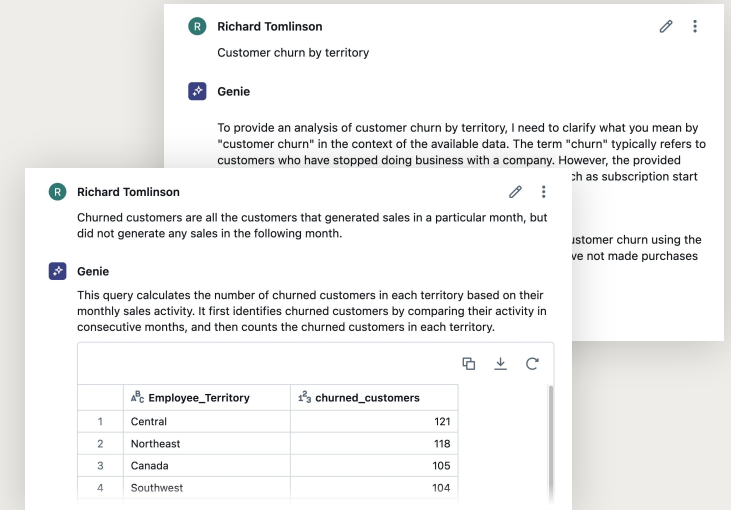
Your Genie is tailored for your data and business



Learns from your
metadata and semantics
in Unity Catalog



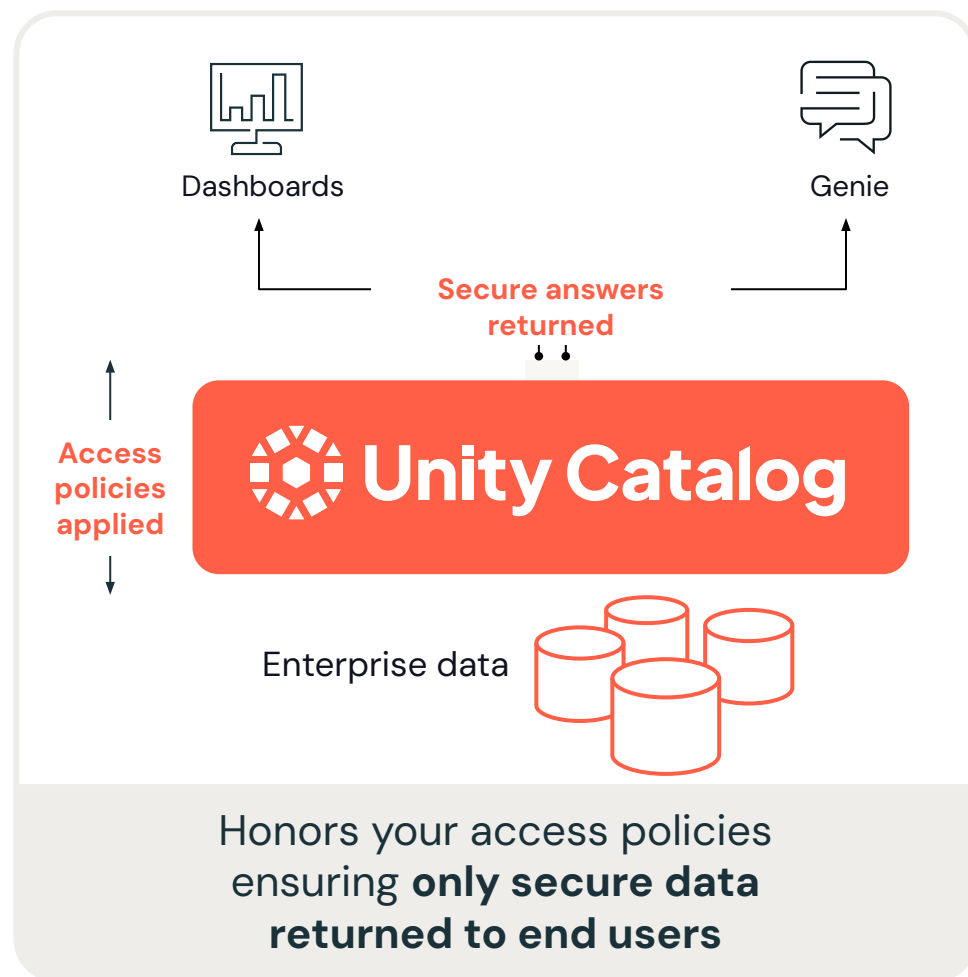
Enhanced context from
space-level instructions
and examples



Continually evolves as
users interact and provide
real-time feedback



Governed and secure AI-generated insights



Sales & Marketing Performance ☆ > **Monitoring** + New chat History Configure Monitoring Share

History of the activities within your space. Understand the errors, feedback, and results the space is returning.

2025-01-01 → 2025-05-05 Rating Request User Status [Reset filters](#)

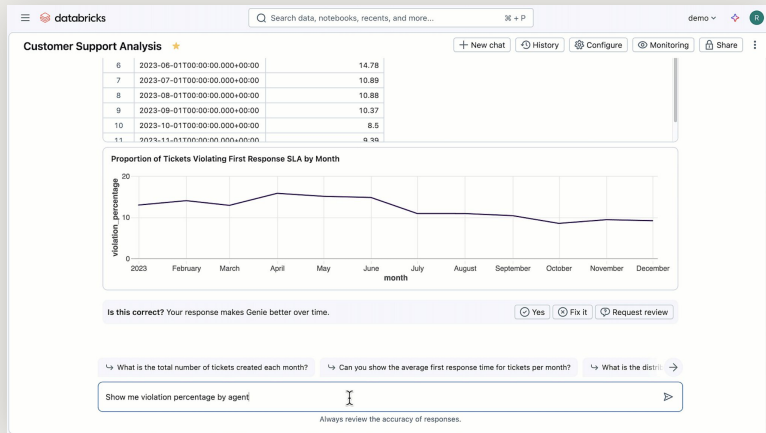
Question	Rating	Request	Comment	User	Created	Last updated
✓ Show sales and avg price by FQ ⓘ	👎 Bad	🔔 Requested	Was expecting t...	Richard Toml...	Jan 03, 2025, 1...	Jan 23, 2025, 17...
✓ add profit to this ⓘ	👍 Good			Mary Drake	Jan 03, 2025, 1...	Jan 23, 2025, 17...
✓ add average discount to this ⓘ	👍 Good	🔔 Requested	Might need to...	Mary Drake	Jan 03, 2025, 1...	Jan 23, 2025, 17...
✓ Hows my business doing? ⓘ	👍 Good			Richard Toml...	Jan 03, 2025, 1...	Jan 23, 2025, 17...
✓ Show total spend by territory in FY24 ⓘ	👍 Good			Richard Toml...	Jan 03, 2025, 1...	Jan 23, 2025, 17...
✓ Show lost revenue due to customer churn in ... ⓘ	👍 Good			Richard Toml...	Jan 03, 2025, 1...	Jan 23, 2025, 17...
✓ Churned customers are customers that gene... ⓘ	👎 Bad			Richard Toml...	Jan 03, 2025, 1...	Jan 23, 2025, 17...
✓ Show by customer but only for Southwest te... ⓘ	👍 Good	🔔 Requested	Looks like the...	Richard Toml...	Jan 03, 2025, 1...	Jan 23, 2025, 17...
✓ Show lost revenue by customer ⓘ	👍 Good			Richard Toml...	Jan 03, 2025, 1...	Jan 23, 2025, 17...

< Previous Next > 100 / page

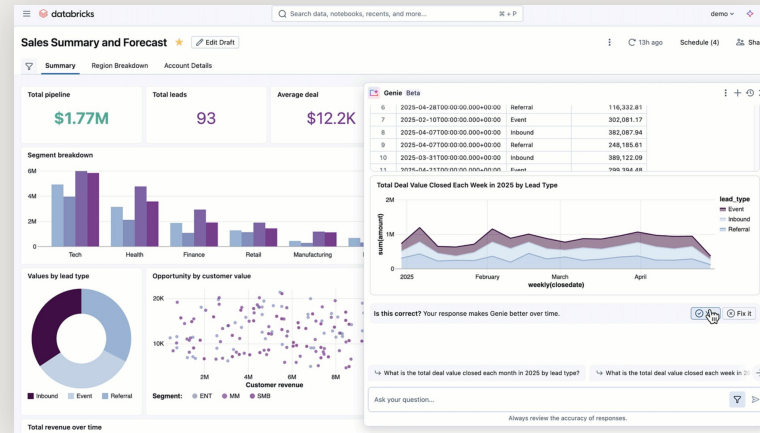
Built-in monitoring to review the every question. Evaluate thumbs up/down ratings to **improve accuracy for future questions**



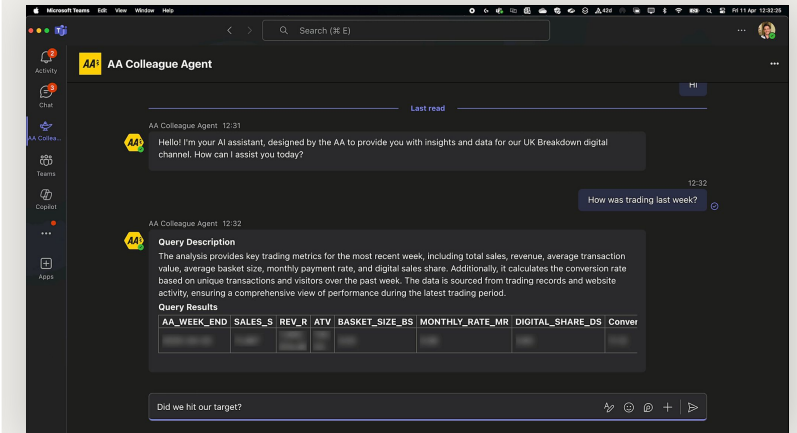
Genie is accessible from anywhere



Dedicated
Genie Spaces
for focused Q&A



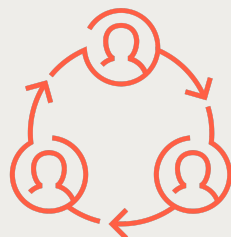
Integrated in every
Dashboard for follow
up questions



Available in external
tools via Genie
Conversation APIs

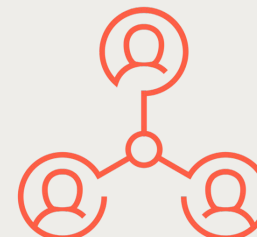


Securely share insights **with any audience**



Your organization

Securely share actionable insights
across your organization



Customers and partners

Deliver new value to your stakeholders
with AI-powered analytics embedded in
your applications



Governed by Unity Catalog



Genie takes Dashboards to the next level

Go beyond canned answers

Stay in the flow

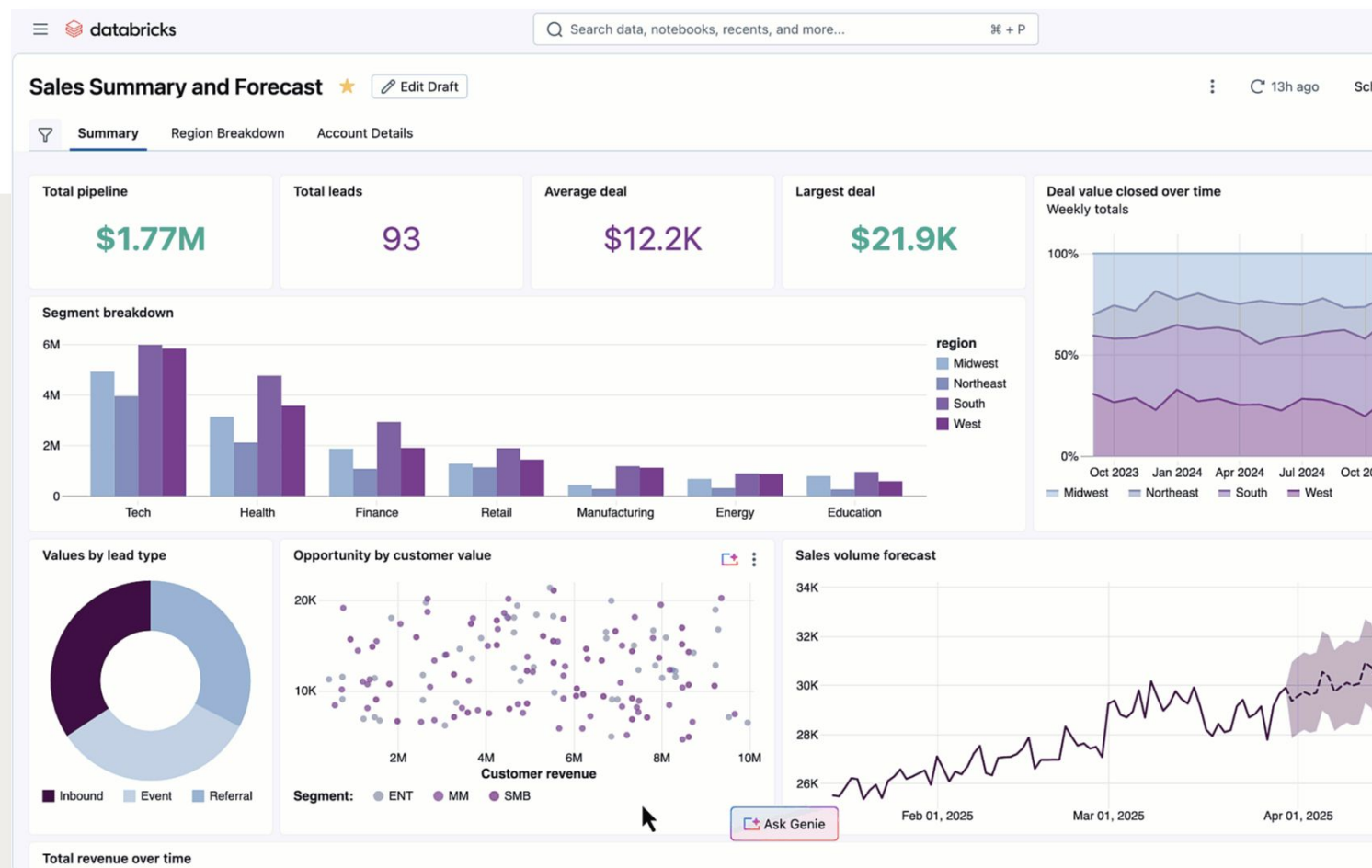
Ask questions and get answers based on what you see without navigating away and losing context.

Uncover “What if” and “Why”

Forecast based on historical trends and identify key drivers of an observed increase/decrease.

Short circuit the feedback loop

Fill any dashboard content gaps yourself and get all your analysis done in one session.



Genie Research Agent.

Tackle complex “why” questions

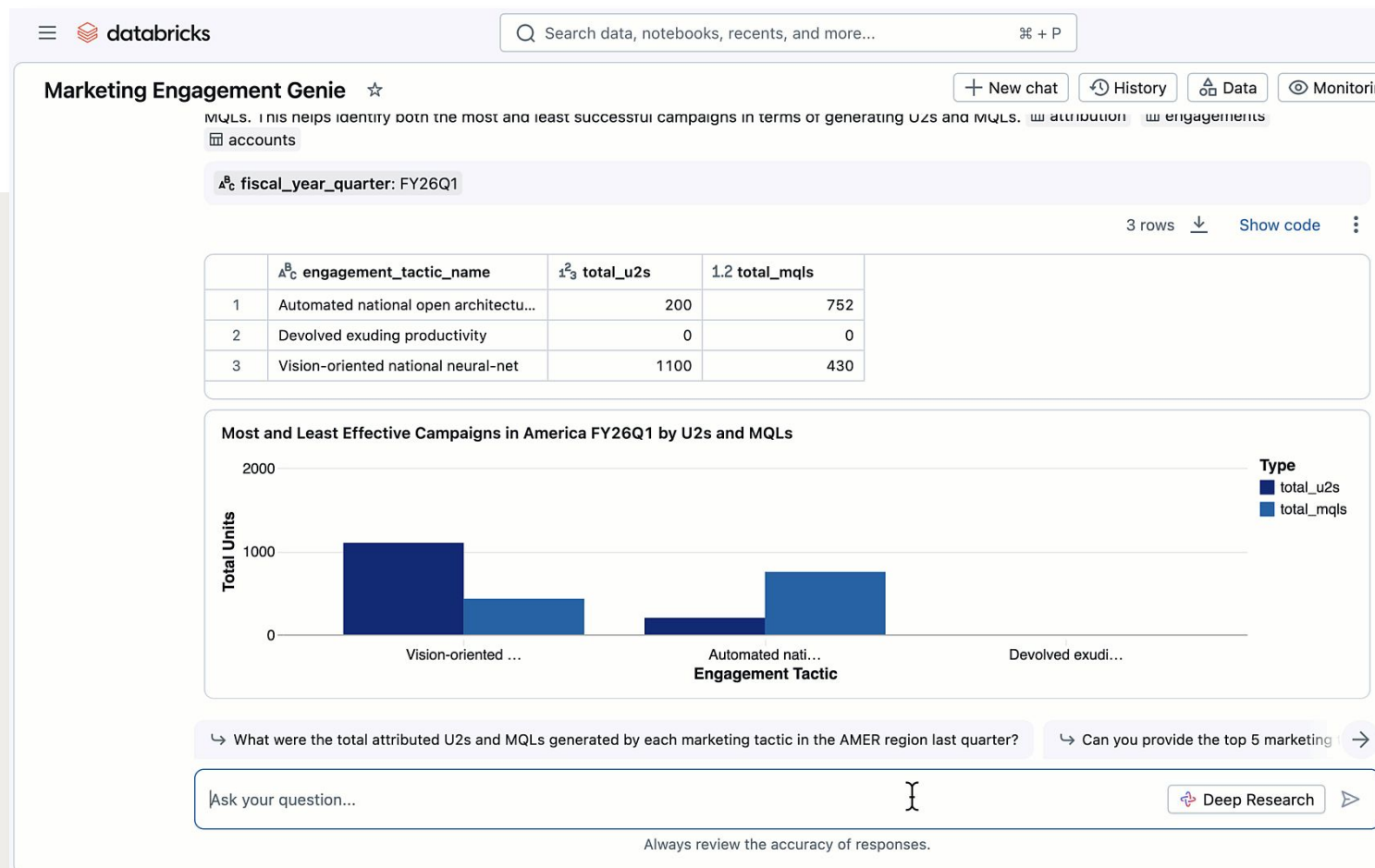
Automatically generates multi-step research plans to explore root causes and explain business anomalies.

Backs every answer with proof

Genie shows its work, referencing exactly where the insight came from via citations with supporting data.

Deep analysis at scale

Emulates how a skilled analyst would investigate complex data questions, accelerating insight without sacrificing accuracy.



Genie – Tips, Tricks and Best Practices



Genie: Conceptually

Best practices for Genie space setup

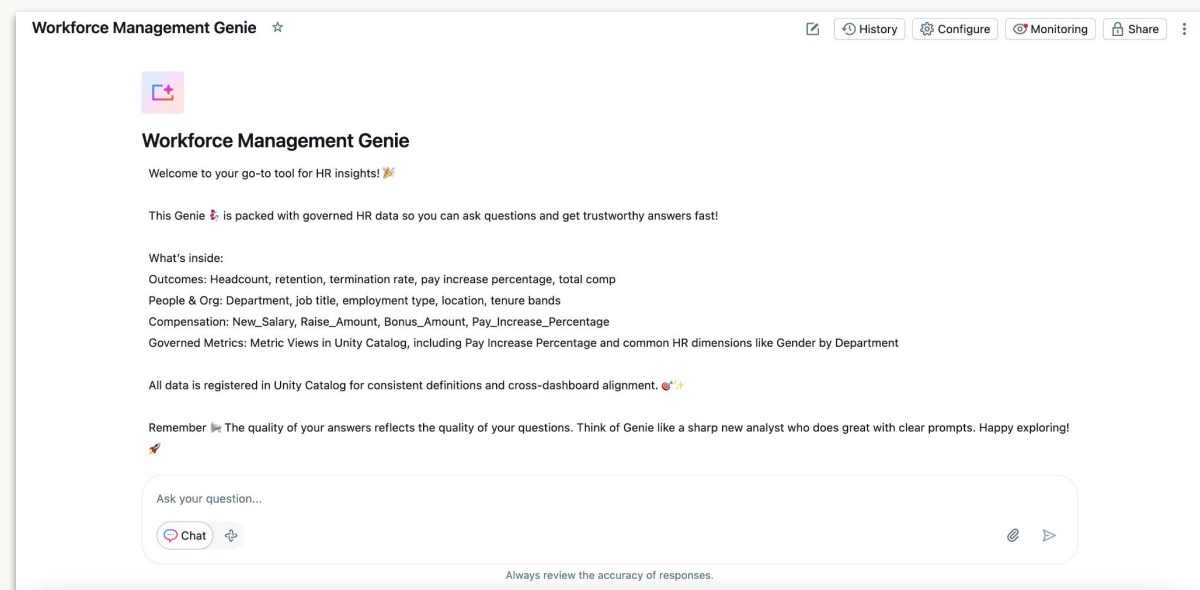
- Think of Genie as a **brand new analyst** joining your organization as their first job post university
- This analyst is going to be **pretty good at writing SQL but will have no pre-existing knowledge about your business** – not high level concepts, not low level jargon
- The only knowledge this analyst will have about your business is the context provided in that particular space
 - **If context is vague or contains conflicting guidance, expect Genie to get confused**, just like a human analyst would be



Genie: Topic selection

Best practices for Genie space setup

- Create a Genie space that maps to a single, focused topic
 - Just like there's no single dashboard for the organization, there's no single Genie space
- A topic should be broad enough to encapsulate a meaningful dataset but narrow enough to provide focused insights
 - Examples include product line performance, marketing campaign analysis, in stock inventory, regional supply chain
- Gather a representative **set of 10–15 questions end users typically ask** of this data



Genie: Benchmarks

Best practices for Genie space setup

- Create a set of questions that you expect users to ask as benchmarks
- Create “correct SQL” for each benchmark question; it may take some iteration to make sure the question fits the answer
- Run benchmark as you iterate on your instructions

The screenshot displays the Databricks Genie Benchmarks interface. On the left, a list of benchmark questions is shown, with the first one selected: "which mid market accounts have...". The main panel shows the details for this benchmark. The question is "which mid market accounts have generated the most sales globally?". The assessment is "Good". The response section shows the "Model output SQL" and the "Ground truth SQL". Both SQL queries are identical, showing a SELECT statement that filters for '5. Closed Won' and 'MM' company size segments, groups by account name, and orders by total sales in descending order, limited to 5 results. Below the SQL, a table shows the results for the account 'Pixope' with a total sales of 189,887.21.

Sales Opportunities ★ > **Benchmarks** + New chat History Configure Monitoring Share ⋮

which mid market accounts have...

top 10 customers this year?

top 10 customers by sales in 2024?

Give me top 10 customers by revenue ...

which mid market accounts have...

Which accounts have we generated th...

Who is my largest customer in europe?

who are my most active sales reps by...

Who is my best rep in asia by sales?

Who are my best 5 salespeople by...

which mid market accounts have generated the most sales globally?

Assessment: ✨ Good ✎

Question

which mid market accounts have generated the most sales globally?

Response

Model output SQL Copy

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rocks_schema.accounts a
  JOIN acme_co_sales.big_rocks_schema.big_rock_opportuni
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment__c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

	account_name	total_sales
1	Pixope	189,887.21

Ground truth SQL Copy

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rocks_schema.accounts a
  JOIN acme_co_sales.big_rocks_schema.big_rock_opportuni
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment__c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

	account_name	total_sales
1	Pixope	189,887.21



Genie: Data preparation

Best practices for Genie space setup

- **UC metadata is the first and best option** for providing Genie with the context it needs to answer end user questions
- Focused set of UC tables with **defined PK/FK relationships and populated table/column descriptions**
 - Flatten into views if needed, dropping extraneous columns
- Include **value dictionaries** on key string columns

The screenshot displays the Databricks Genie interface within the 'Sales Opportunities' workspace. The 'Benchmarks' tab is active, showing a list of questions on the left. The selected question is 'which mid market accounts have generated the most sales globally?'. The interface shows an assessment of 'Good' and the question text in a search bar. Below, the 'Response' section contains two panels: 'Model output SQL' and 'Ground truth SQL', both displaying identical SQL queries. The queries select account names and total sales, rounded to 2 decimal places, from the 'acme_co_sales.big_rock_schema.accounts' table, joined with 'acme_co_sales.big_rock_schema.big_rock_opportunities' on account ID. The results are filtered by 'o.stagename = '5. Closed Won'' and 'a.company_size_segment__c = 'MM'', grouped by account name, ordered by total sales descending, and limited to 5 results. Below the SQL, two tables show the results. Both tables have columns 'account_name' and 'total_sales'. The first table shows 'Pixope' with '189,887.21' in sales. The second table also shows 'Pixope' with '189,887.21' in sales.

Model output SQL

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rock_schema.accounts a
  JOIN acme_co_sales.big_rock_schema.big_rock_opportunities o
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment__c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

	account_name	total_sales
1	Pixope	189,887.21

Ground truth SQL

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rock_schema.accounts a
  JOIN acme_co_sales.big_rock_schema.big_rock_opportunities o
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment__c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

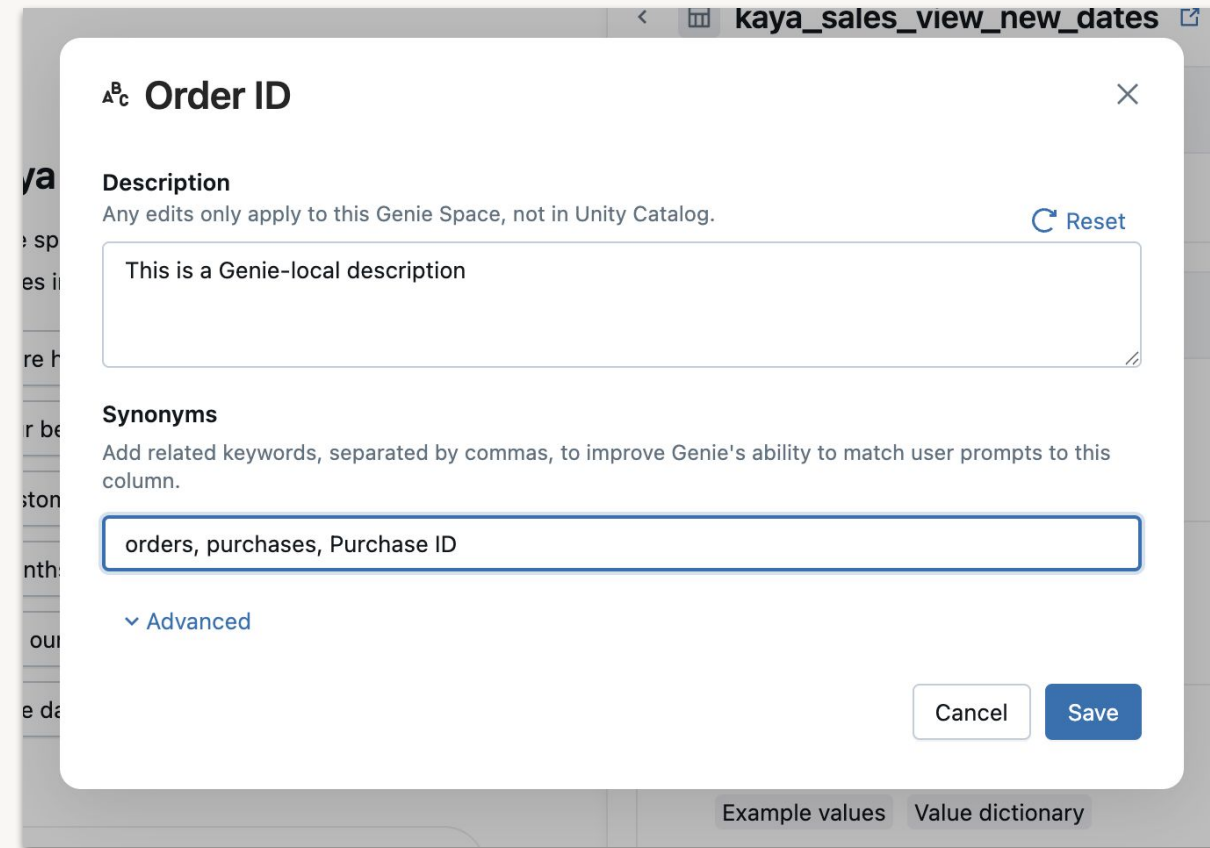
	account_name	total_sales
1	Pixope	189,887.21



Genie: Data preparation

Best practices for Genie space setup

- **Knowledge Store** provides Genie local semantics for when you can't/don't want to update UC metadata
 - Space-level table/column descriptions
 - Column synonyms
 - Show/hide columns
 - Joins
- Include **value dictionaries** on key string columns



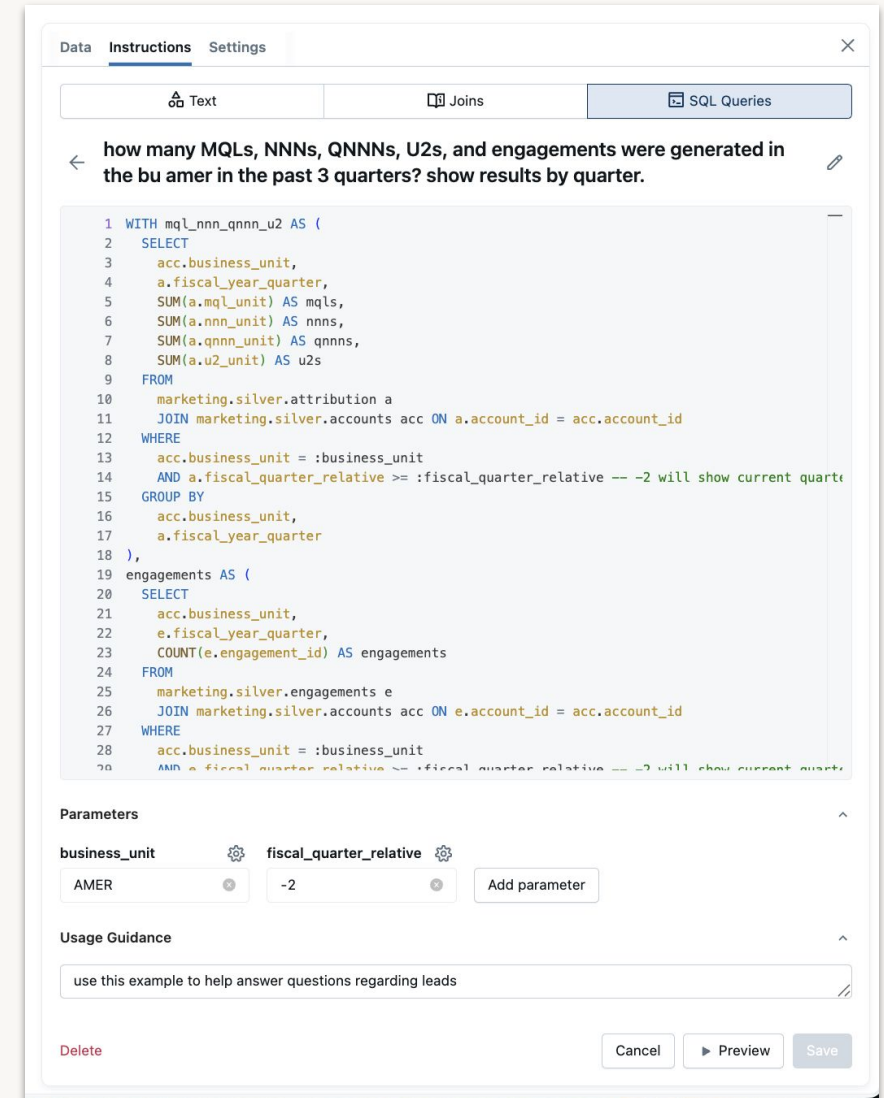
The screenshot shows a web interface for configuring a Genie space. The browser tab is labeled 'kaya_sales_view_new_dates'. The main dialog is titled 'Order ID' and has a close button (X) in the top right corner. Below the title, there is a 'Description' section with the text 'Any edits only apply to this Genie Space, not in Unity Catalog.' and a 'Reset' button. A text input field contains the text 'This is a Genie-local description'. Below this is a 'Synonyms' section with the text 'Add related keywords, separated by commas, to improve Genie's ability to match user prompts to this column.' and a text input field containing 'orders, purchases, Purchase ID'. At the bottom of the dialog, there is a 'Cancel' button and a 'Save' button. Below the dialog, there are two tabs: 'Example values' and 'Value dictionary'.



Genie: Example SQL

Best practices for Genie space setup

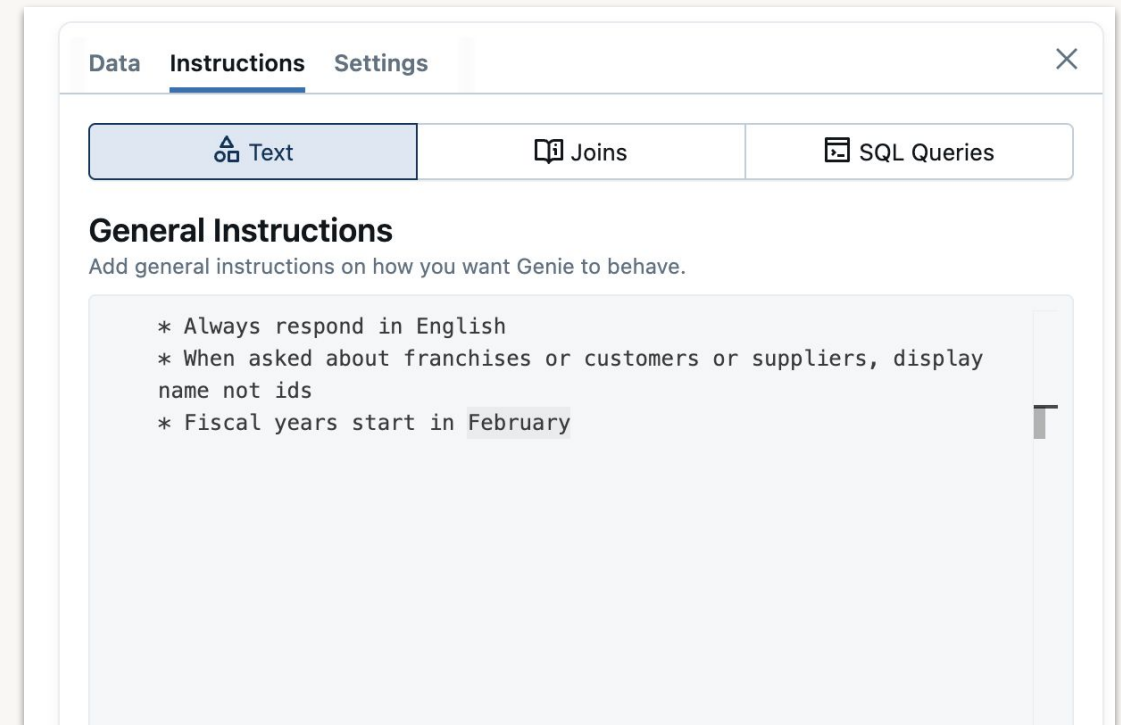
- After UC metadata and knowledge store, this is your **second best option** providing Genie with context
- Leverage **parameterized example SQL** wherever possible to illustrate the representative questions you gathered
- **Leverage Usage Guidance to include additional guidance or alternate phrasing** that should map to that same SQL
 - The question title will have more impact on Genie's behavior than Usage Guidance so include the most common phrasing as the title
- **Bias towards Example SQL over UC Functions when possible**



Genie: General text instructions

Best practices for Genie space setup

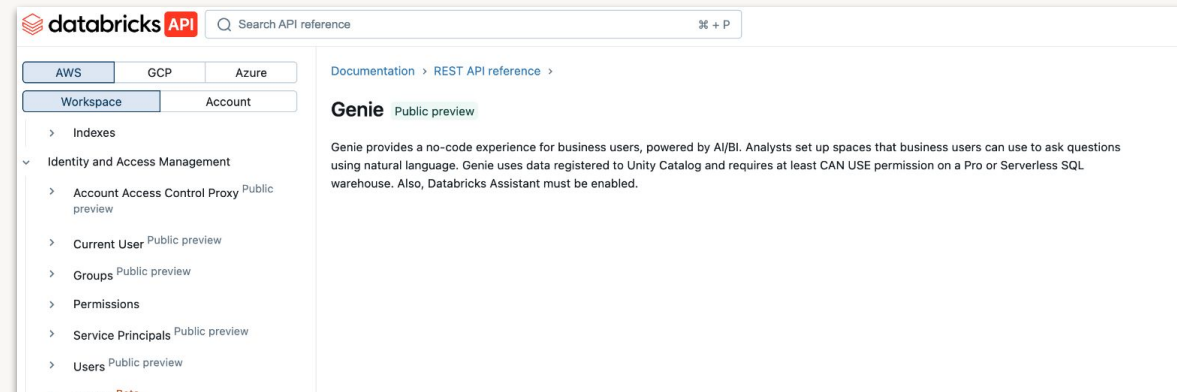
- **General text instructions are the last option** – if something could be expressed as table metadata or Example SQL, it should. E.g.
 - Use example SQL to illustrate how to calculate margin
 - Illustrate join relationships via Knowledge Store Joins
- Best for context that should apply to every single prompt & response, e.g.
 - Format all numeric results to two decimal places
 - All questions related to sales must include a time range that has a specific year; if that is omitted refuse to answer and ask the user to specify
 - Fiscal years start in February
- There is not any filtering on General text instructions so occupies a lot of context



Genie: Application best practices

Best practices for Genie space setup

- REST API documentation
 - Genie conversation API is stateful – conversation history is stored
 - To use in stateless way, start a new conversation for each question
- Implement retry or queuing logic in your application to smooth the user experience during peak concurrency
- Cache answers to common questions to skip Genie API. Doing so will increase determinism and decrease throughput required



Genie: Rollout

Best practices for Genie space setup

- Once you feel your space is ready, **use it yourself for a few weeks to answer new questions** that come in
 - Adjust instructions as needed
- Identify a group of 5–10 initial test users. Ahead of rollout to that group:
 - Update description & starter questions
 - Establish **benchmarks**
 - Configure **alerts** on user feedback
- Expand to a broader audience after success with the initial group

The screenshot displays the Databricks Genie interface within a workspace named 'Sales Opportunities'. The 'Benchmarks' tab is active, showing a list of questions on the left and a detailed view of a selected benchmark on the right. The benchmark question is 'which mid market accounts have generated the most sales globally?'. The assessment is 'Good'. The question text is 'which mid market accounts have generated the most sales globally?'. The response section shows the 'Model output SQL' and the 'Ground truth SQL', both of which are identical. Below the SQL, there are two tables showing the results of the query.

Model output SQL

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rocks_schema.accounts a
  JOIN acme_co_sales.big_rocks_schema.big_rock_opportuni
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment_c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

	account_name	total_sales
1	Pixope	189,887.21

Ground truth SQL

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rocks_schema.accounts a
  JOIN acme_co_sales.big_rocks_schema.big_rock_opportuni
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment_c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

	account_name	total_sales
1	Pixope	189,887.21



Alert on user feedback

Best practices for Genie space setup

- The audit system table will typically reflect all events, including Genie user feedback, within ~15 minutes
- You can query that system table and then user that query in an alert (documentation)

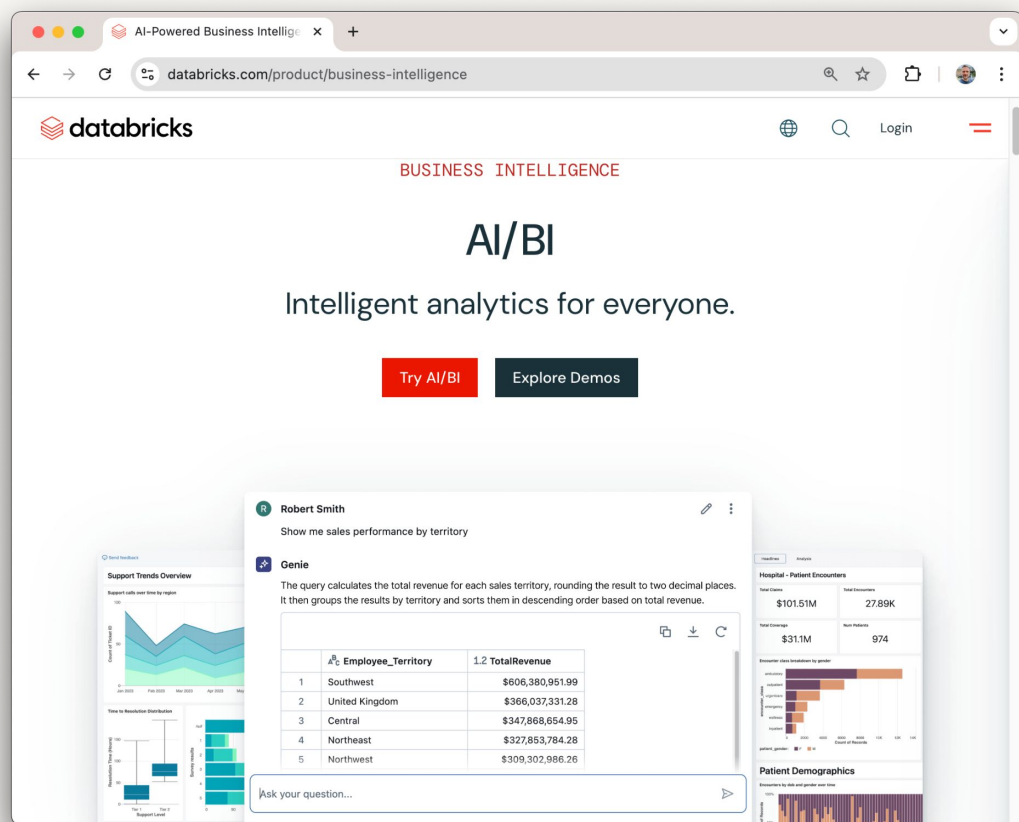
Feedback via API

```
SELECT
  user_identity.email as user_email,
  action_name,
  request_params.space_id,
  request_params.feedback_rating,
  *
FROM
  system.access.audit
WHERE
  service_name = 'aibiGenie'
  AND action_name =
    'genieSendMessageFeedback'
  AND event_date >= current_date() -
    interval 30 days
```



Want to know more about Genie?

Get started today



How to learn more about Genie

- Try Genie in the Databricks [demo center](#)
- Download [Databricks Free Edition](#)
- Take the free [training course](#)
- Read the [announcement blog](#) and [product documentation](#)
- Understand Genie [best practices](#)

Visit our web page

www.databricks.com/product/business-intelligence



Genie Onboarding Plan



Onboarding Plan

Ensure successful completion of your business team goals

- Achieve a validated, high-quality Genie space that reaches business users effectively
- Focuses on structured, phased rollout for successful adoption and end-user reliability
- Designed to maximize insights while minimizing confusion or failure points



Tracking Onboarding Steps

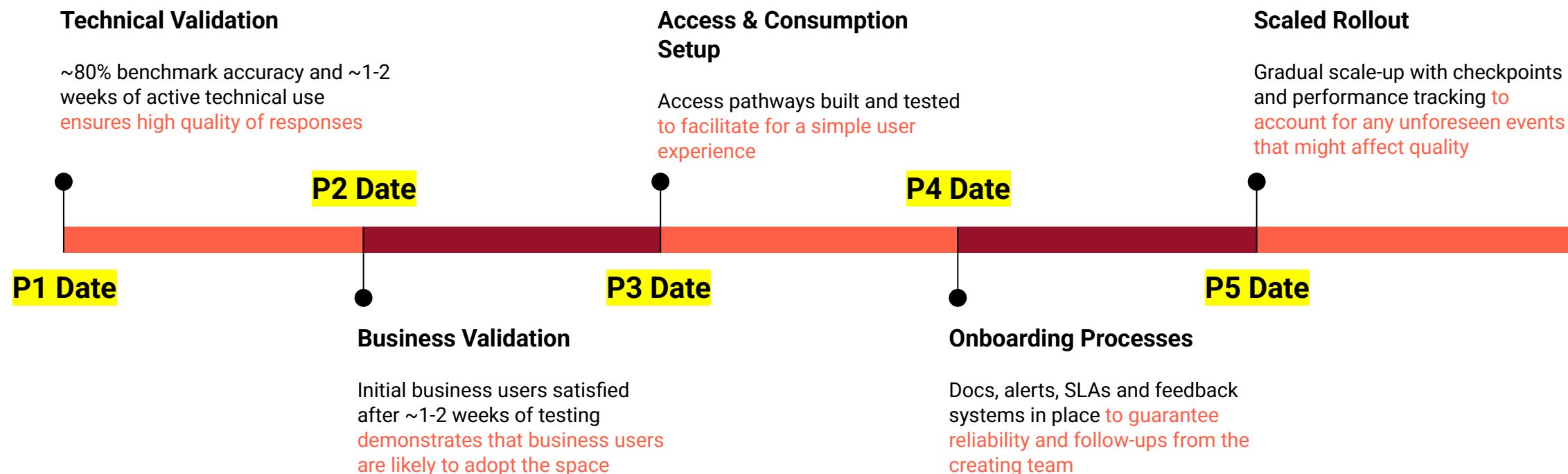
Assign clear owners and due dates for progression

Milestone Tracker				
✓ Milestone	Owner	 Due Date	Status	Notes / Blockers
Phase 1 - Technical User Validation of the Genie Space	Person	Date	Not started	
Phase 2 - Business User Validation of the Genie Space	Person	Date	Not started	
Phase 3 - Prepare Access and Consumption Layer	Person	Date	Not started	
Phase 4 - Put Onboarding & Monitoring Processes in Place	Person	Date	Not started	
Phase 5 - Stage the Rollout to More Users	Person	Date	Not started	



Tracking Onboarding Steps

Timeline and recommended exit criteria



Scale Out

- Genie Build Workshop
- Genie hackaton



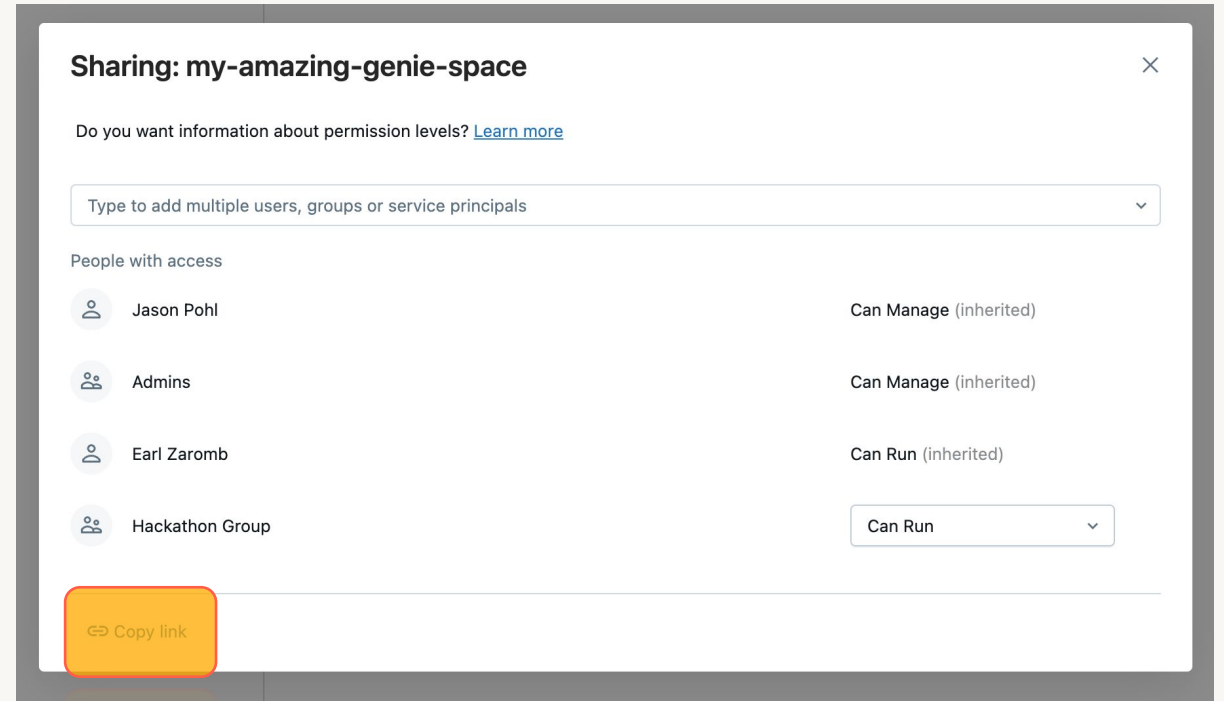
How to Use Genie



Login to Genie

Bookmark for Later

1. [Admin] Click Share and then “Copy Link”
2. [Admin] Share this link with hackathon users
3. [Users] Login & Bookmark this page



Explain the data set

Click this (or ask the question)



Explain the data set

Will summarize and describe
all the tables in the Genie
Space



Genie

There are 10 tables available, each serving different purposes:

1. `system.billing.usage`: Contains account-wide billable usage data.
2. `system.access.audit`: Logs access and audit events.
3. `system.billing.list_prices`: Historical log of SKU pricing changes.
4. `system.compute.clusters`: History of compute configurations over time.
5. `system.compute.warehouse_events`: Captures events related to SQL warehouses.
6. `system.compute.node_types`: Information about available node types and their hardware specifications.
7. `system.lakeflow.job_run_timeline`: Tracks job run times and metadata.
8. `system.lakeflow.job_task_run_timeline`: Tracks job task run times and metadata.
9. `system.lakeflow.job_tasks`: History of job task configurations.
10. `system.lakeflow.jobs`: History of job configurations.

Connections:

- `usage` and `list_prices` are related through `sku_name`.
- `clusters` and `usage` can be linked via `cluster_id` in `usage_metadata`.
- `warehouse_events` and `usage` can be linked via `warehouse_id` in `usage_metadata`.
- `job_run_timeline`, `job_task_run_timeline`, `job_tasks`, and `jobs` are related through `job_id` and `run_id`.

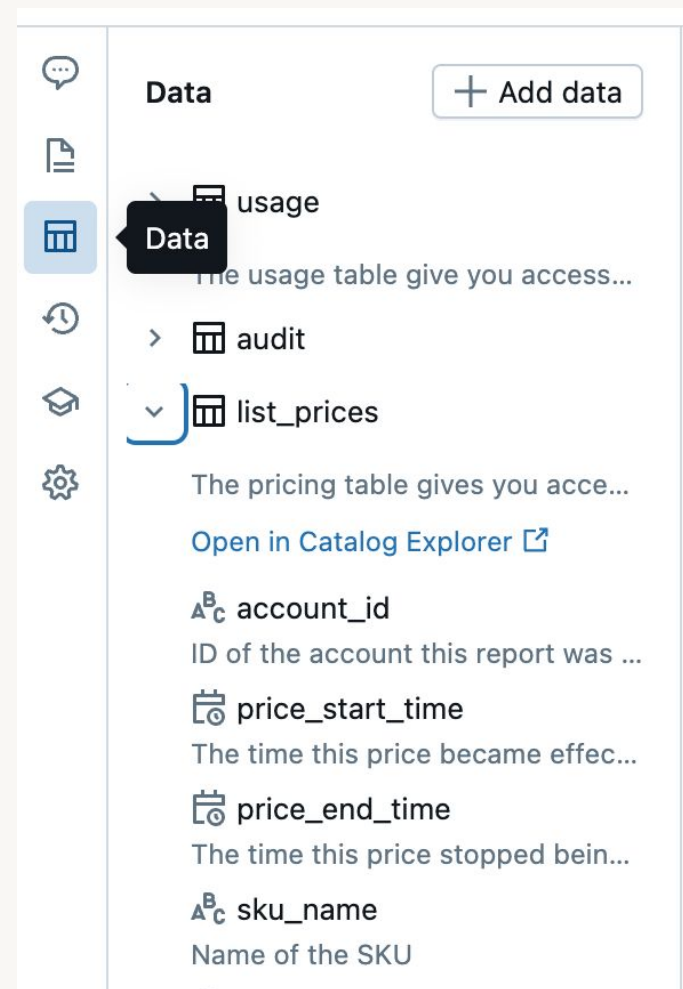


Explore the data set

Click the Data Icon

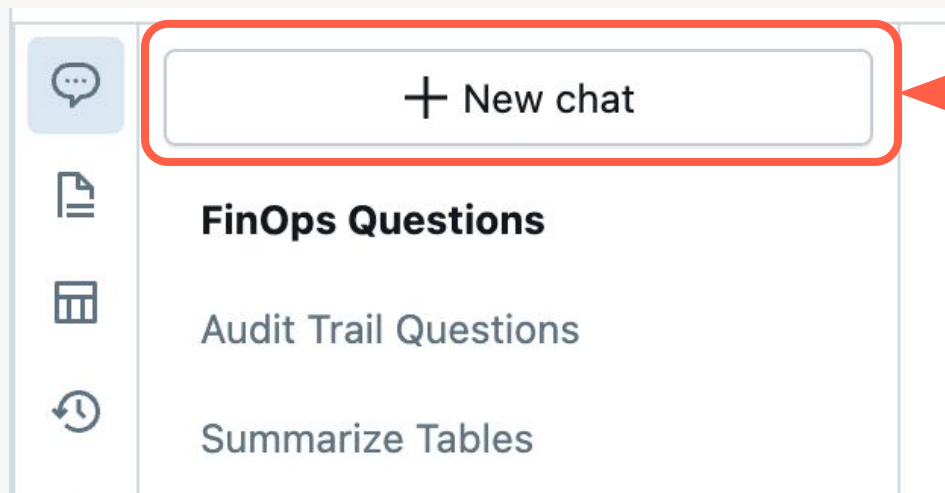


Allows you to explore the tables, views, columns, and descriptions



Multiple Chats

Click on “+ New Chat” to start a new chat conversation.

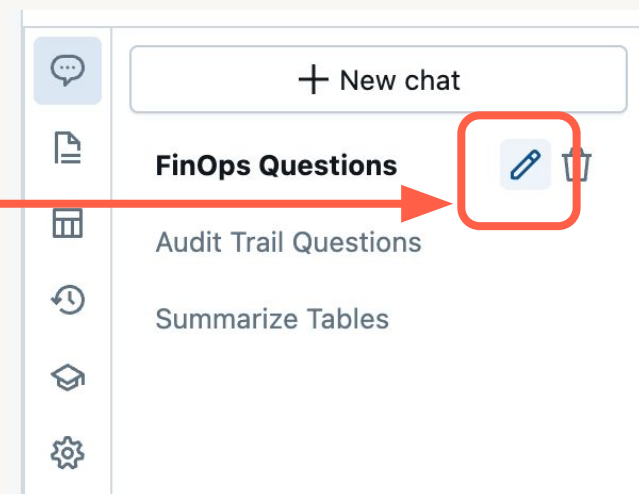


Each chat will use conversation history when answering new questions

Other users cannot see your chats

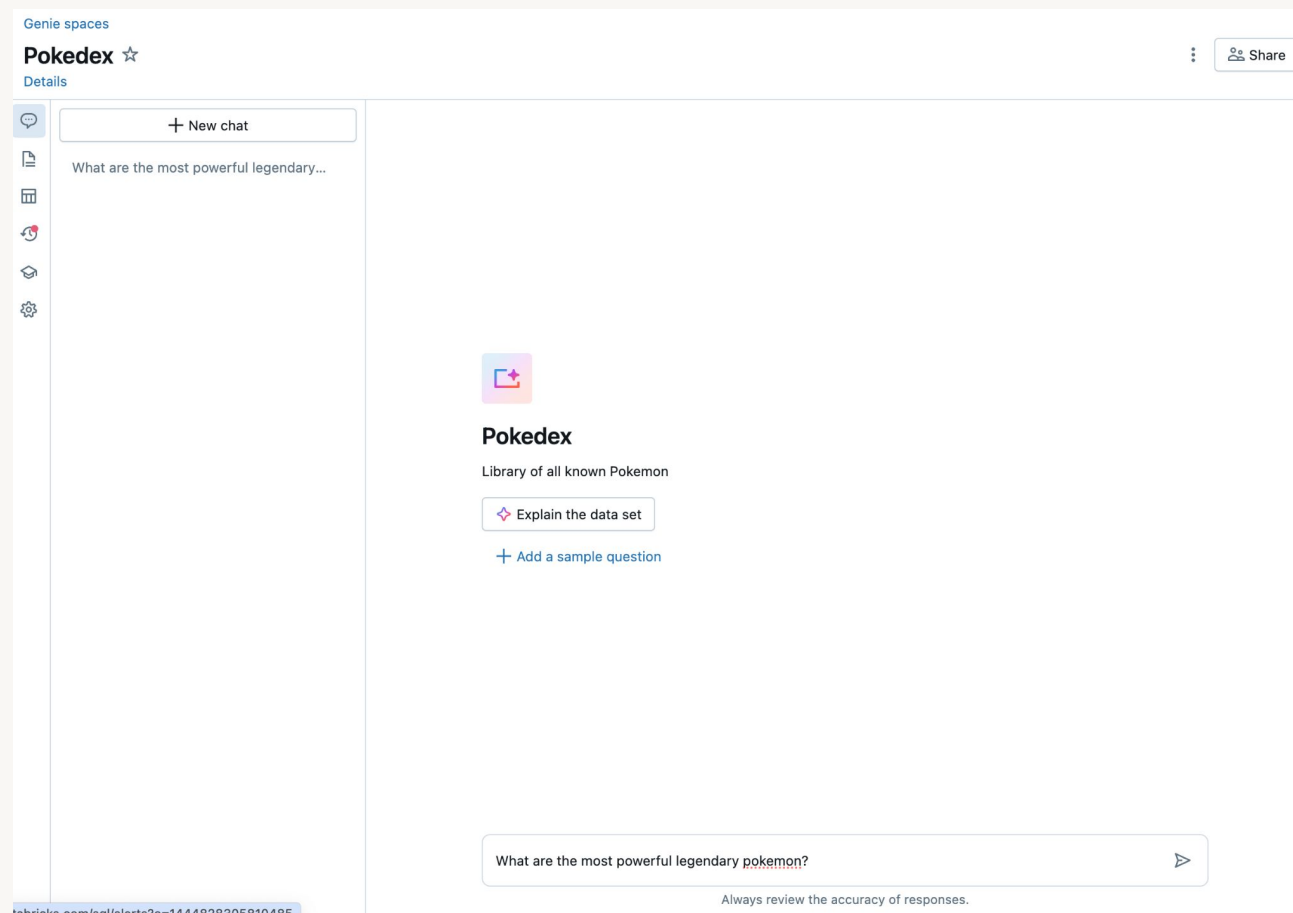
Hover and click on the pen icon to edit the name of the chat

Allows you to group your frequent topic together



Ask a Question

[Admin] Demonstrate with a successful question from the Benchmark



Evaluate the Answer

- Look at the answer and gauge if it feels correct based on your domain knowledge.
- Did it include the columns and rows you expected?
- Indicate as such with the feedback buttons [next slide]

Jason Pohl What are the most powerful legendary pokemon?

Genie This query retrieves the names and total base points of the top 10 legendary Pokémon, sorted by their total base points and then by name.

	^A _C name	¹ ₂ ³ base_total
1	Mewtwo	780
2	Rayquaza	780
3	Groudon	770
4	Kyogre	770
5	Arceus	720
6	Zygarde	708
7	Diancie	700
8	Kyurem	700
9	Latias	700
10	Latios	700

10 rows

 [Add as instruction](#)  [Show generated code](#)

Quick actions

 Auto visualize  Bar chart  Line chart  Pie chart

Ask your question... 

Always review the accuracy of responses.

End user feedback

3

Monitor & Improve

Is this correct? Your response makes Genie better over time.

☐ Yes ☒ Fix it ☐ Request review

Is this correct? Your response makes Genie better over time.

☐ Yes ☒ Fix it ☐ Request review

(Optional) Explain what is wrong with the response

Is this correct? Your response makes Genie better over time.

☐ Yes ☒ Fix it ☐ Request review

Add a comment for the space admin (optional)

Note: This end user feedback does not automatically change the model's behavior. Instead, it surfaces the issue to the space author so they can determine how to update space context (if at all) in response.

Evaluate the Code

Advanced: Only if you understand SQL

Click on “Show generated code” to view the SQL that Genie created

J Jason Pohl

What are the most powerful legendary pokemon?

Genie

This query retrieves the names and total base points of the top 10 legendary Pokémon, sorted by their total base points and then by name.

📄 ⬇️ ↺

	^A _C name	¹ ² ₃ base_total
1	Mewtwo	780
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7	Diancie	700
8	Kyurem	700
9	Latias	700
10	Latios	700

10 rows

👍 📄 🗨️ Add as instruction

Show generated code

Quick actions

🌟 Auto visualize

📊 Bar chart

📈 Line chart

📊 Pie chart

Ask your question...

➤

Always review the accuracy of responses.

Evaluate the Code

Can Edit & Rerun SQL

SQL should reflect what you would have written on your own without Genie to get the right answer

Jason Pohl

What are the most powerful legendary pokemon?

Genie

This query retrieves the names and total base points of the top 10 legendary Pokémon, sorted by their total base points and then by name.

	name	base_total
1	Mewtwo	780
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5	Arceus	720
6	Zygarde	708
7	Diancie	700
8	Kyurem	700
9	Latias	700
10	Latios	700

10 rows



Add as instruction



Hide generated code

```
1 SELECT
2   `name`,
3   `base_total`
4 FROM
5   `users`.`jason`.`pokemon`
6 WHERE
7   `is_legendary` = 1
8 ORDER BY
9   `base_total` DESC,
10  `name`
11 LIMIT
12  10
```

Run

Quick actions

Auto visualize

Bar chart

Line chart

Pie chart



Visualize the Answer

Genie can chart the answer for you

Click on “Auto Visualize” to view the answer in a chart

J Jason Pohl

What are the most powerful legendary pokemon?

Genie

This query retrieves the names and total base points of the top 10 legendary Pokémon, sorted by their total base points and then by name.

📄

📄

🔄

	name	base_total
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10 rows

👍

🗨️

?

Add as instruction

🔄

> Show generated code

Quick actions

🌟 Auto visualize

📊 Bar chart

📈 Line chart

🥞 Pie chart

Ask your question...

➤

Always review the accuracy of responses.

Visualize the Answer

Simple Visualizations are a click away

Be sure to give feedback on the visualizations too

Jason Pohl

What are the most powerful legendary pokemon?

Genie

This query retrieves the names and total base points of the top 10 legendary Pokémon, sorted by their total base points and then by name.

	A ^B _C name	1 ² ₃ base_total
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6	Zygarde	708
7	Diancie	700
8	Kyurem	700
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10	Latios	700

10 rows

Remove as instruction

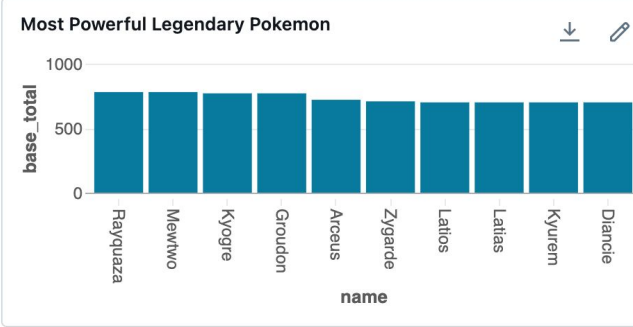
> Show generated code

Jason Pohl

Visualize

Genie

Most Powerful Legendary Pokemon



base_total

name

Where does the feedback go?



Monitor Your Space

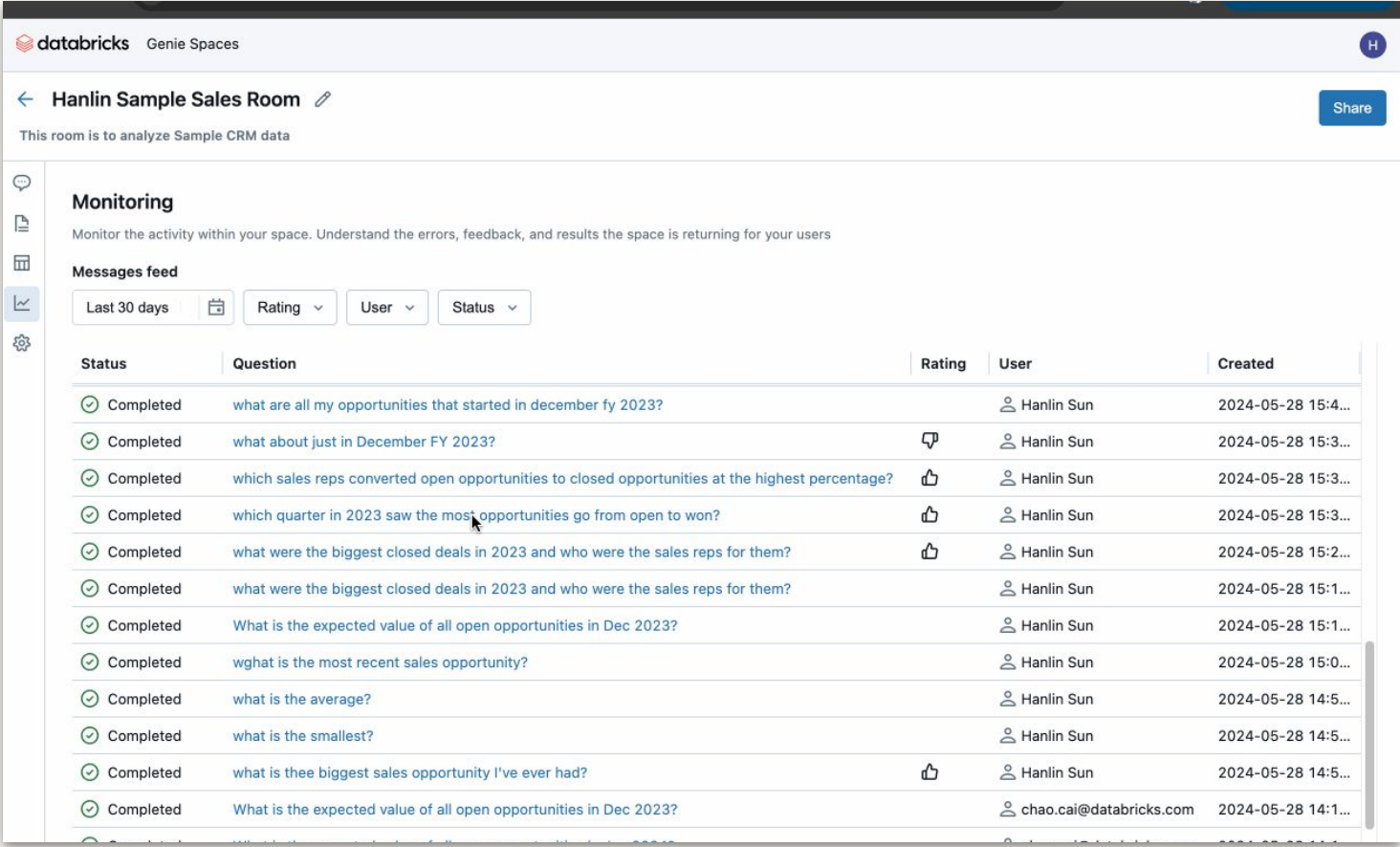
3

Monitor & Improve

Only users with Can Manage permission on the space can view this screen [Technical Stakeholder]

Review this page regularly (daily / weekly)

Review in real-time with Business Stakeholder



The screenshot displays the 'Monitoring' interface within a Databricks Genie Space. The space is titled 'Hanlin Sample Sales Room' and is intended for analyzing Sample CRM data. The 'Monitoring' section provides a 'Messages feed' showing a list of user questions, their status (all 'Completed'), and the user who asked them. The feed includes filters for 'Last 30 days', 'Rating', 'User', and 'Status'. The table below lists the questions and their details.

Status	Question	Rating	User	Created
Completed	what are all my opportunities that started in december fy 2023?		Hanlin Sun	2024-05-28 15:4...
Completed	what about just in December FY 2023?		Hanlin Sun	2024-05-28 15:3...
Completed	which sales reps converted open opportunities to closed opportunities at the highest percentage?		Hanlin Sun	2024-05-28 15:3...
Completed	which quarter in 2023 saw the most opportunities go from open to won?		Hanlin Sun	2024-05-28 15:3...
Completed	what were the biggest closed deals in 2023 and who were the sales reps for them?		Hanlin Sun	2024-05-28 15:2...
Completed	what were the biggest closed deals in 2023 and who were the sales reps for them?		Hanlin Sun	2024-05-28 15:1...
Completed	What is the expected value of all open opportunities in Dec 2023?		Hanlin Sun	2024-05-28 15:1...
Completed	wghat is the most recent sales opportunity?		Hanlin Sun	2024-05-28 15:0...
Completed	what is the average?		Hanlin Sun	2024-05-28 14:5...
Completed	what is the smallest?		Hanlin Sun	2024-05-28 14:5...
Completed	what is thee biggest sales opportunity I've ever had?		Hanlin Sun	2024-05-28 14:5...
Completed	What is the expected value of all open opportunities in Dec 2023?		chao.cai@databricks.com	2024-05-28 14:1...



Monitor Your Space

Monitor & Improve



Are there questions that look high value?

- Follow Up with the user to understand the impact this answer had to them
- Broadcast these examples to the users on a regular basis to showcase the art of the possible
- Share success stories with Databricks team



Which answers missed the mark?

- Check comments for why
- Add this question to the benchmark
- Refine space until correct
- If Genie still can't answer correctly, share details with Databricks



Request for Review

Monitor & Improve

Technical Stakeholder: Filter History for “Requested”

History

History of the activities within your space. Understand the errors, feedback, and results the space is returning.

Last 30 days

Rating

Request: Requested

User

Status

Reset filters

Question	Rating	Request	Comment	User	Created	Last updated
✓ Yes ⓘ		ⓘ Requested	We should create ...	👤 Jason Pohl	2024-12-17 15:17:...	2024-12-17 15:18:11

Technical Stakeholder: Click into requests and review

DONE - created function

Cancel request

Mark as reviewed



Share the Genie Space

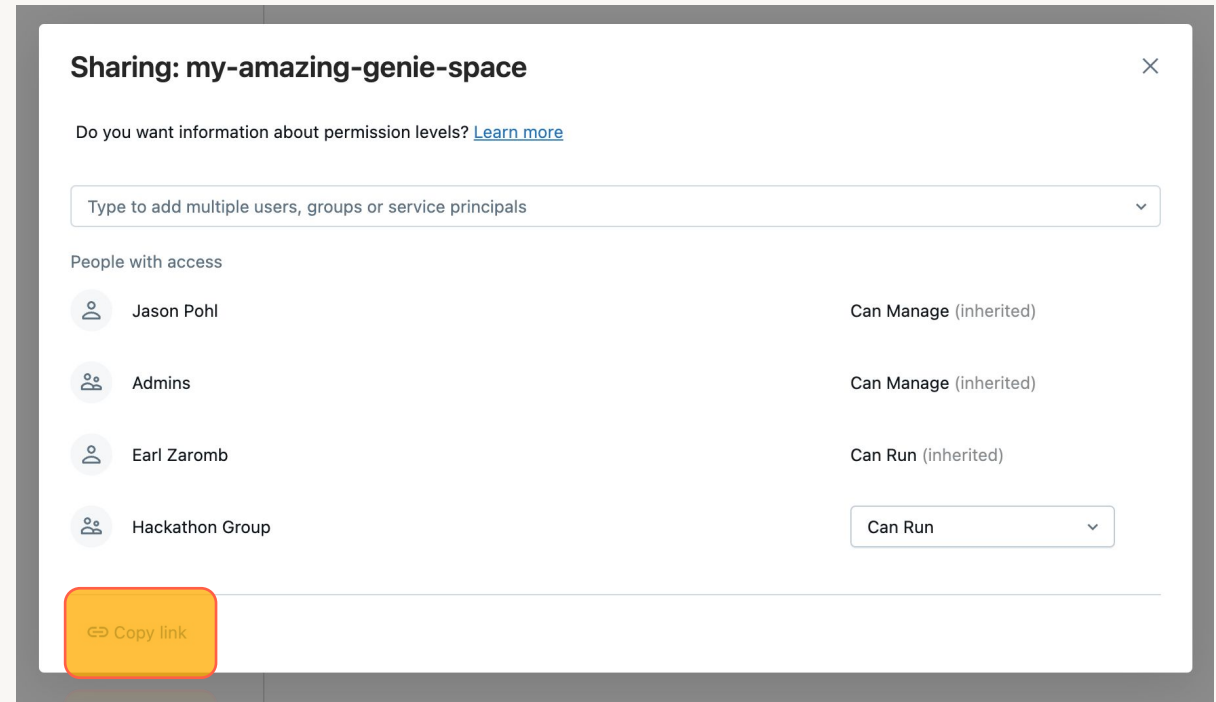


Share your Space

User just needs to be registered in Databricks Account

1. Click Share and then “Copy Link”
2. Email / Slack / Teams this link with someone on your team not in the hackathon today

If the user does not have access to the underlying tables, then their queries will fail with “unauthorized”



Best Practices



Ask Questions in the Right Way

- Keep questions **focused and clear**
 - *Ambiguous questions may lead to irrelevant answers.*
- Use specific business terms that match the column comments and descriptions within the Genie space.
 - *Ask Admin to include missing business terms via “request a review”*
- Avoid overly complex questions in one question
 - *instead, break them down for more precise answers*
 - *Genie will remember the context of your chat*
- DO ask WHAT HAPPENED style questions
 - *What were my Sales in EMEA last quarter?*
- DON'T ask WHY style questions
 - *Why are my sales down?*





Dashboards – Core Concepts



Dashboards

Native, no-extract, low-code dashboards

What it is

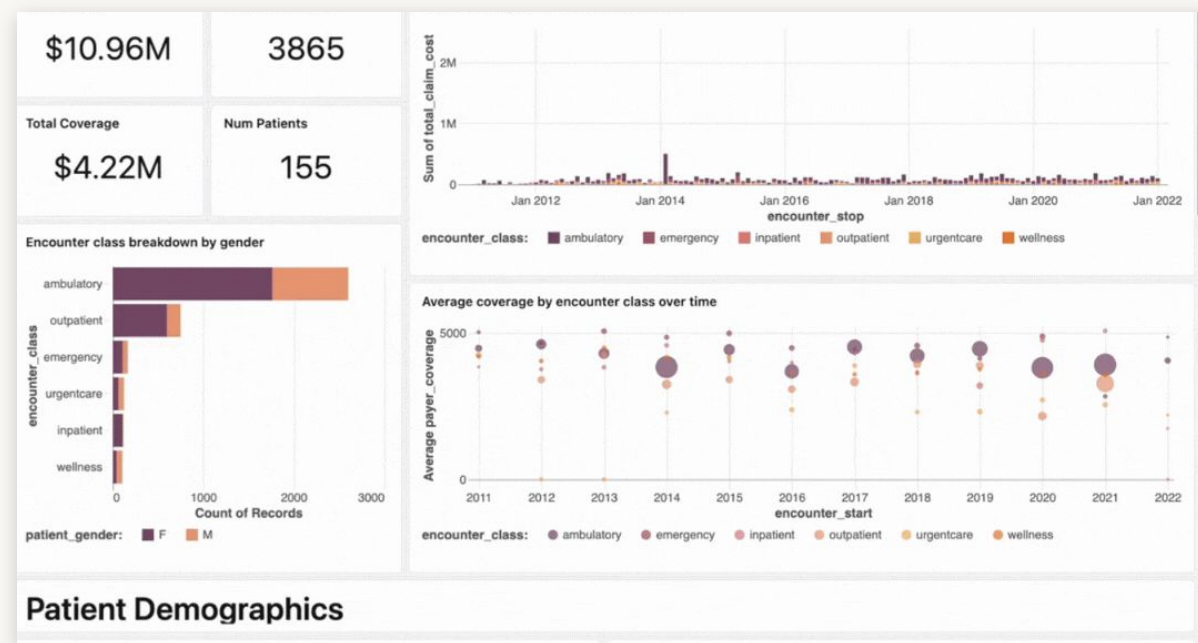
- A fully native, no-extract, low-code dashboard interface enabling users to build powerful business visualizations directly on top of Lakehouse data

Why it matters

- No lag between data landing and insight
- No replication of security model
- Unified semantics and governance flow throughout

Key Benefits

- Built-in AI assistant
- Self-service data access
- Fully featured unified analytics experience
- Governed by Unity Catalog



Built-In AI Assistant

AI-powered assistant embedded directly in the dashboard canvas

What it is

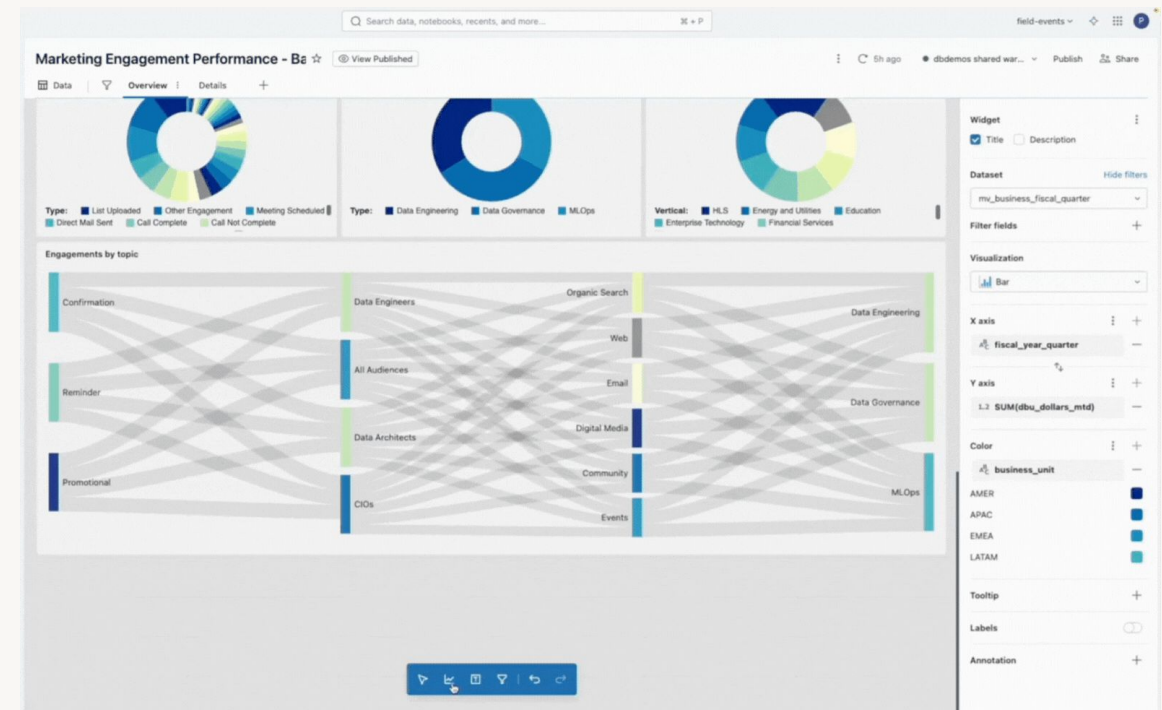
- AI assistant that explains datasets, auto-generates SQL, and suggests visuals within the dashboard canvas

Why it matters

- Accelerates dashboard building and enables teams to maximize curated, documented data assets

Key Benefits/Use Cases

- Fast authoring, fewer errors
- Better adoption of curated assets
- Includes automatic forecasting and top driver analysis out of the box



Lightning Fast Interactivity

Parameterization, Cross-Filtering and Drilling Through

What it is

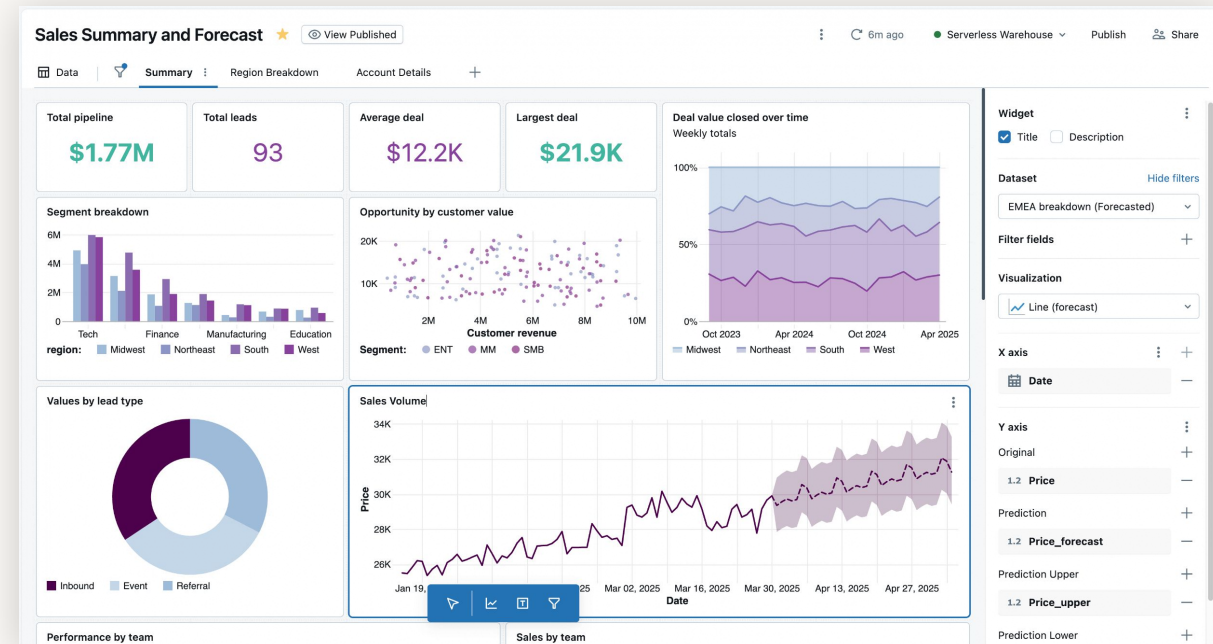
- Dashboards offer robust support for cross-filters, parameter widgets, and seamless sharing, creating an interactive analytics environment

Why it matters

- Users can intuitively explore and manipulate data across multiple visualizations, customize views from a single place

Key Benefits/Use Cases

- Instantly drill into insights by clicking any visual without manual filter setup
- Easily customize and share dashboards for team-wide or executive reporting
- All interactivity, sharing, and controls are native to the platform so no plug-ins, external tools, or complex configuration required



End-to-end Governance

Security, lineage, and audit visibility through Unity Catalog

What it is

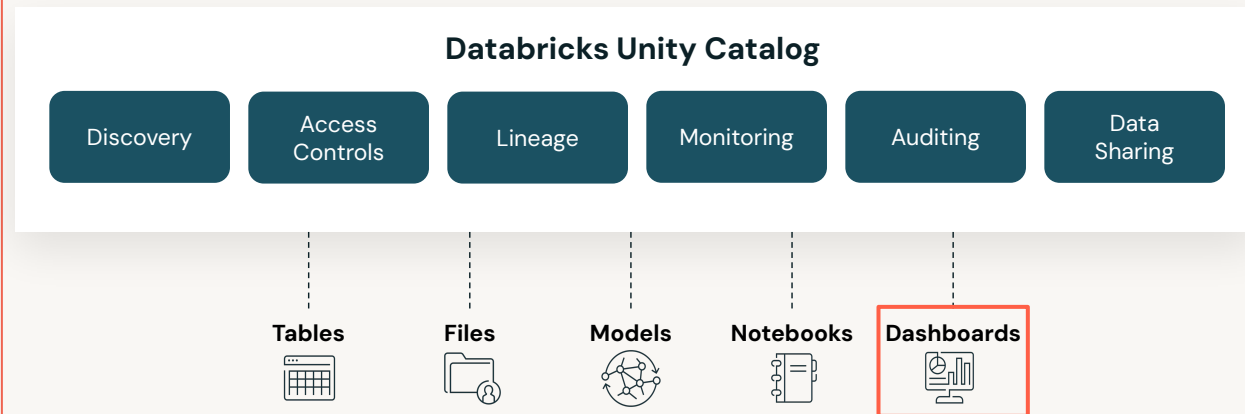
- Dashboards are governed through Unity Catalog, inheriting consistent data access, lineage, and audit policies

Why it matters

- Provides unified visibility into who accessed what data and how it was used in analytics
- Ensures dashboards participate in the same governance framework as all other Databricks assets

Key Benefits/Use Cases

- Eliminates shadow DW
- Eliminates governance silo between your data and your BI tool



Dashboard – Authoring Workflow



Creating an Dashboard

Defining your datasets

The screenshot shows the Databricks interface for a 'Workforce Management Dashboard'. On the left, a 'Datasets' sidebar lists various data sources, including 'compensation', 'Employee_Raise', 'terminations', 'filtered_employees', 'filtered_employees (Forecasted)', 'recruitment', 'snowflake_perf_view', 'hiring_metric_view', 'recruitment mv', 'employees', 'hiring_mv (Forecasted)', and 'hiring_metric_view_2'. The main area displays a SQL query: `1 SELECT * FROM bakehouse_hr.hr_data.compensation`. Below the query, a 'Result Table' is shown with the following data:

	Employee_ID	1.2 Raise_Amount	1.2 Bonus_Amount	1.2 New_Salary	Rating	1.2 Salary
1	1a3d1fa7	8534.53	17069.06	196294.14	Good	170690.55
2	b2b9437a	0	0	331303.44	Poor	331303.44
3	a9488d99	0	0	170761.33	Poor	170761.33

Below the table, there is a note: 'This dataset contains 6 fields and 2 measures. Results table displaying fields only.' and a link to 'See performance (1)'. The interface also includes a search bar at the top, a 'Run' button, and a 'Create from SQL' button at the bottom left.

- **Datasets can be defined using:**
 - The inline assistant to generate new queries against tables or views
 - Existing UC tables or views
 - Custom calculations allow you to perform calculations without modifying your dataset query
 - Metric views can be added as datasets

Dashboards – MVs, Serverless, and Caching



Materialized Views

Precomputed Views

What it is

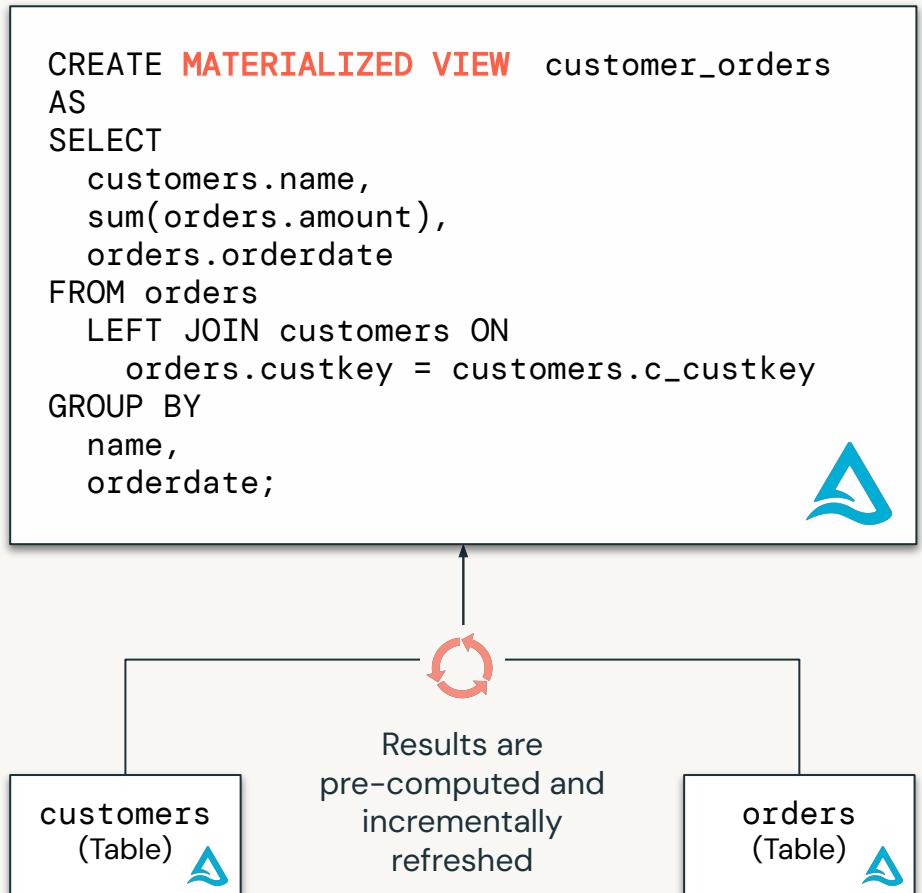
- Automatically refreshed views that speed up dashboards by serving fresh data from pre-aggregated results

Why it matters

- Query smaller, pre-processed data instead of massive source tables

Key Benefits/Use Cases

- Incrementally update with incoming source data and support scheduled refreshes
- Faster dashboards for high-volume and complex queries



Serverless Warehouses and Autoscaling

Performance and cost efficiency across varied workloads

What it is

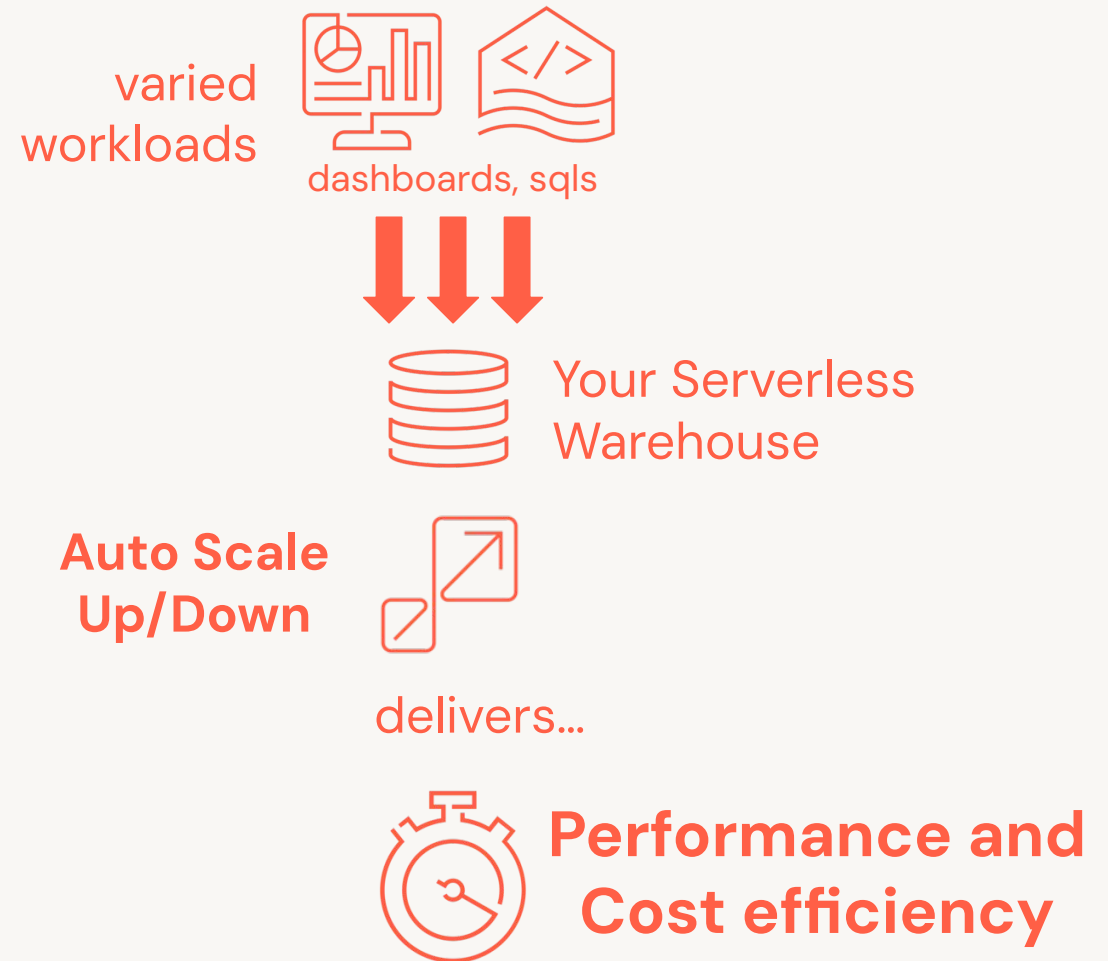
- Serverless warehouses that elastically scale to match dashboard concurrency and workload demands

Why it matters

- Ensures minimal startup latency and efficient resource use for any size team or workload pattern
- Automatically scales warehouse size and compute power based on expected concurrency, burst traffic, and business SLAs

Key Benefits/Use Cases

- Align warehouse size and type closely with workload characteristics and business SLA requirements



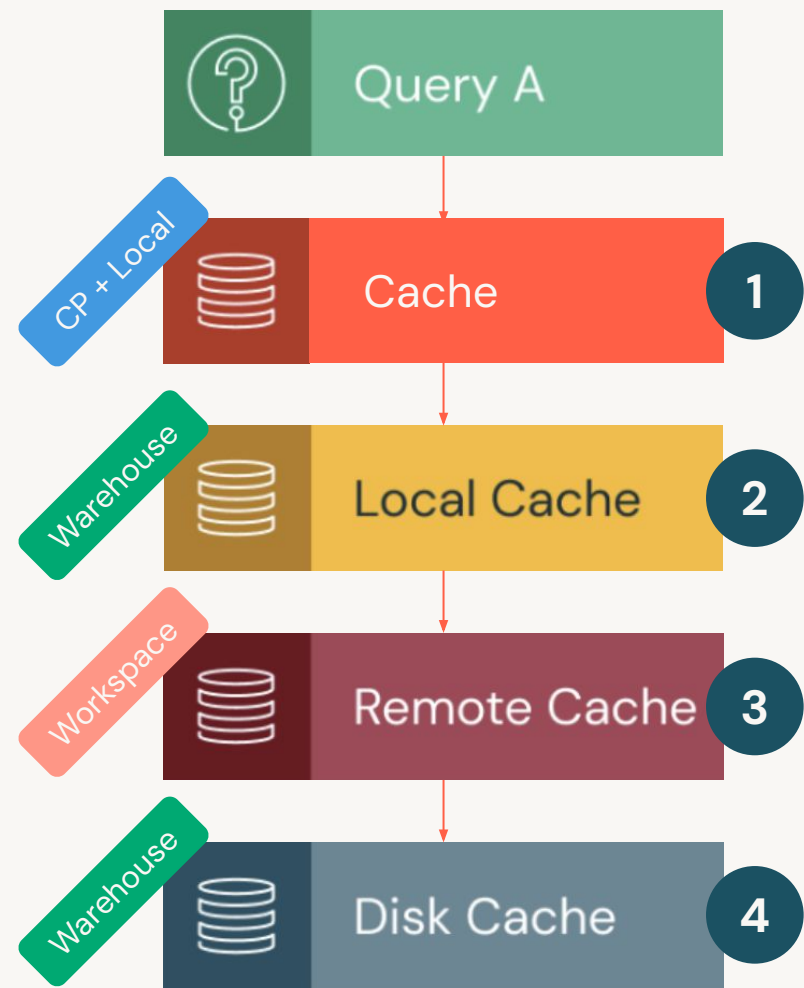
Types of Caching



Caching Layers

What caching mechanisms do queries have?

- An initial load dashboard cache + a local result cache for in-browser operations
- An in-memory cache at the warehouse level that stores query results
- A serverless-only cache that retains query results as workspace data
- A local SSD cache for data read from data storage



Caching Layers

Cache for initial loads / small dataset operations

What it is

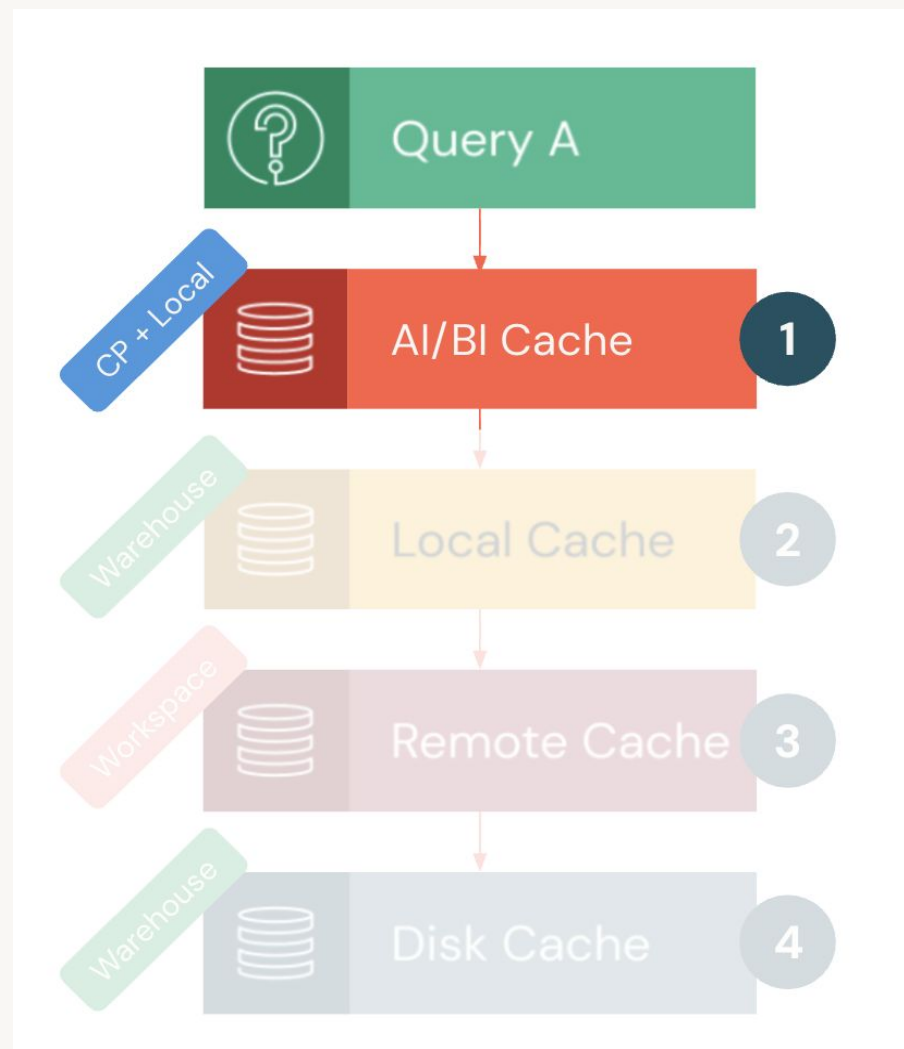
- A front-end result cache used by Databricks dashboards and visualizations to accelerate query rendering

Why it matters

- Enhances user experience for business intelligence and visualization workflows by reducing the time needed for initial dashboard loads or repeated queries

Key Benefits/Use Cases

- Caches recent results for up to 24 hours to improve responsiveness
- Performs browser-side aggregation for small datasets ($\leq 100K$ rows or $\leq 100MB$) and backend processing for larger ones
- Merges similar visualization queries (same filters or GROUP BY) to minimize redundant computation



Caching Layers

Local Cache for speeding up repetitive queries

What it is

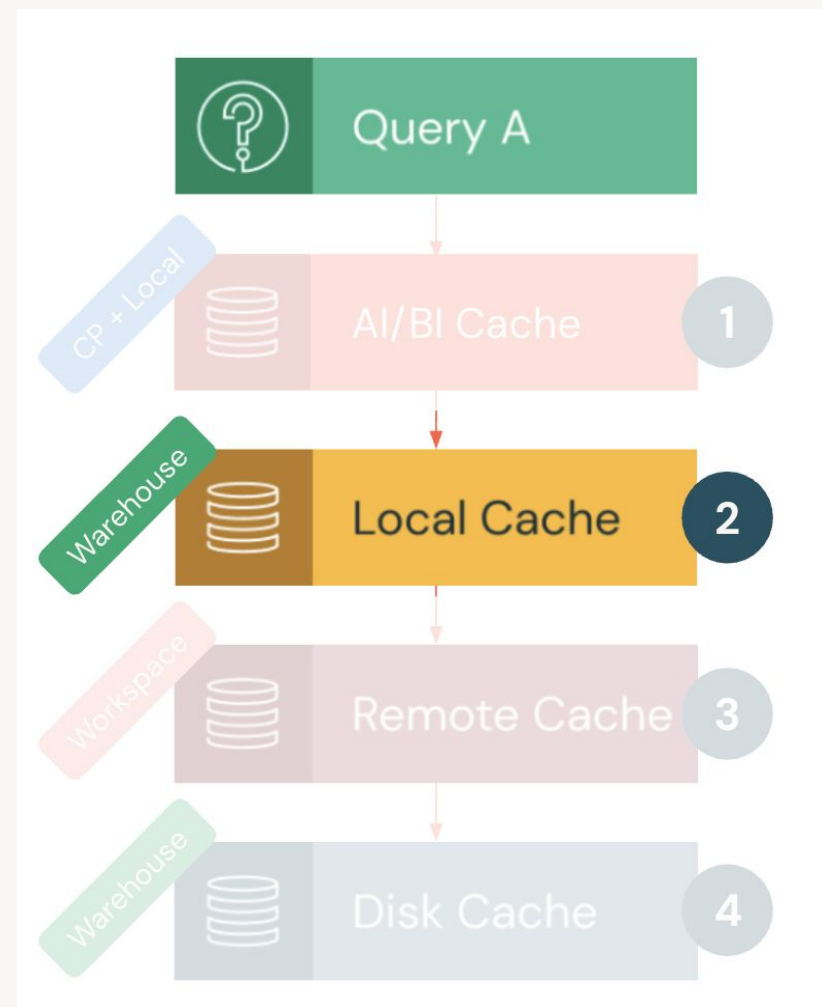
- in-memory cache that stores query results for the lifetime of a SQL warehouse cluster, or until the cache reaches capacity

Why it matters

- Eliminates the need to recompute results for repeated queries

Key Benefits/Use Cases

- Reduces compute overhead by avoiding redundant query execution
- The local cache temporarily keeps query outputs close to the compute resources for fast access



Caching Layers

Remote Cache persistent across warehouses after compute is stopped

What it is

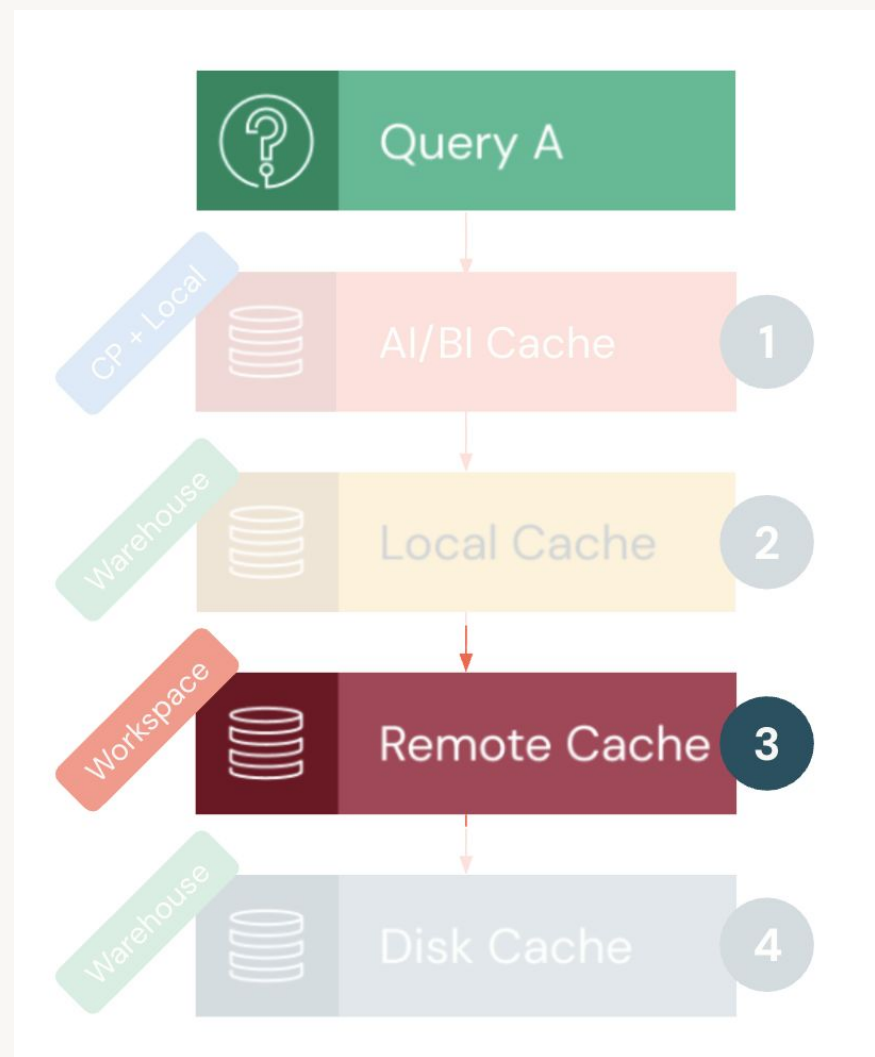
- A serverless, persistent cache layer that stores query results as workspace system data

Why it matters

- Remote result cache ensures consistent performance and cost efficiency by allowing previously executed queries to return results instantly, without the need to re-run computations

Key Benefits/Use Cases

- Reduces redundant query execution, minimizing warehouse usage and associated costs
- Eliminates cache invalidation on restart, ensuring data remains accessible



Caching Layers

Disk Cache designed to enhance query performance

What it is

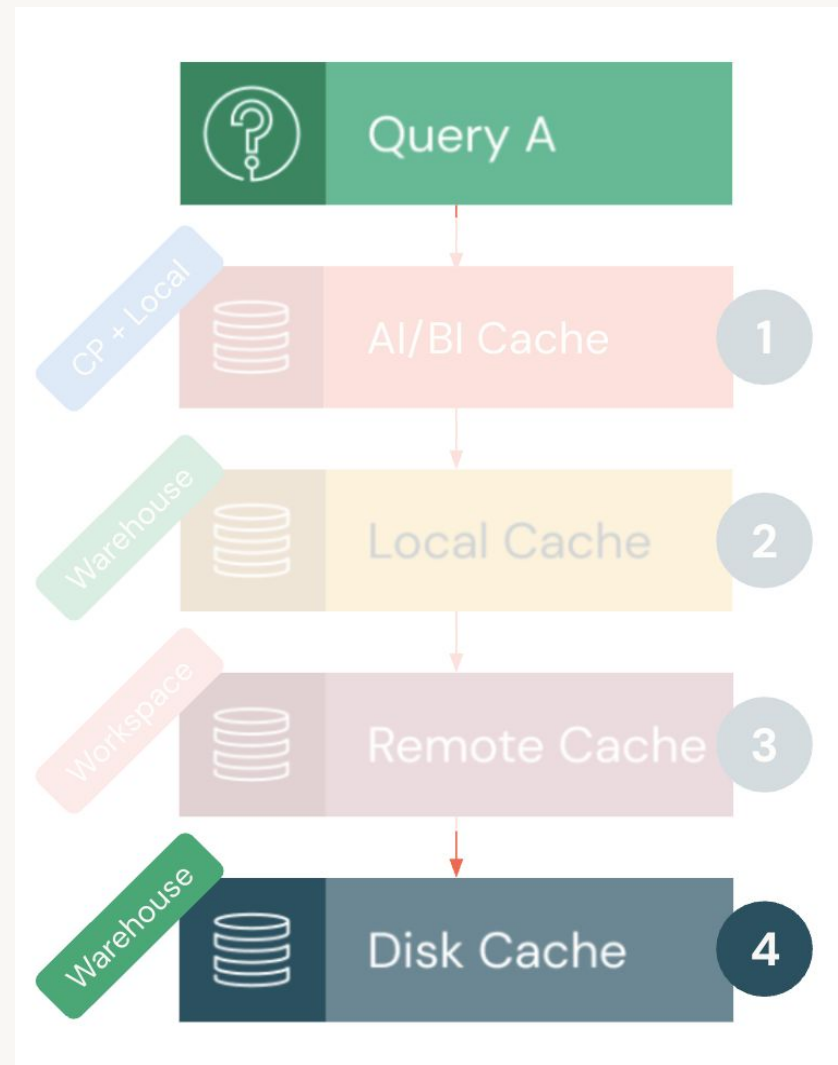
- Local SSD caching layer for data read from storage during SQL warehouse queries. The disk cache automatically stores frequently accessed data files on the local storage of compute nodes in an optimized intermediate format

Why it matters

- By keeping data physically closer to the compute resources, disk cache minimizes data transfer time from remote storage and accelerates subsequent reads

Key Benefits/Use Cases

- Enables accelerated query execution by serving data directly from local disks instead of remote cloud storage
- Data is automatically cached when first fetched so no manual configuration required
- Reduces query latency and enhances overall warehouse performance



Dashboards – Governance and Security



Full Policy Inheritance

Inherit all data access, filtering, and masking policies from Unity Catalog

What it is

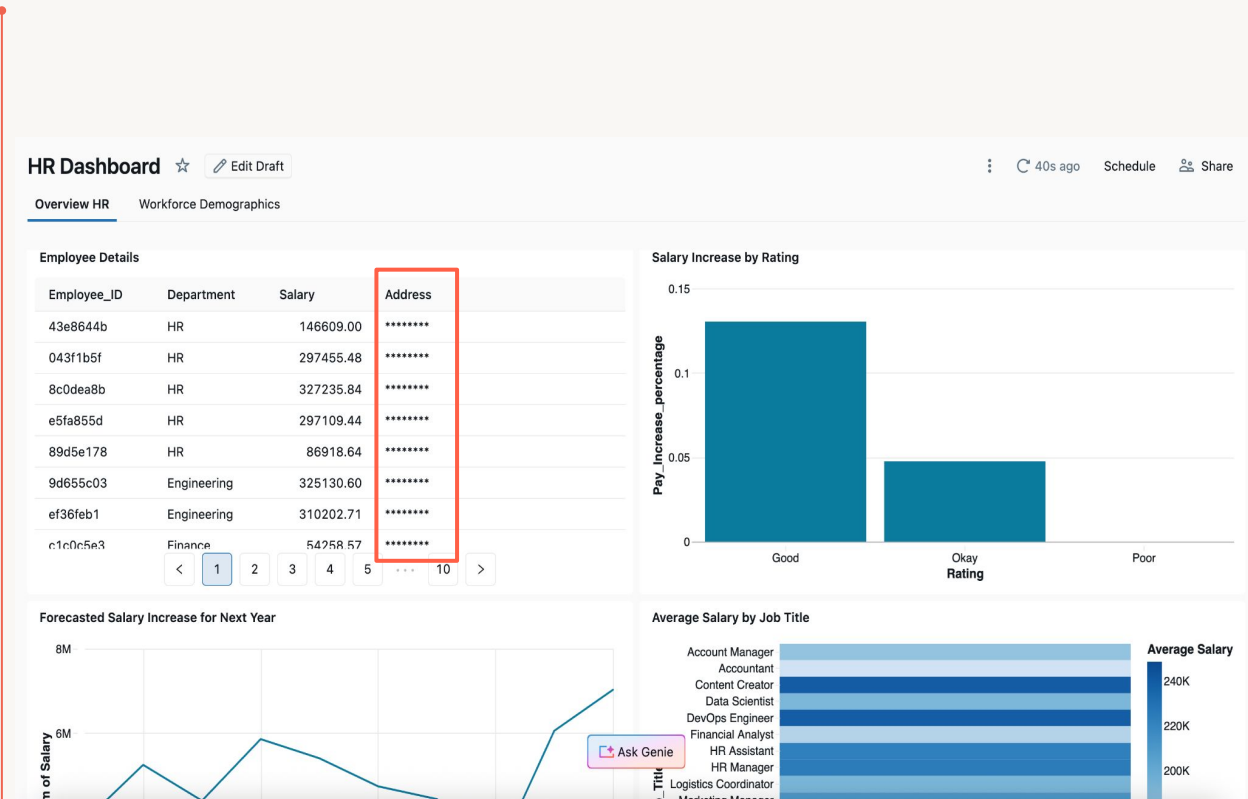
- When dashboards run in the viewer's context and use Unity Catalog data, they automatically apply inherited data policies such as masking and filtering

Why it matters

- Enforces consistent table, row, and column-level access control no matter how the dashboard is shared or embedded
- Roles and group assignments are managed in UC, and data tags (like "PII") trigger dynamic protection in dashboards

Key Benefits/Use Cases

- Guarantees secure, compliant analytics for every user, across all sharing modes
- Data masking, row filters, and access control are evaluated per user, per query



Audit, Compliance & Data Lineage

Fully auditable and traceable dashboards

What it is

- Every dashboard action such as queries, access, sharing, and export is fully logged for auditing

Why it matters

- Ensures transparency, regulatory readiness, and simplifies compliance by tracking every interaction and data flow at granular detail

Key Benefits/Use Cases

- All dashboard queries and activities are captured in platform audit logs, including who did what, when, and how
- End-to-end lineage makes every dashboard cell traceable to its source and transformation steps, supporting strict audit standards



Access Control

Run As...

What it is

- Dashboards can be monitored, access-controlled, and securely shared using “Run As” modes

Why it matters

- Supports both broad sharing and strict entitlements, while keeping audits and permissions in sync

Key Benefits/Use Cases

- Run As Publisher” enables read-only access for broader audiences; “Run As Viewer” enforces strict RBAC and row-level security

RUN AS MODES

RUN AS PUBLISHER



Broad access
read-only

RUN AS VIEWER



Strict RBAC
Row-level security



Consumer Entitlement

For Dashboards

What it is

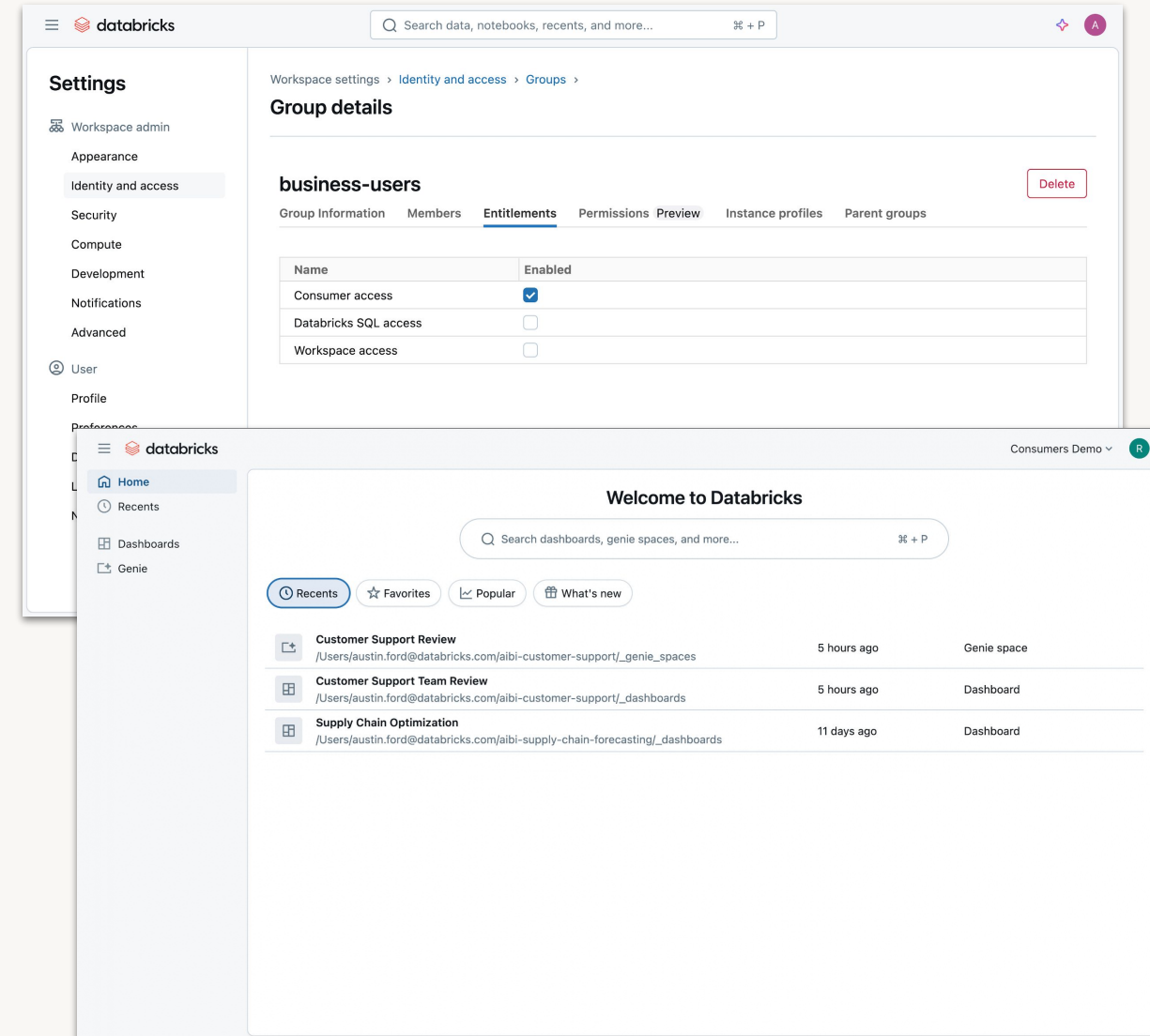
- Designed for business users and consumers who only need to view and interact with content, not create or author objects

Why it matters

- Enables secure dashboard sharing with non-technical or external business users while maintaining strict governance

Key Benefits/Use Cases

- For Admins:** Easily manage dashboard access and ensure users stay within governed permissions
- For Business Users:** A clean, simplified Databricks interface focused only on viewing shared dashboards
- For IT/Security:** Assurance that Consumer Access users cannot create, modify, or delete any workspace objects



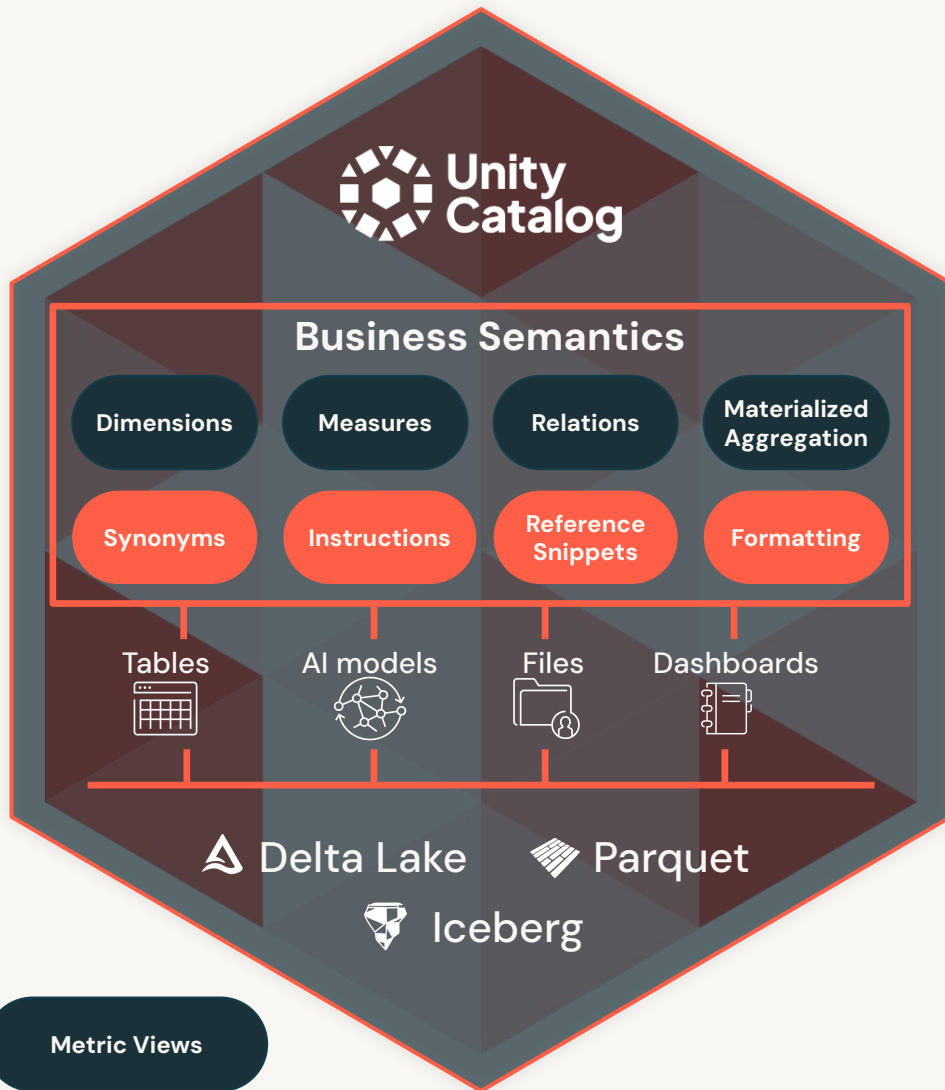
Dashboards – Semantics



Shift Semantics Left with Unity Catalog

The first unified solution for managing operational metadata **and** business semantics

- Supports both traditional BI and Agentic AI semantic concepts with **Metric Views** and **Agent Context Metadata**
- **Open** access through REST and SQL
- **Native integration** with leading ecosystem providers including Tableau, Sigma, Thoughtspot, Hex, Alation, and many others.



Metric Views

Agent Context
Metadata



Metric Views

Centralized Business Metrics in Unity Catalog

What it is

- Centralized abstraction in Unity Catalog that defines reusable business KPIs by specifying measures and dimensions

Why it matters

- Ensure metric definitions are consistent, governed, and reused organization-wide
- One metrics layer to unify many tools including external BI tools
- Derivative metrics (like ratios) are easy to get wrong when writing SQL manually

Key Benefits/Use Cases

- Empower both technical and business users to explore metrics flexibly without rewriting complex SQL
- Unlike normal views, metric views separate aggregate definitions (measures) from groupings,

"What's the average revenue per customer per segment for fulfilled orders since Jan 1, 2024?"

Traditional SQL

```
SELECT
  c.customer_segment,
  SUM(o.revenue) / COUNT(DISTINCT
  c.customer_id) AS
  revenue_per_customer
FROM orders o
JOIN customer c ON o.c_id=c.id
WHERE order_date > '2024-01-01'
AND o.status = 'fulfilled' AND
o.check IS NOT NULL
GROUP BY c.customer_segment;
```

UC Metric View Query

```
SELECT
  'Customer Segment',
  MEASURE('Revenue per Customer')
FROM customer_metrics
WHERE 'Order Category' = "Valid
Fulfilled"
GROUP BY ALL
```



Dashboards – Sharing and Embedding



Internal Sharing

Sharing across workspaces, users, and groups

What it is

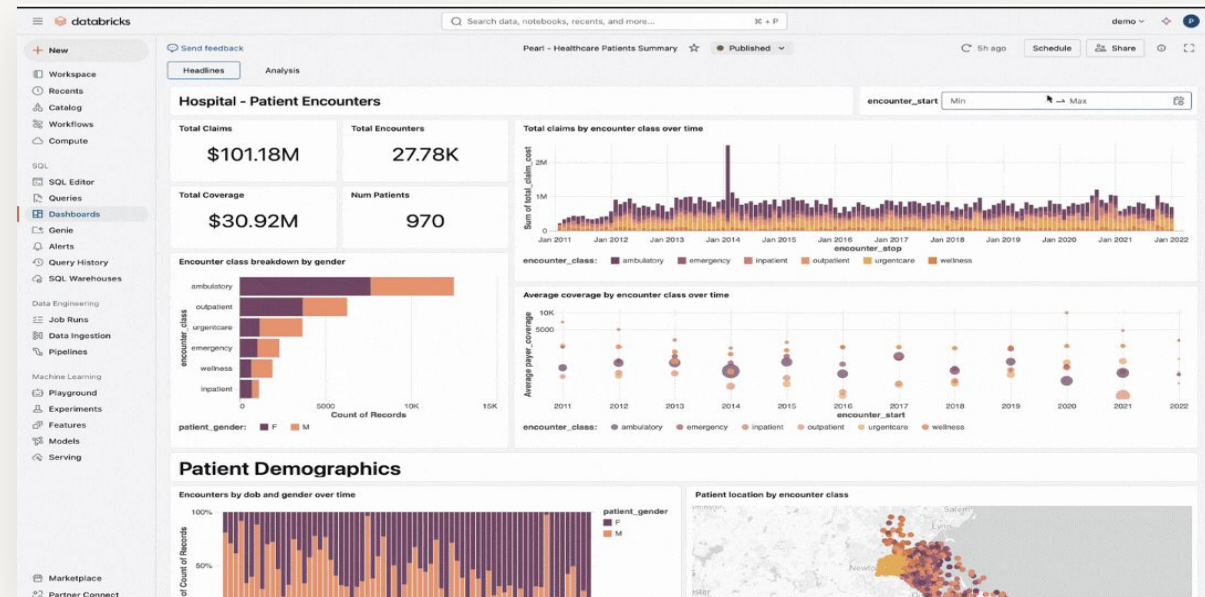
- Dashboards can be shared directly with workspace users or groups, with each user's data access enforced by Unity Catalog entitlements and security controls per session

Why it matters

- Everyone sees only the data they're permitted to see

Key Benefits/Use Cases

- Share dashboards with users/groups in the same workspace, into another workspace, or with Databricks users not tied to a workspace
- Schedule- and subscription-based dashboard delivery always honors each recipient's Unity Catalog entitlements for full compliance



External Sharing

Dashboard access for external users

What it is

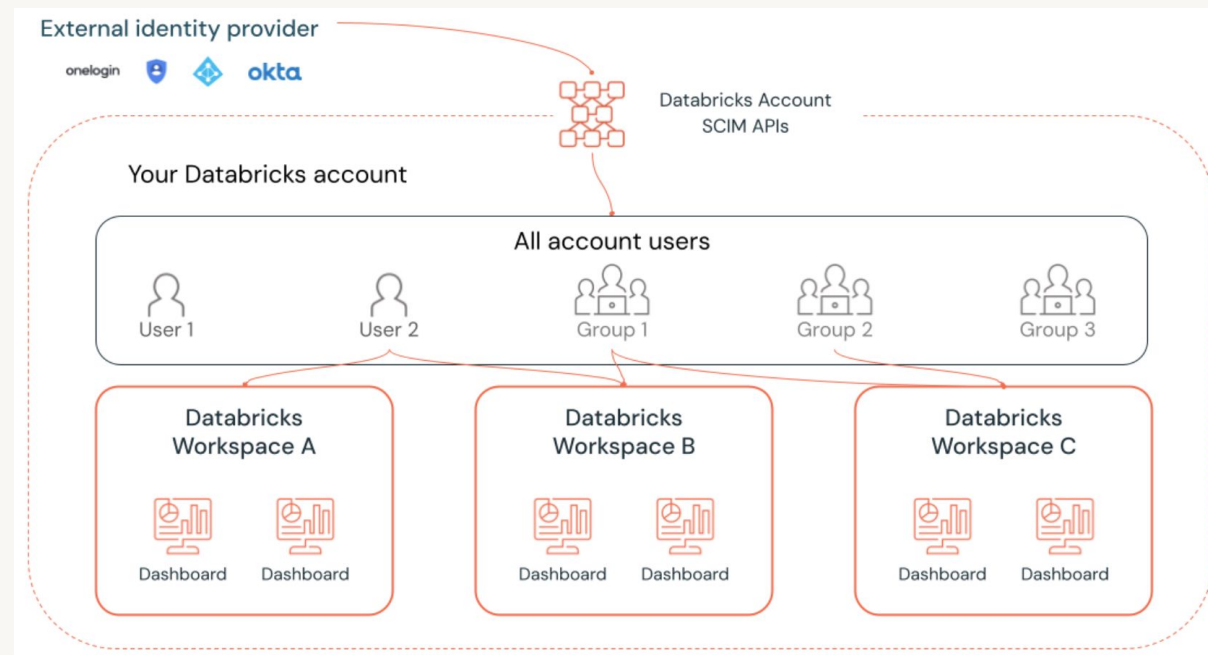
- “View Only” dashboard access for external users via embedded authentication

Why it matters

- Enables partners and end users to view insights without accessing raw data or needing a Databricks account

Key Benefits/Use Cases

- Grant view-only access through embedded authentication using access tokens or SSO
- App-delegated authentication enables embedding dashboards directly in custom portals or partner apps



Embedding Dashboards

Your Dashboard Anytime, Anywhere

What it is

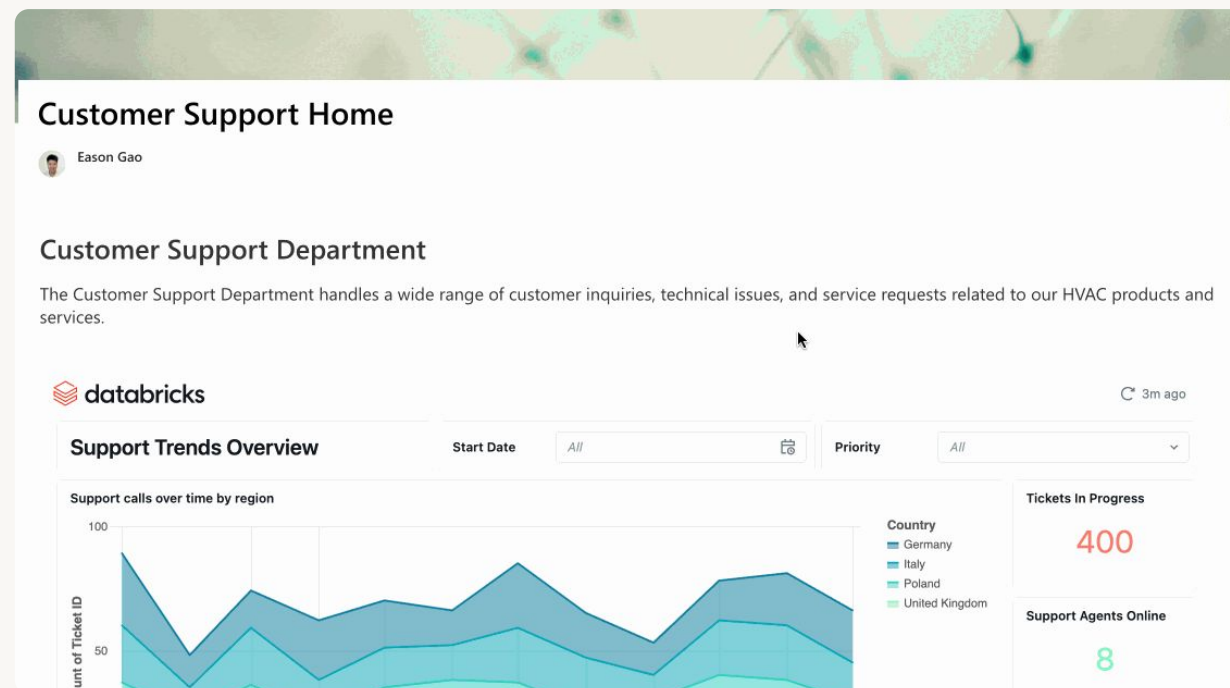
- Embed Databricks dashboards into third-party tools (like Confluence, SharePoint, or internal portals) to share live insights
 - Basic Embedding via iFram is for internal users already logged in to Databricks (SSO session required)
 - Embedding for External Users is for external audiences outside your workspace. Uses service principal + OAuth tokens for secure API-based access

Why it matters

- Seamless analytics experiences within external or internal platforms while maintaining data security and context

Key Benefits/Use Cases

- Dashboards always reflect the security permissions and data access of the viewing user from Unity Catalog



What's Supported by Embedding Type

Capability	Basic Embedding	SSO Embedding	External Embedding
Dashboard interactivity (filters etc)	✓	✓	✓
Dashboard caching	✓	✓	✓
Consumption only pricing (no per seat, per view charges)	✓	✓	✓
No manual login required	✗	✓	✓
No developer skills required	✓	✗	✗
Automatic auditing	✓	✓	⚠ Manual
Native genie integration	✓	✓	✗ (Requires Genie Conversation API)
No rate limits	✓	✓	✗ (Limit is 20 concurrent dashboard loads)



Dashboards – Interactive Features for Business Teams



Cross-Filtering

Interactive Analysis

What it is

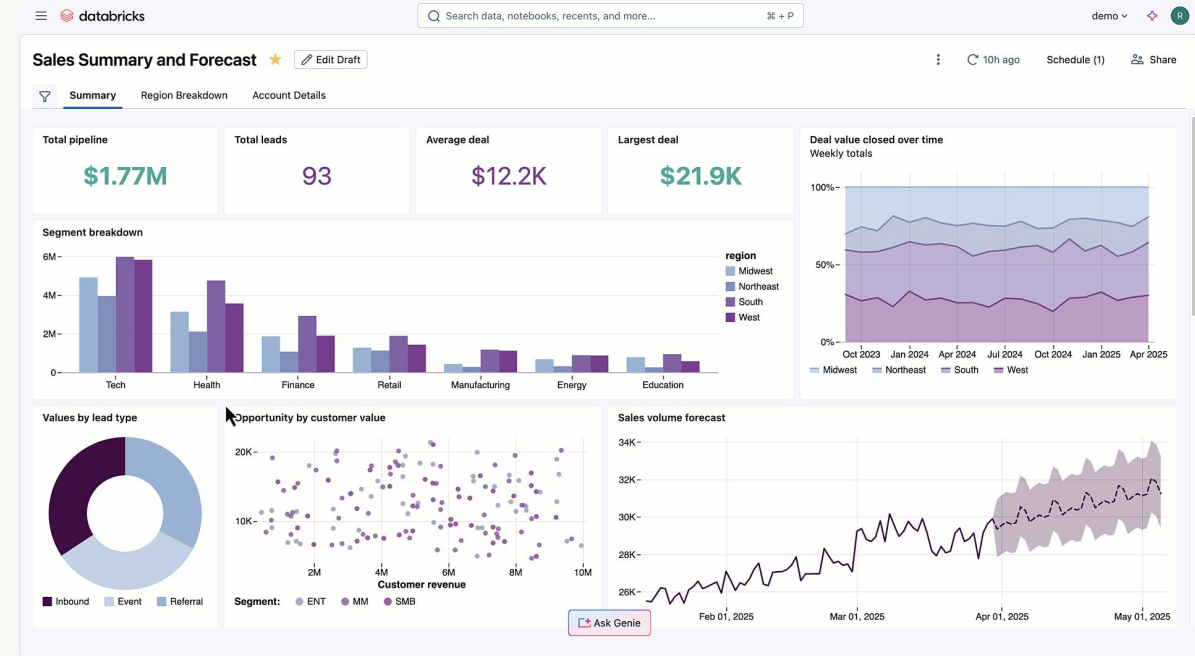
- Filters all visuals based on user selection in one chart

Why it matters

- Dynamic, cohesive data exploration and analysis without writing SQL or advanced queries

Key Benefits/Use Cases

- Ideal for trend analysis, root cause investigation, and drilldowns with context
- Intuitively investigate data relationships across multiple visuals in real time



Drilling

Explore deeper levels of data granularity

What it is

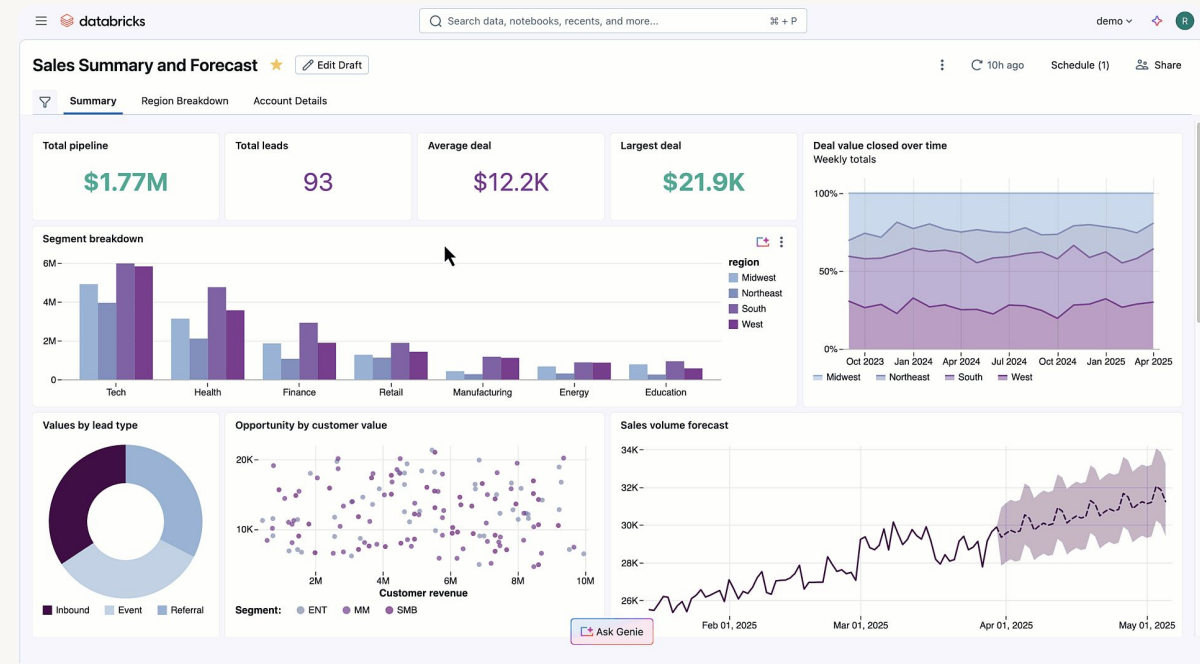
- Lets users move from summarized or high-level visuals to more detailed data within the same report or dashboard

Why it matters

- Multi-level exploration of metrics without leaving the dashboard or writing new queries

Key Benefits/Use Cases

- Quickly break down aggregates to see underlying patterns or exceptions
- Usually hierarchical like expanding “Year” to view “Months,” then “Days”



Forecasting and Functions

Using Built in AI Functions

What it is

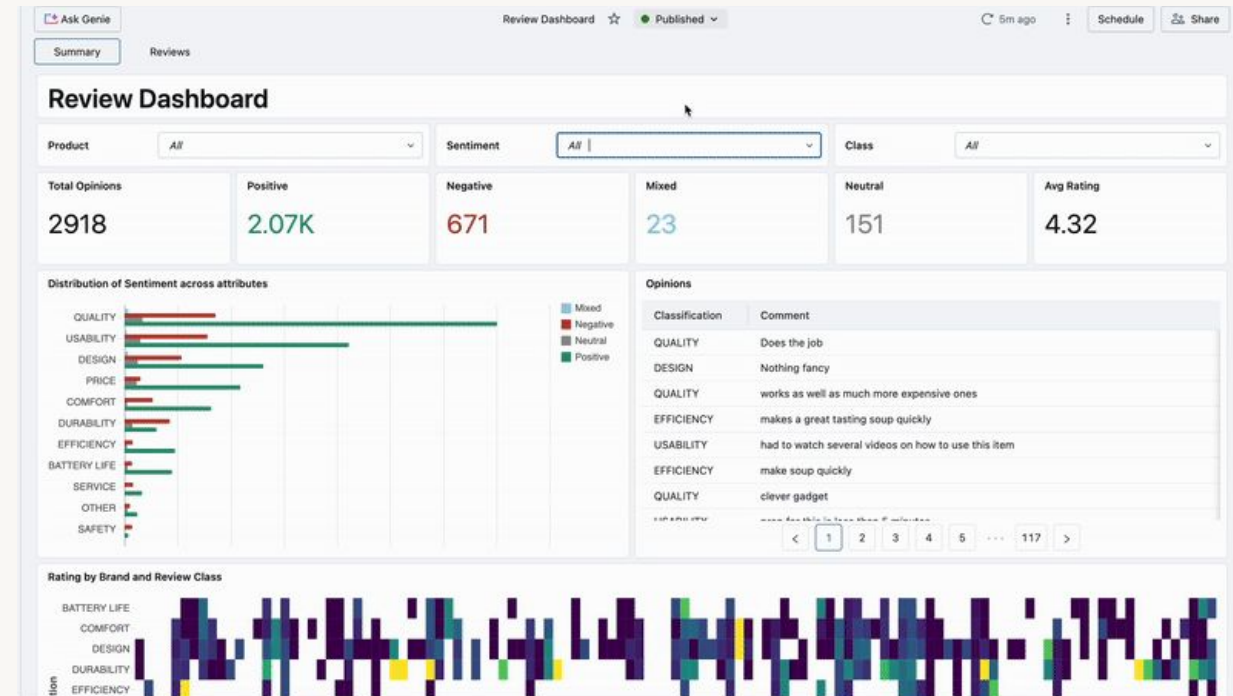
- Automatically apply machine learning and generative AI capabilities directly in your Dashboard

Why it matters

- Makes advanced analytics available to analysts and business users, not just data scientists

Key Benefits/Use Cases

- You don't have to leave your dashboard environment; predictions are integrated into the same workflow as exploration and reporting



Calculated Dimensions and Measures

Custom fields and metrics

What it is

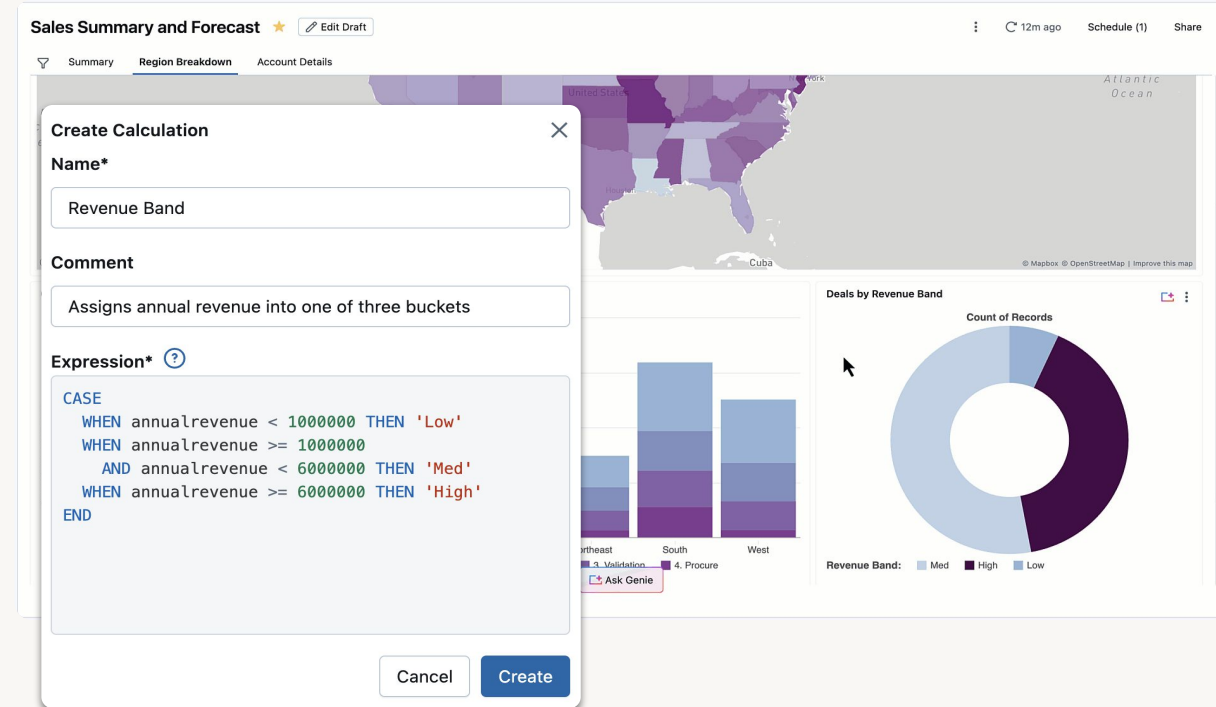
- Custom, reusable fields or metrics defined by formulas

Why it matters

- Users can create tailored, business-specific insights like ratios, classifications, and derived KPIs.

Key Benefits/Use Cases

- Calculated dimensions transform existing fields into new categories (e.g., create "Customer Age Group" by binning age)
- Calculated measures compute KPIs and aggregates with custom formulas (e.g., Profit Margin = (Sales – Costs)/Sales)
- All calculations update automatically as underlying data or dashboard filters change



Rich Interactive Visualizations

Visually compelling charts with real-time, intuitive interactivity

What it is

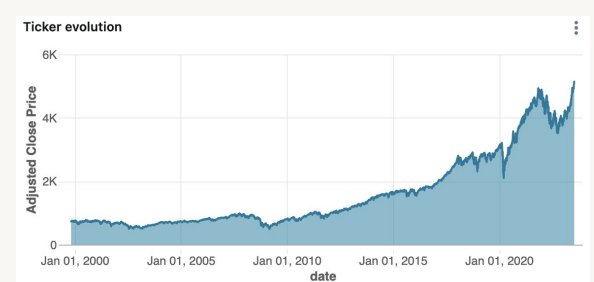
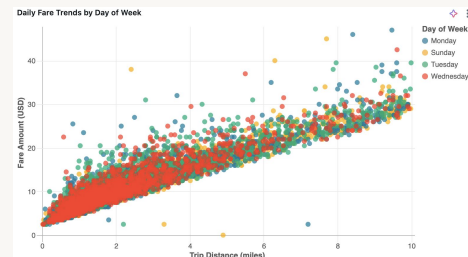
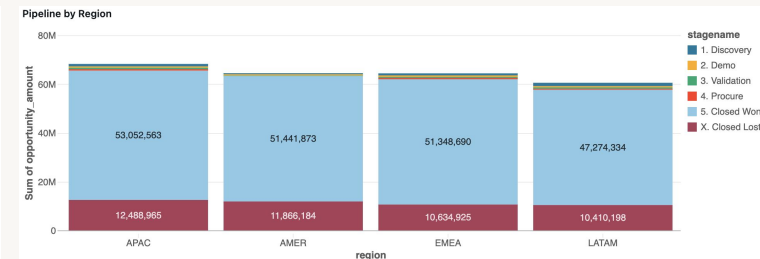
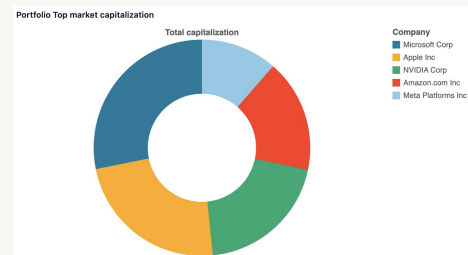
- Blend advanced graphical representations with responsive, intuitive user interactions

Why it matters

- Allow users to investigate data, uncover patterns, relationships, or outliers, and gain insights

Key Benefits/Use Cases

- Dashboards support a wide range of interactive visualization types for rich analytical experiences:
 - Counters
 - Scatterplots
 - Pie charts
 - Bar charts
 - Point maps and more



Total Trips

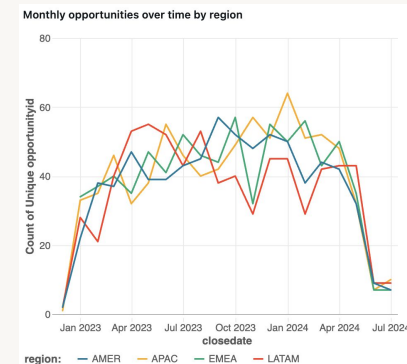
20.69K

Most Valuable Customers

All Time

Customer ID #	Customer Segment
4	MACHINERY
5	HOUSEHOLD
13	BUILDING
14	FURNITURE
17	AUTOMOBILE

Franchise Locations



Multipage Reporting

Dashboards structured into multiple pages

What it is

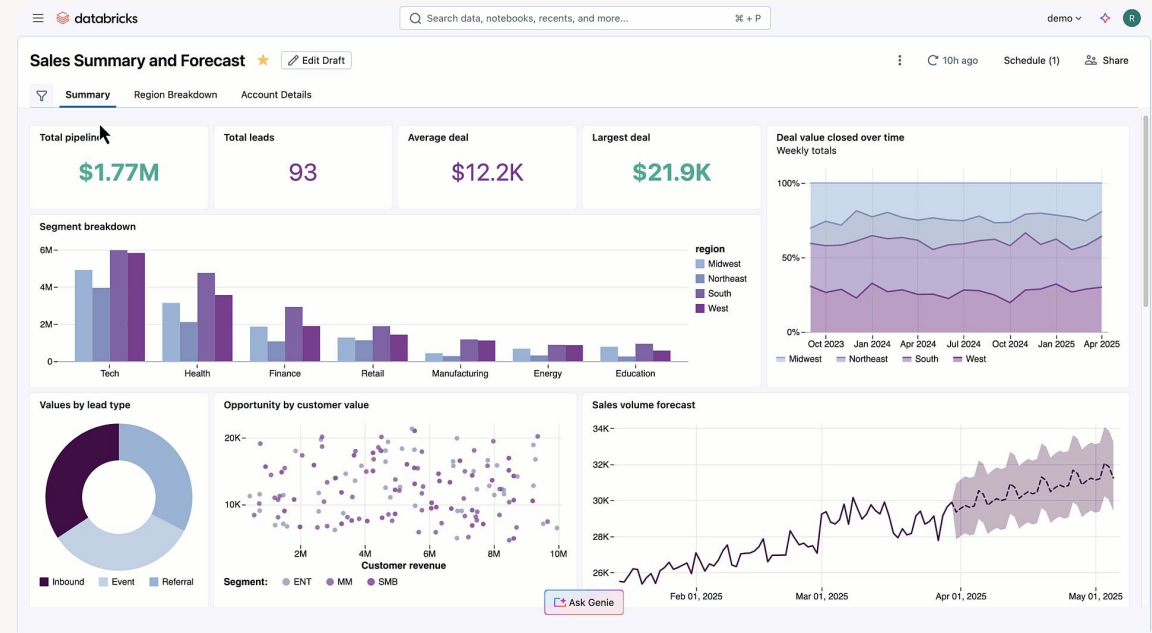
- Structures reports or dashboards across multiple pages, tabs, or sections

Why it matters

- Prevents information overload and allows targeted views for different teams or needs

Key Benefits/Use Cases

- Clear storytelling, modular reporting, and tailored experiences for different stakeholder groups



Themes

Consistent and Branded Design

What it is

- Predefined or custom visual design settings for dashboards and reports

Why it matters

- Govern global style such as fonts, colors, backgrounds, borders, gridlines, and overall layout aesthetic of dashboards

Key Benefits/Use Cases

- Improve professional appearance for stakeholders or external sharing
- Standardize design across teams



Parameterized Widgets

Dynamic, No-Code Filters

What it is

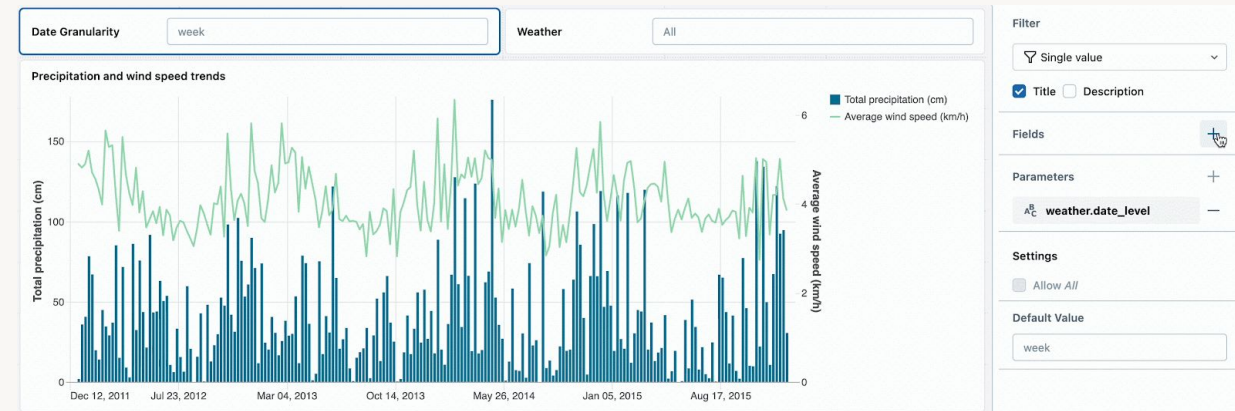
- Interactive controls (dropdowns, date pickers, text inputs) that update dashboard queries

Why it matters

- Empowers business users to filter and customize dashboards based on their needs without editing SQL

Key Benefits/Use Cases

- Dynamic, self-service exploration to answer business-specific questions on demand



"Ask Genie"

AI-Powered Insights on Demand

What it is

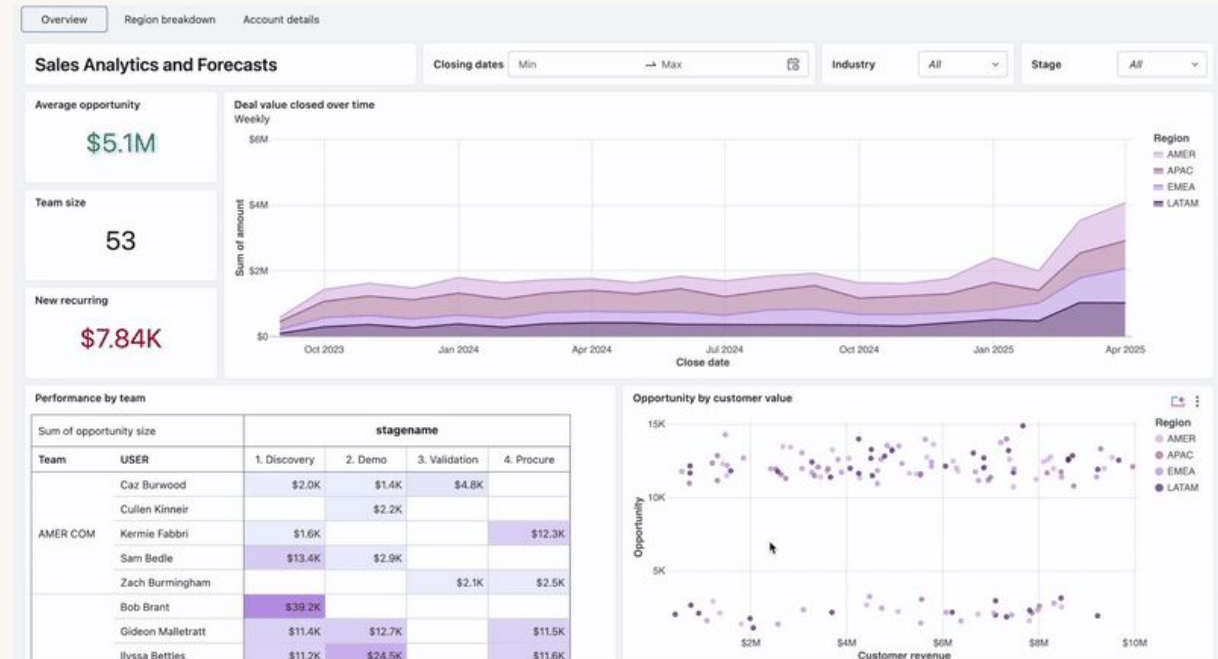
- Lets users pose natural language questions directly inside a dashboard

Why it matters

- Provides instant answers, visualizations, or SQL snippets by leveraging dashboard visuals and the underlying dataset

Key Benefits/Use Cases

- Empowers business users to get instant, self-service answers to ad-hoc or exploratory questions
- Accelerates decision-making by removing analytics bottlenecks and democratizing AI-powered BI



Dashboards – Tips, Tricks and Best Practices



Client-Side Compute Optimization

Keep results $\leq 100\text{K}$ rows and $\leq 100\text{ MB}$ to stay fast

What it is

- Client-side filtering and processing for small result sets ($\leq 100\text{K}$ rows, $\leq 100\text{MB}$) that occurs entirely in the browser instead of the warehouse.

Why it matters

- Eliminates unnecessary server queries and warehouse load for basic filtering operations, resulting in zero execution costs and instant response times.

Key Benefits/Use Cases

- Zero queries executed = zero warehouse costs
- Instant filtering with no server round-trips
- Perfect for interactive dashboards with drill-down capabilities
- Ideal for executive dashboards where users frequently filter by date, region, or category

✗ Don't do this

Return $> 100\text{ K}$ rows or $> 100\text{ MB}$

Engine must rerun queries for every filter

Large payloads to download

✓ Do this instead

Trim result set to $\leq 100\text{ K}$ rows and $\leq 100\text{ MB}$

Filtering & aggregation run in the browser

Sub-second, spreadsheet-like UX

Smaller payload downloads quickly



Custom Calculations for Dataset Management

Define metrics once—re-use them everywhere, with faster loads

What it is

- Defining metrics and calculations once in the dataset level and reusing them across multiple dashboard widgets instead of recreating them separately.

Why it matters

- Prevents dataset sprawl and ensures consistency while maintaining performance benefits of client-side processing for small result sets.

Key Benefits/Use Cases

- One dataset powers multiple widgets with synchronized cross-filters
- Faster dashboard loads through fewer warehouse queries
- Consistent metric definitions across all visualizations

✗ Don't do this

Create a new *dataset* every time you need a new calculation

✓ Do this instead

Add a Custom Calculation to your core dataset:

$$\text{profit_margin} = (\text{SUM}(\text{revenue}) - \text{SUM}(\text{cost})) / \text{SUM}(\text{revenue})$$

Fewer datasets → easier maintenance

Fewer datasets → faster dashboards



Selective Column Querying

Reduce I/O by Selecting Only What You Need

What it is

- Explicitly selecting only required columns instead of using `SELECT *` to minimize data transfer and improve query performance.

Why it matters

- `SELECT *` ships unnecessary columns, creating larger payloads and longer load times, while explicit column selection reduces I/O and keeps SQL resilient to schema changes.

Key Benefits/Use Cases

- Faster dashboards through smaller data transfers
- Cheaper cache storage costs
- More readable and maintainable SQL code

Don't do this

```
SELECT * FROM wide_table
```

Pulls whole table

Pulls unnecessary columns

Breaks if schema changes

Do this instead

```
SELECT a, b FROM wide_table
```

Tiny payload

Easier to read and debug

Safer against schema evolution



Filter Before You Join Large Tables

Reduce Shuffles by Filtering Early

What it is

- Applying WHERE clauses before JOIN operations on large tables to reduce the amount of data being processed during joins.

Why it matters

- Filtering early reduces data shuffles and network traffic, significantly improving query performance on large datasets.

Key Benefits/Use Cases

- Dramatic join performance improvement on large datasets
- Reduced compute costs through less data processing
- Lower memory usage during query execution

Don't do this

```
SELECT *  
FROM users u  
JOIN orders o  
  ON u.id = o.user_id;
```

Do this instead

```
SELECT *  
FROM users u  
JOIN orders o  
  ON u.id = o.user_id AND o.user_id = '123';
```



Prefer Filters to Parameters when filtering

Reduce file scans and speed up dashboards by narrowing wide tables

What it is

- Strategic choice between using filters (for row inclusion/exclusion) versus parameters (for structural query changes) to optimize dashboard performance.

Why it matters

- Proper filter usage reduces file scans and speeds up dashboards by narrowing wide tables before processing, while parameters should only be used when query structure needs to change.

Key Benefits/Use Cases

- Faster dashboard loads through reduced data scanning
- Use filters for: date ranges, category selections, status filtering
- Use parameters for: changing tables, columns, or complex expressions

✗ Don't do this

```
SELECT a, b FROM table WHERE a =:param
```

Any value change rewrites the SQL and reruns on the server

✓ Do this instead

Add a dashboard filter on A

Instant client-side filtering, zero extra compute



Query Structure Optimization

Flatten Logic Instead of Using Nested Subqueries

What it is

- Using CTEs (Common Table Expressions) or simplified query structures instead of deeply nested subqueries for complex logic.

Why it matters

- Flattened logic improves query readability, makes debugging easier, and often results in better query execution plans by the optimizer.

Key Benefits/Use Cases

- Better query optimizer performance and execution plans
- Easier debugging and maintenance for complex queries
- Improved team collaboration on shared SQL code
- Ideal for multi-step data transformations and complex business logic

❌ Don't do this

```
SELECT *  
FROM orders  
WHERE customer_id IN (  
  SELECT customer_id  
  FROM returns  
  WHERE reason = 'damaged');
```

✅ Do this instead

```
WITH damaged_returns AS (  
  SELECT customer_id  
  FROM returns  
  WHERE reason = 'damaged' )  
SELECT *  
FROM orders  
JOIN damaged_returns  
  USING (customer_id);
```



Index-Preserving Data Processing

Avoid Functions on Indexed Columns to Keep Queries Fast

What it is

- Preprocessing data transformations outside of query predicates to maintain index effectiveness on filtered columns.

Why it matters

- Functions applied to indexed columns in WHERE clauses prevent index usage, forcing full table scans and dramatically slowing queries.

Key Benefits/Use Cases

- Maintains fast index seeks instead of slow table scans
- Significantly better performance on large indexed tables
- Lower compute costs through efficient query execution
- Essential for date calculations, text formatting, and numeric transformations

Don't do this

`WHERE UPPER(name) = 'JOHN'`

Results in full table scans

Do this instead

`WHERE name_upper = 'JOHN'`

Preserve index scan ability

Normalize the data during ingestion to maintain indexed lookups



Use Views for Simplification

Strategy for simplifying analysis

What it is

- Use documentation-rich views to present only business-relevant fields to analysts and end users

Why it matters

- Simplifies analytics workflows and minimizes user confusion by reducing noise from raw source data

Key Benefits/Use Cases

- Create curated views on top of raw tables, exposing only the necessary columns and clear documentation for each field
- Use Unity Catalog metric views to standardize metrics and hide unnecessary schema complexity

The screenshot displays the Databricks web interface. On the left, a sidebar shows a list of datasets including 'business fiscal quarter', 'engagements', 'dbu forecast', 'country responses', 'learners and active users', 'quarterly dbu summary b...', 'attribution', and 'marketing impressions'. The main panel shows a SQL query for 'Marketing Engagement Performance' with a 'Run' button. The query is a SELECT statement joining 'marketing.silver2.Consumption c' and 'marketing.silver2.Accounts a' on 'ON'. Below the query, a 'Result Table' and 'Schema' are visible, showing columns like 'bu by vertical' and 'Measure (4)' with values for 'apac_fq_per', 'emea_fq_per', 'latam_fq_per', and 'amer_fq_per'.

On the right, a 'Create Metric View' dialog is open. It shows the 'YAML definition' for a metric view named 'mv_business_fiscal_quarter'. The definition includes a 'name' field, a 'comment' field, and a 'measures' field. The 'measures' field contains a list of metrics, including 'apac_fq_per' and 'emea_fq_per', with their respective SQL expressions. The dialog also has a 'Add metric view to dashboard' checkbox and 'Cancel' and 'Create' buttons.



Document Everything in UC

Define Tables and Columns

What it is

- Add detailed descriptions to tables and columns in Unity Catalog for clear, trusted business context

Why it matters

- Boosts discoverability, enables confident self-service, and improves AI-driven SQL and visual suggestions

Key Benefits/Use Cases

- Enter explicit business definitions and field descriptions directly in the Unity Catalog UI
- AI Assistant and Genie use these descriptions to suggest precise queries and charts

Catalog Explorer > bakehouse_hr > hr_data >

compensation [icon] [star] [Open in a dashboard] [Share] [Create]

No certification set [edit]

Overview Sample Data Details Permissions Policies History Lineage Insights Quality

Description [edit]

The table contains information related to employee compensation, including salary adjustments and performance ratings. It can be used to analyze salary trends, assess the impact of performance ratings on raises and bonuses, and track individual employee compensation over time. The table includes columns for current salary, new salary after adjustments, raise amount, bonus amount, and performance rating.

AI can help suggest comments for columns without comments

Filter columns... [AI generate]

Column	Type	Comment	Tags	Column masking rule
Employee_ID	string	Represents the unique identifier for each employee in the organization, allowing for easy tracking and reference.		
Raise_Amount	double	Indicates the amount of salary increase awarded to the employees, reflecting their performance and contributions.		
Bonus_Amount	double	Represents the bonus amount awarded to the employees, reflecting their performance and contributions.		
New_Salary	double	Represents the updated salary of the employee after adjustments.		

About this table

Owner: Pearl Ubaru [edit]
Type: Managed [icon]
Data source: Delta [icon]
Popularity: [icon]
Last updated: 2 weeks ago
Size: 30.8KiB, 1 file

Quality [Beta]

Healthy [icon] as of 6 days ago
Freshness: As expected [icon]
[See results]

Tags
[Add tags]

Row filter
[Add filter]



Primary and Foreign Keys

Define trusted table relationships

What it is

- Explicitly define primary and foreign key relationships in your tables to formalize how data is joined and related

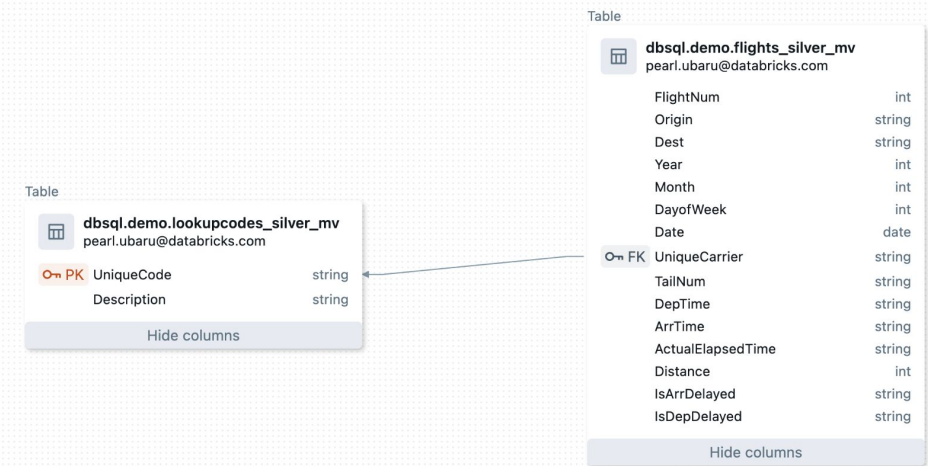
Why it matters

- Enables accurate query interpretation by the AI assistant and Genie

Key Benefits/Use Cases

- Use the RELY keyword to inform Databricks and connected tools to treat the annotated keys as trusted
- Makes relationship diagrams, compliance audits, and advanced analytics more robust and accurate

Entity Relationship Diagram for dbsql.demo.flights_silver_mv



Genie – Core Concepts

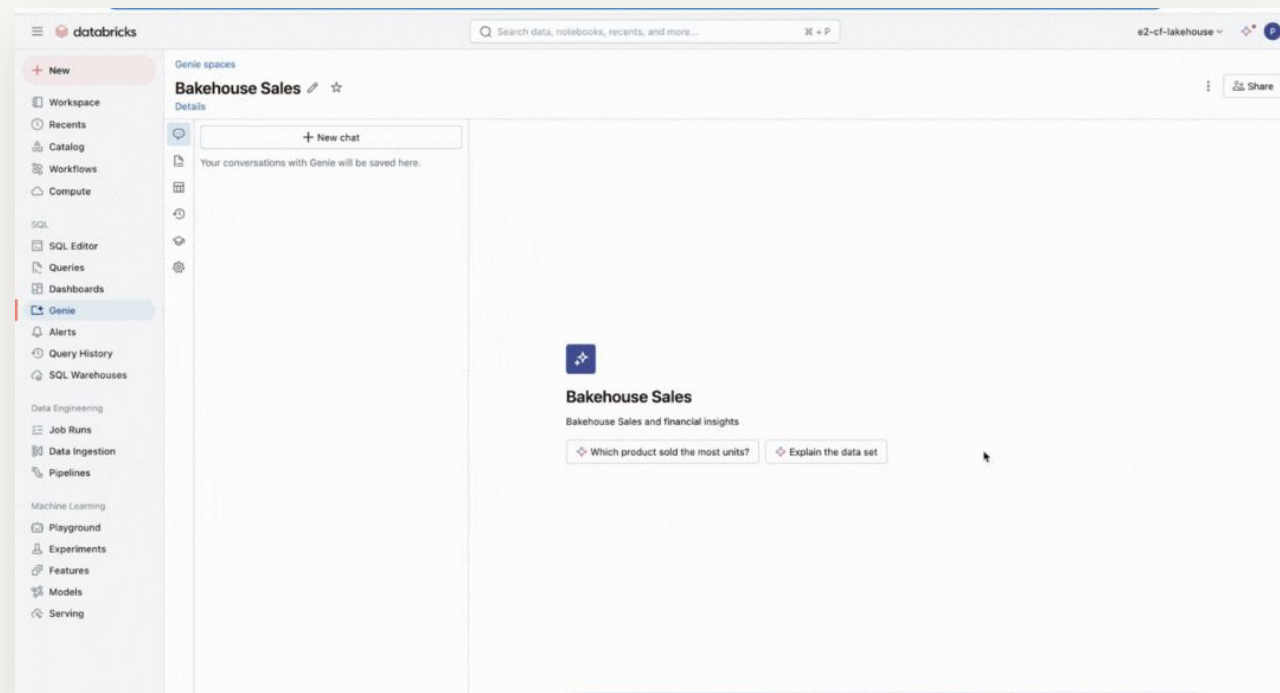


Genie

Assistant with context knowledge

Allow practitioners to get insights without needing any pre-existing knowledge of your data layer

- **Self-service Q&A**—go beyond dashboards and converse directly with your data
- **AI with Data Intelligence**—accurate answers from AI with unique knowledge of your data that improves through user feedback
- **Governed and secure**—fully governed interactions that adhere to your policies established in Unity Catalog



Genie – Semantics



Leverage Metric Views in Genie

Metrics provide determinism while tables augment with flexibility

What it is

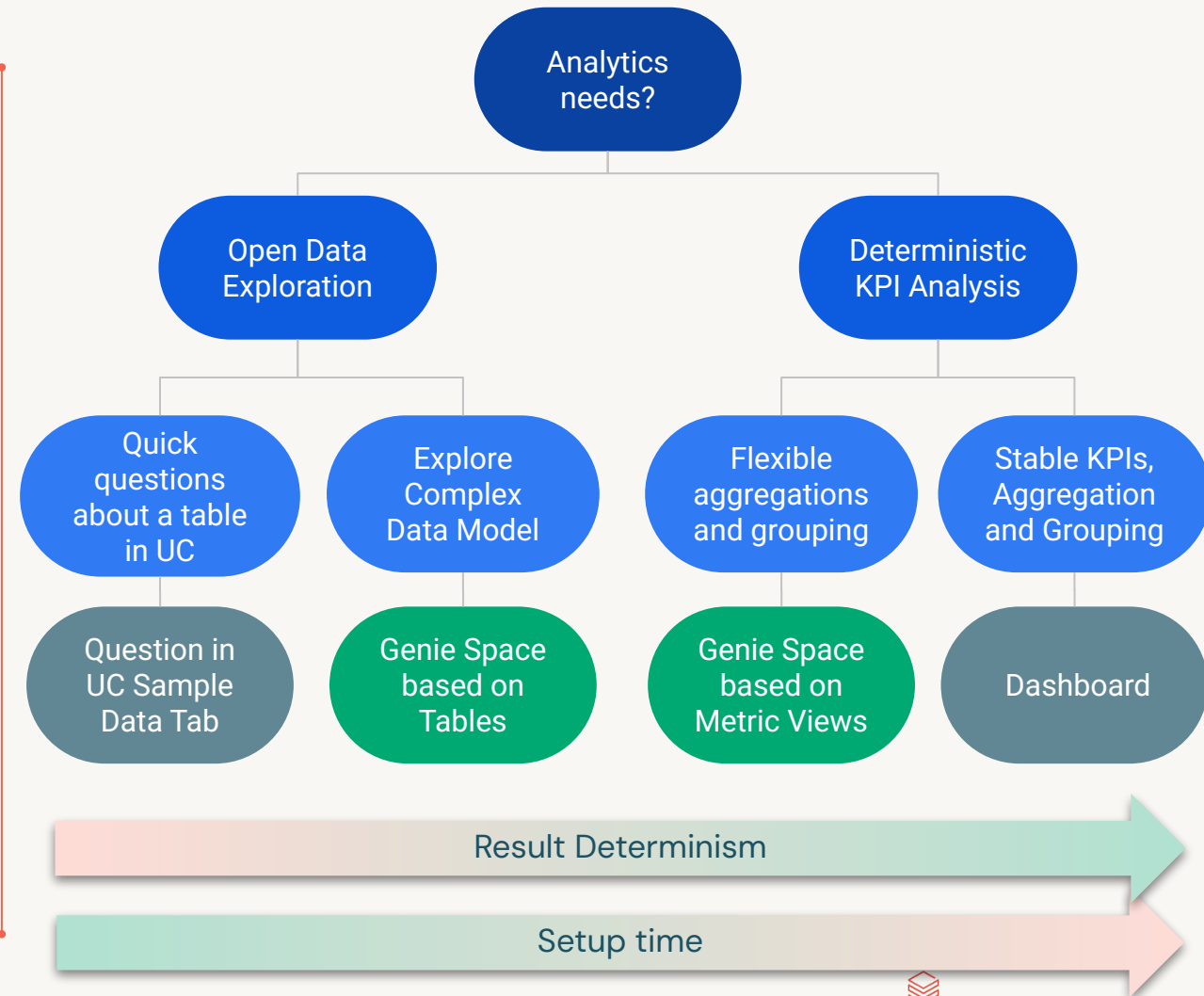
- Use Metric views (reusable KPIs through measures and dimensions) for business analysis, augment it with tables for open data exploration

Why it matters

- Genie will have more **deterministic answers** with Metric Views
- Generated queries will be simpler to interpret and validate

Key Benefits/Use Cases

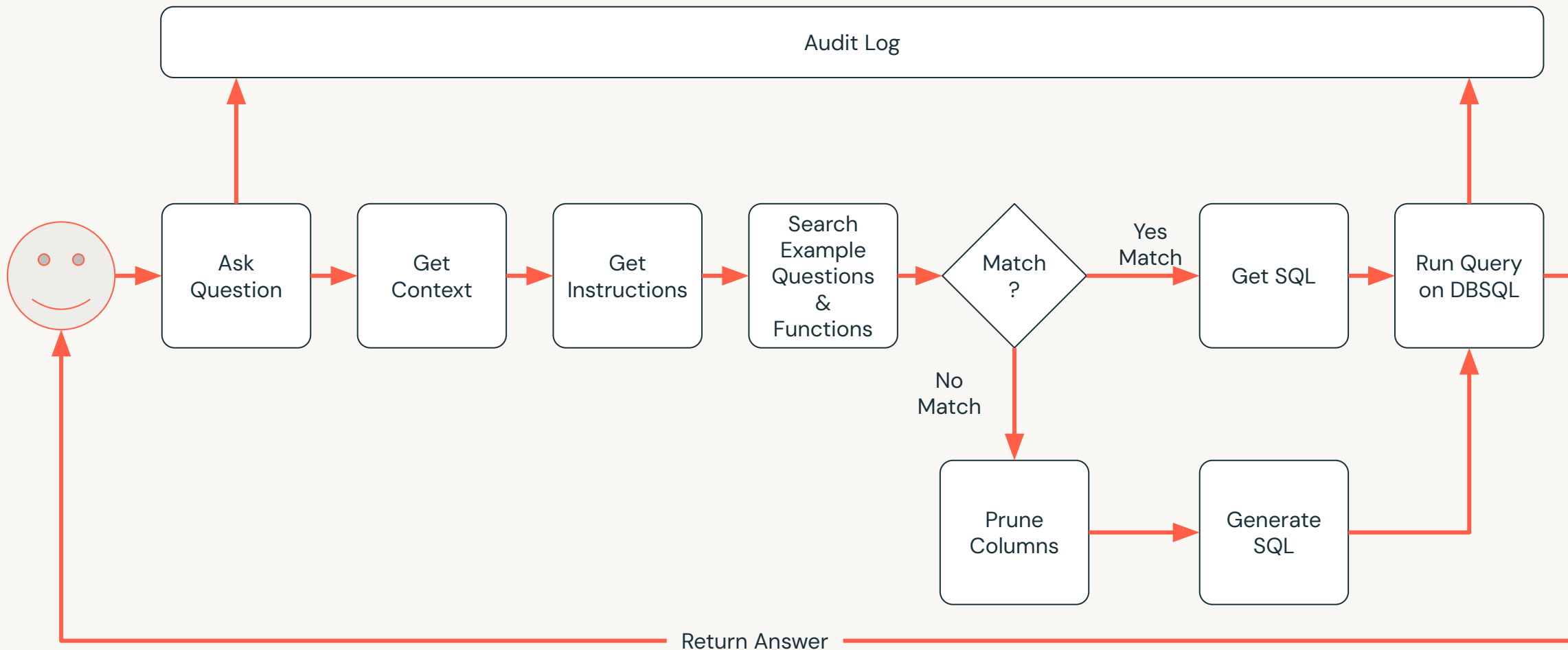
- **Improve Trust** for end-users
- Better reusability of business KPI definitions across spaces and use cases
- Easier maintenance in the long run (one central definition)



Genie Architecture



Genie Architecture

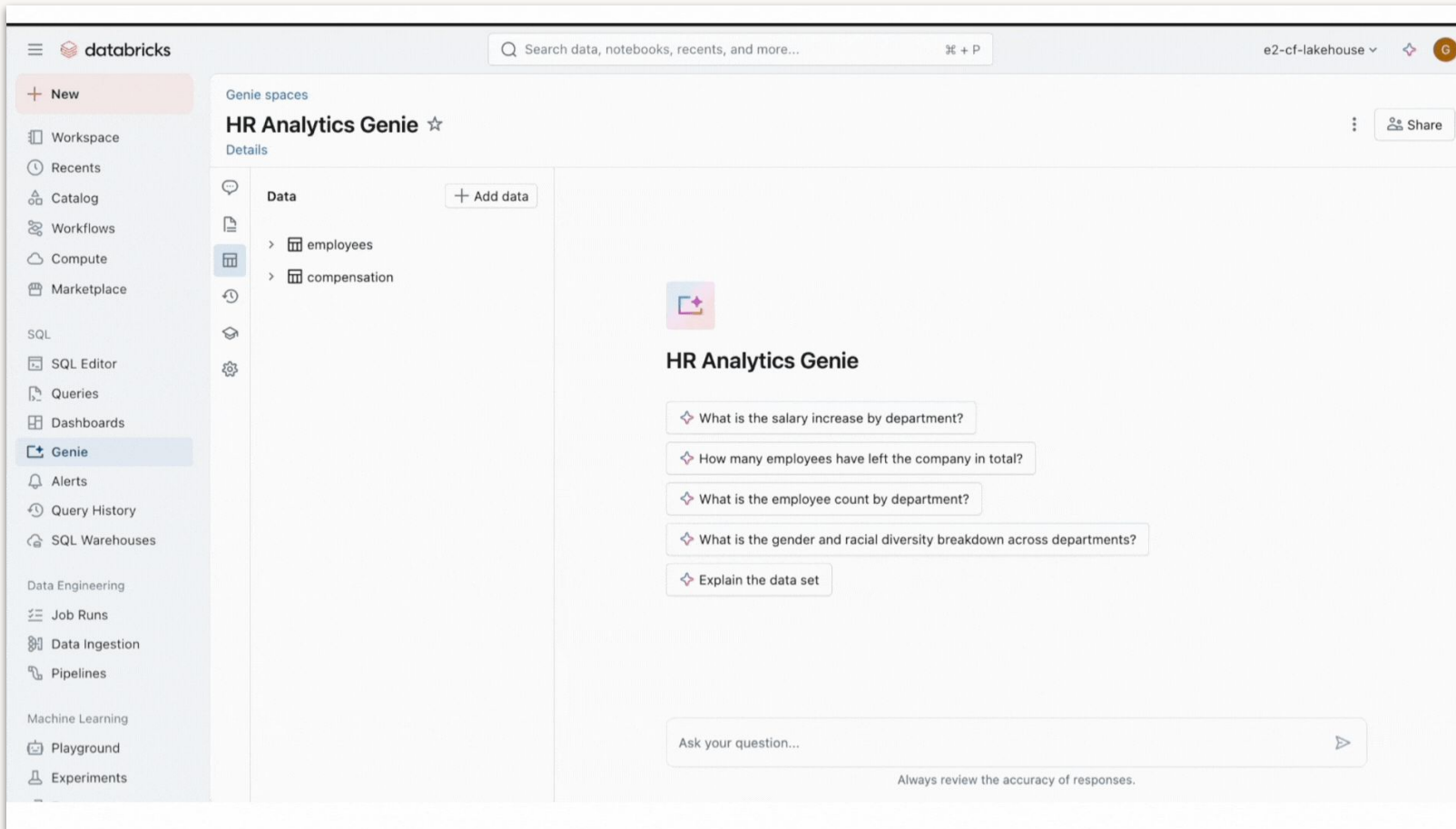


Genie Setup and Lifecycle



Building your Genie Space

Start with well-documented tables



- Begin with UC tables/views and add meaningful comments to columns and table descriptions for clarity
- Clearly define primary keys and foreign keys for better context understanding

Genie – Learning and Improving



Knowledge Mining

Learning semantics from *existing* platform usage

What it is

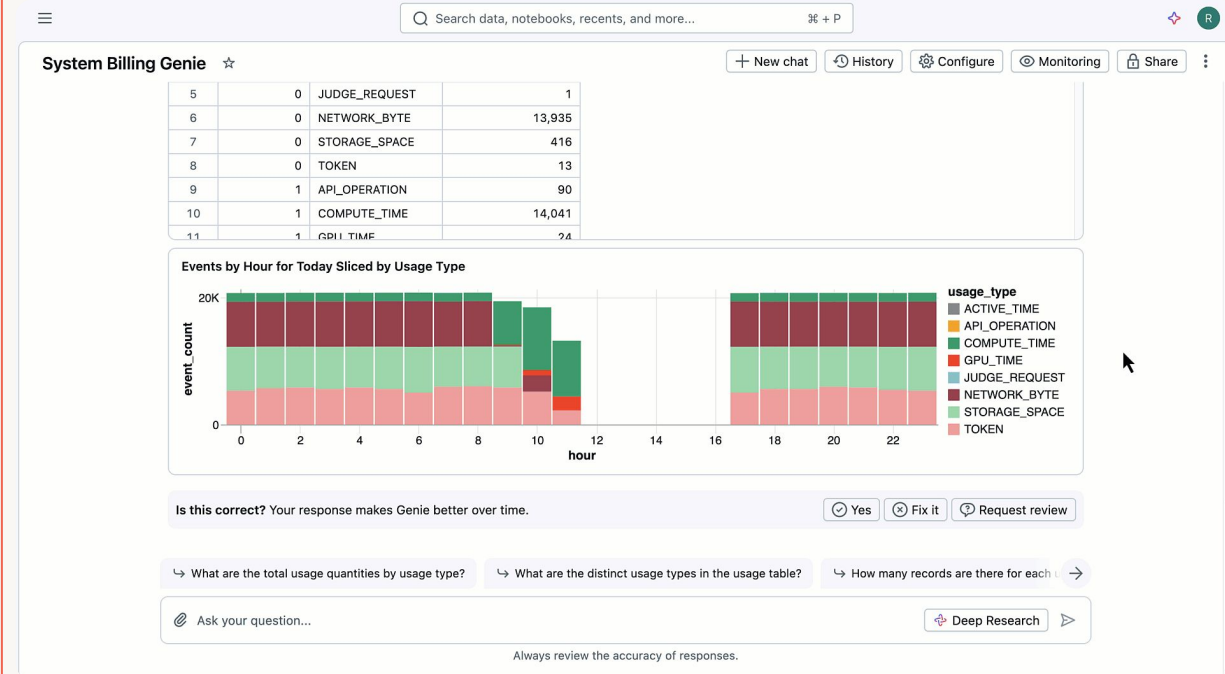
- Discovery and aggregation of relevant information from a variety of data sources across your environment; Genie uses this to surface both technical and business context

Why it matters

- Enables Genie to answer smarter questions

Key Benefits/Use Cases

- Metadata ingestion from Unity Catalog (table names, column descriptions, PK/FK relationships, lineage, access policies)
- General Instructions
- Example SQL Queries and Functions
- Genie builds a semantic map of your data landscape



Knowledge Extraction

Learning semantics from *new* Genie usage

What it is

- Transforms mined information into actionable knowledge, by merging metadata with business context for a unified understanding

Why it matters

- Consistent, accurate, and explainable insights tailored to your organization's data and terminology

Key Benefits/Use Cases

- Reduces confusion, boosts adoption, and builds trust in Genie

The screenshot displays the Genie interface with three overlapping panels. The background panel shows a SQL query editor with the following code:

```
1 WITH filtered_attribution AS (  
2   SELECT  
3     engagement_id,  
4     SUM(mql) as mqls  
5   FROM  
6     `marketing`  
7   WHERE  
8     fiscal_year = 2023  
9     AND mql > 0  
10  GROUP BY  
11    engagement_id  
12 ),  
13 joined_data AS (  
14   SELECT  
15     e.engagement_id, a.attribution, o.outcome, ac.accounts, c.contacts  
16   FROM filtered_attribution f  
17   JOIN engagement e ON f.engagement_id = e.engagement_id  
18   JOIN attribution a ON e.engagement_id = a.engagement_id  
19   JOIN outcome o ON e.engagement_id = o.engagement_id  
20   JOIN accounts ac ON e.engagement_id = ac.engagement_id  
21   JOIN contacts c ON e.engagement_id = c.engagement_id  
22 )  
23 SELECT  
24   region, mqls, attribution, outcome, accounts, contacts  
25 FROM joined_data  
26 GROUP BY region, attribution, outcome, accounts, contacts  
27 ORDER BY region, mqls DESC
```

The middle panel shows a 'Joins' configuration table:

Left table	Right table	Condition
attribution	contacts	`attribution`.`attribution_id` = `contacts`.`contact_id`
engagement	contacts	`engagement`.`engagement_id` = `contacts`.`contact_id`
attribution	engagement	`attribution`.`engagement_id` = `engagement`.`engagement_id`
outcome	contacts	`outcome`.`contact_id` = `contacts`.`contact_id`
attribution	outcome	`attribution`.`outcome_id` = `outcome`.`outcome_id`
attribution	accounts	`attribution`.`account_id` = `accounts`.`account_id`
consumption	accounts	`consumption`.`account_id` = `accounts`.`account_id`

The right panel shows 'General Instructions' for Genie:

General Instructions
Add general instructions on how you want Genie to behave.

FY = Fiscal Year

Region = business unit = BU = business_unit

Comms, Media & Entertainment and CME both mean to filter where account_vertical = "CME"

"Federal Government or Public Sector" means to filter where account_vertical = "PubSec & Fed"

"AMER" business unit means to include: "AMER", "AMER Enterprise & Emerging" and "AMER Industries"

For "delivered emails", "sent emails", or "opened emails" – do not include a filter where is_attr_engagement = true

For all other queries against the engagement table, include this filter: WHERE is_attr_engagement true

This is very important: When looking at

Save



User Feedback

Thumbs Up/Thumbs Down and Request Review

What it is

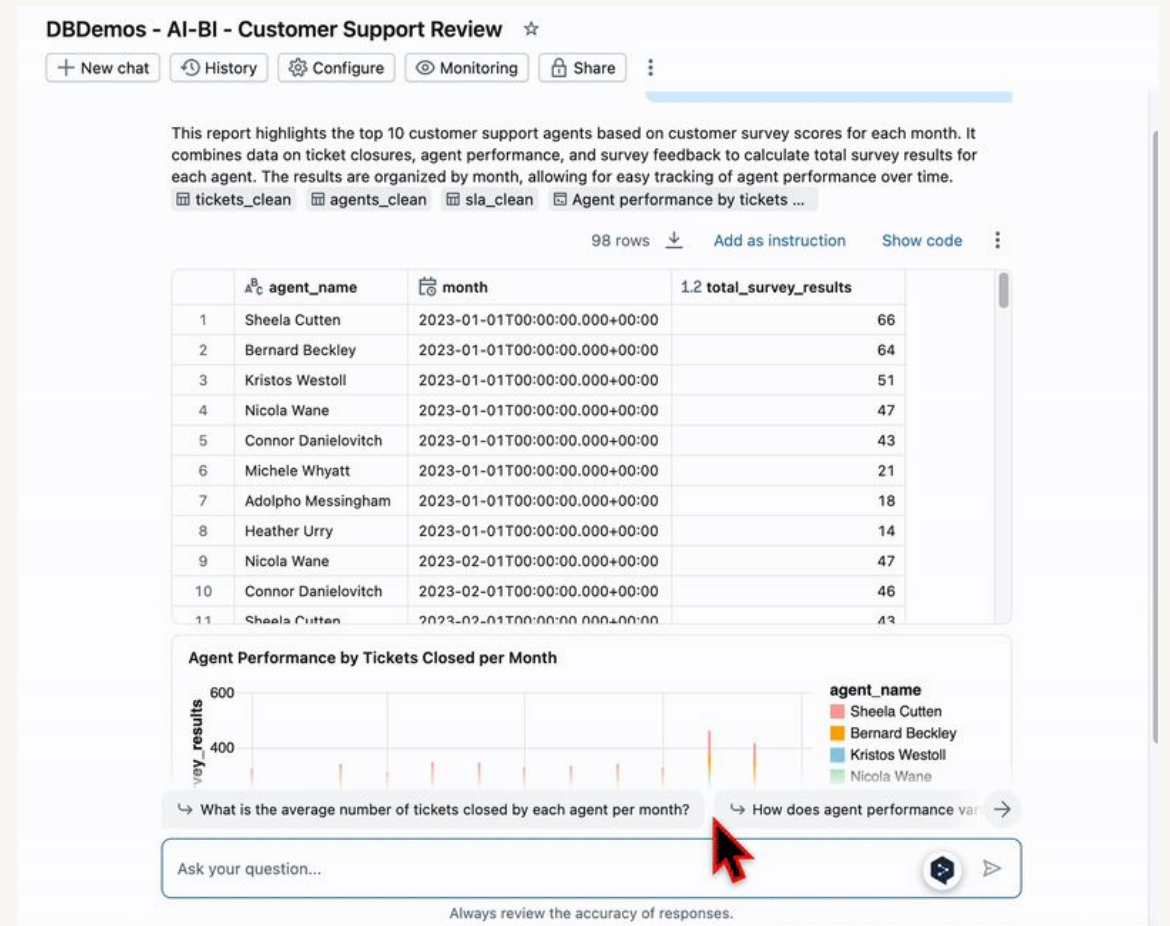
- A built-in feedback system where users can give explicit signals (thumbs up/down, "Request Review," comments) on Genie outputs

Why it matters

- Continuously improves Genie's quality and accuracy by capturing real-world user input and routing it to authors for action

Key Benefits/Use Cases

- Improves model responses with human expertise
- Surfaces gaps in understanding or business logic
- Rapid correction of mistakes and adaptation to business needs



Conversation Context

Session history and state persistence for contextual understanding

What it is

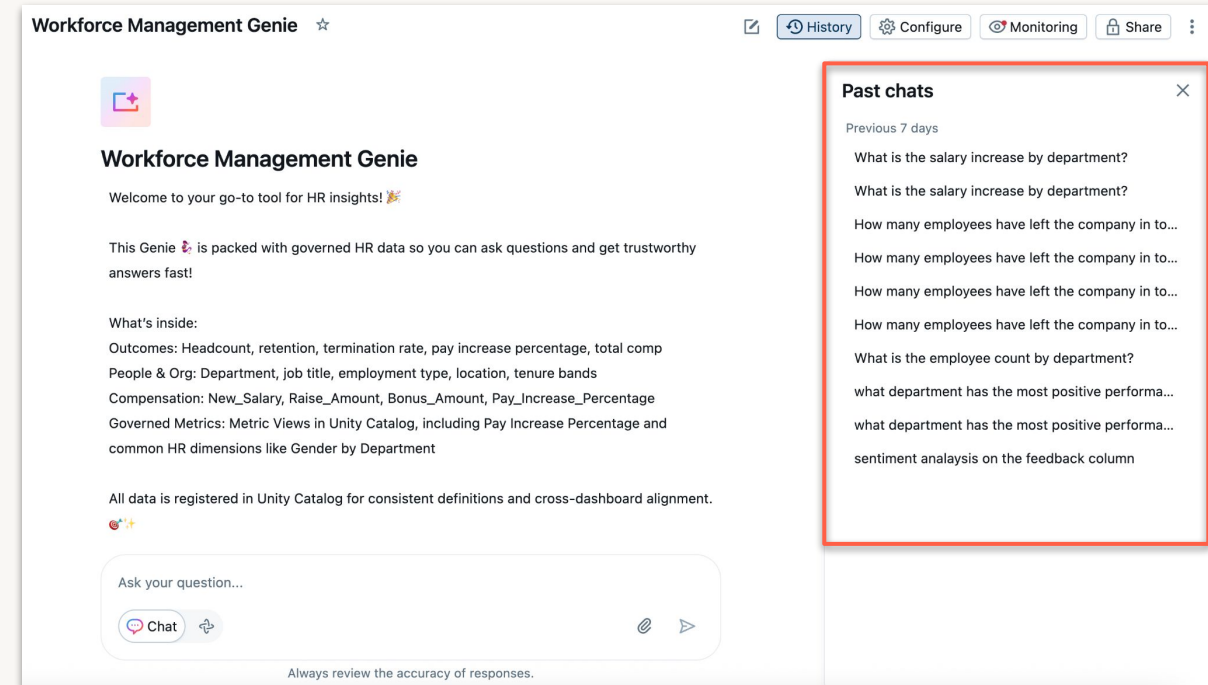
- Genie retains the state and context of ongoing conversations, enabling context-aware Q&A

Why it matters

- Genie can interpret nuanced follow-up questions and adjust responses dynamically based on the user's session history and prior clarifications

Key Benefits/Use Cases

- Session context is continually updated, so Genie's answers reflect everything shared or clarified



Proactive Clarification

Agentic Reasoning in Genie

What it is

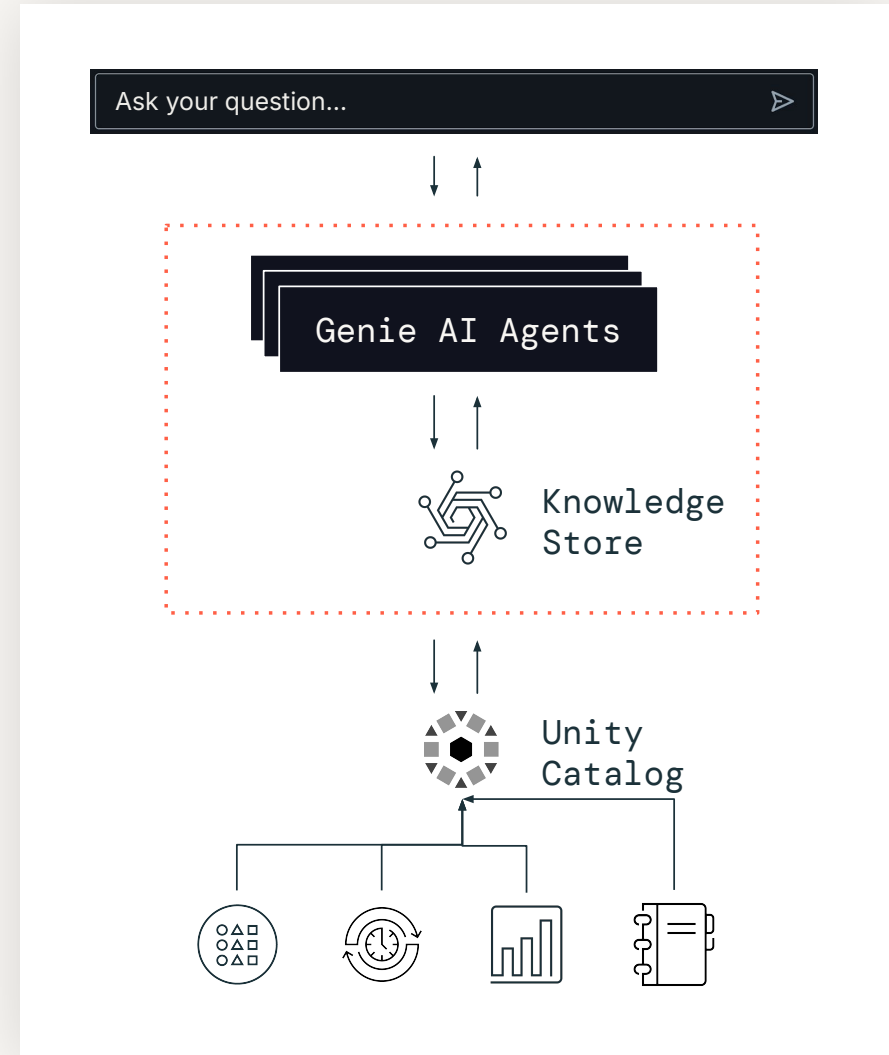
- Genie detects ambiguity or uncertainty in user requests and asks follow-up questions, rather than making low-confidence predictions

Why it matters

- Reduces the risk of incorrect answers and increases user trust by ensuring Genie only responds when intent is clear

Key Benefits/Use Cases

- Clarifications are remembered and shape Genie's future replies within the analytic session



Benchmarking

Measuring the quality and reliability of Genie answers

What it is

- A set of user defined, space-specific questions with known expected results

Why it matters

- Ensure accurate, reliable outputs
- Verify alignment with business logic
- Help build trust in AI-generated insights

Key Benefits/Use Cases

- Detects drift or inconsistencies over time
- Surfaces regressions or bugs early and before org-wide adoption
- Maintains enterprise-grade reliability in natural language analytics

The screenshot displays the 'DBDemos - AI-BI - Customer Support Review' interface with a 'Benchmarks' tab selected. A modal window titled 'Add benchmark' is open, allowing users to define a new benchmark. The modal includes a 'Question' field with the text 'Top agents by number of tickets closed per month' and a small icon. Below this is an 'SQL Answer (optional)' section with a 'Run' button and a text area containing instructions: 'Enter a ground truth SQL statement that correctly answers your benchmark question. This statement will be executed, and the query results will be compared to Genie's results to determine accuracy. Questions without a ground truth SQL statement will be marked for manual review.' At the bottom of the modal are 'Cancel' and 'Add benchmark' buttons. The background interface shows navigation links like '+ New chat', 'History', 'Configure', 'Monitoring', and 'Share', along with '+ Add benchmark' and 'Run benchmarks' buttons.



Genie – Governance and Security



End-to-End Security Model

Genie enforces full access controls

What it is

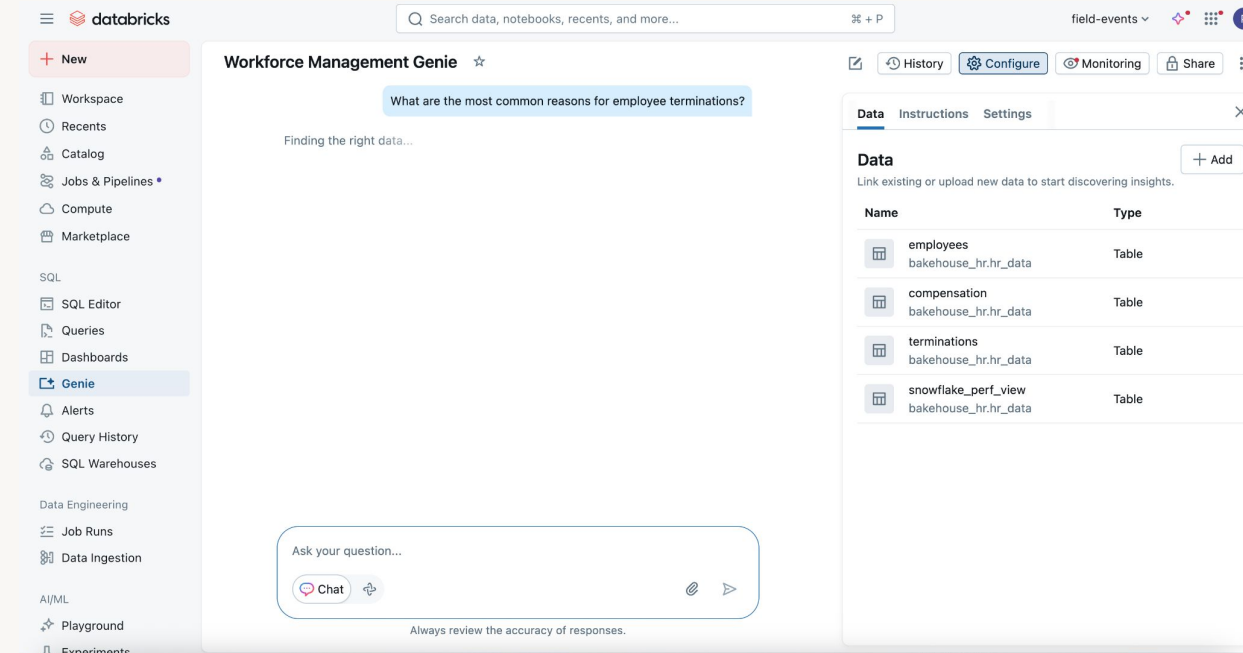
- Every Genie query runs in the exact permission and security context as if the user manually ran SQL in a Databricks notebook, dashboard, etc.

Why it matters

- Guarantees that all data access, filtering, and masking are strictly enforced for each individual

Key Benefits/Use Cases

- Users only see and query data they are entitled to so no risk of accidental overexposure or leaks
- Uniform security compliance across Genie, dashboards, notebooks, and all user interfaces



Audit-Logging

Every question, clarification, and answer

What it is

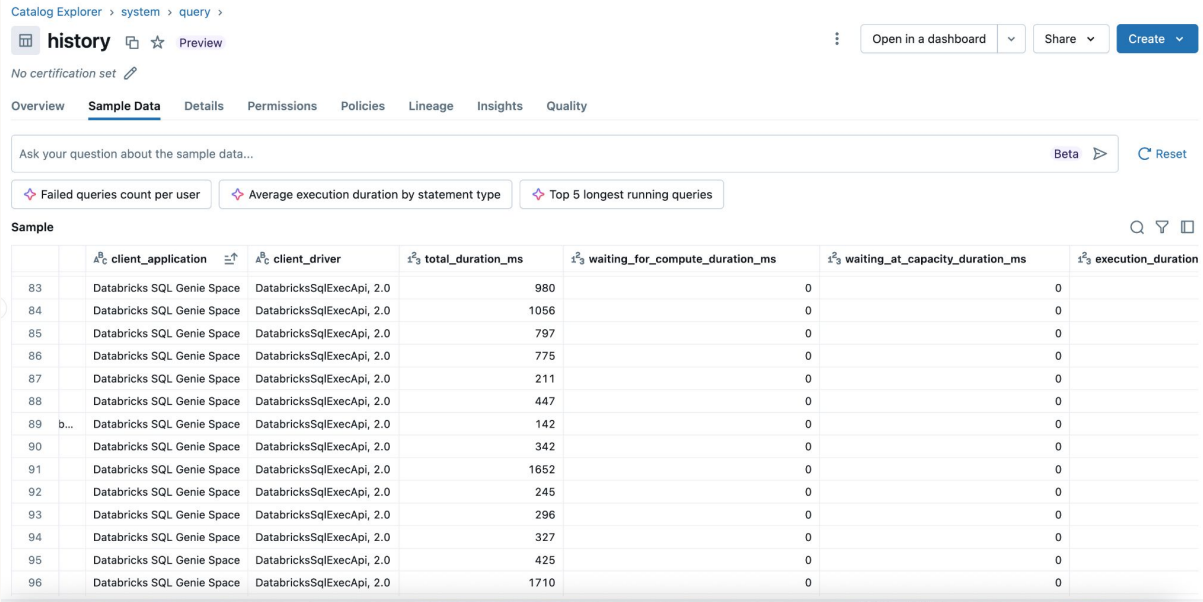
- Genie logs each event with details such as user identity, timestamp, workspace is done through System Tables

Why it matters

- Full transparency, compliance, and traceability for all Genie activities across users and workspaces

Key Benefits/Use Cases

- Supports compliance and governance
- Enables post-hoc review of user queries
- Audits AI behavior for transparency and trust



The screenshot shows the 'history' page in the Databricks Catalog Explorer. The page has a navigation bar with 'Overview', 'Sample Data', 'Details', 'Permissions', 'Policies', 'Lineage', 'Insights', and 'Quality'. The 'Sample Data' tab is active. Below the navigation bar is a search bar with the text 'Ask your question about the sample data...'. To the right of the search bar are buttons for 'Open in a dashboard', 'Share', and 'Create'. Below the search bar are three tabs: 'Failed queries count per user', 'Average execution duration by statement type', and 'Top 5 longest running queries'. The 'Sample Data' tab is selected, and it displays a table with the following columns: 'id', 'client_application', 'client_driver', 'total_duration_ms', 'waiting_for_compute_duration_ms', 'waiting_at_capacity_duration_ms', and 'execution_duration'. The table contains 17 rows of data, with the first row having an 'id' of 83 and a 'total_duration_ms' of 980. The table is titled 'Sample' and has a 'Beta' label and a 'Reset' button in the top right corner.

	client_application	client_driver	total_duration_ms	waiting_for_compute_duration_ms	waiting_at_capacity_duration_ms	execution_duration
83	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	980	0	0	
84	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	1056	0	0	
85	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	797	0	0	
86	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	775	0	0	
87	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	211	0	0	
88	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	447	0	0	
89	b...	DatabricksSqlExecApi, 2.0	142	0	0	
90	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	342	0	0	
91	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	1652	0	0	
92	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	245	0	0	
93	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	296	0	0	
94	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	327	0	0	
95	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	425	0	0	
96	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	1710	0	0	



Genie – Usage Patterns & API



Key API Endpoints

Genie Conversation Management

What it is

- Core REST API endpoints to launch, manage, and organize conversations in Genie spaces

Why it matters

- Enables applications and users to programmatically control Genie chat sessions, manage their life cycles, and list accessible Genie workspaces

Key Benefits/Use Cases

- Automate creation and interaction with natural language analytics conversations from any application
- Enables multi-turn conversations with follow-up questions

Functionality	Description	Type	Endpoint
Start conversation	Starts a new chat thread with an initial question	POST	/api/2.0/genie/spaces/{space_id}/start-conversation
Create new conversation message	Ask a follow up question to an existing chat thread	POST	/api/2.0/genie/spaces/{space_id}/conversations/{conversation_id}/messages
Get conversation message	Track the status of your question + retrieve the generated SQL & description	GET	/api/2.0/genie/spaces/{space_id}/conversations/{conversation_id}/messages/{message_id}
Get SQL query results	Retrieve the query results from the executed SQL	GET	/api/2.0/genie/spaces/{space_id}/conversations/{conversation_id}/messages/{message_id}/query-result
Re-execute SQL query	Re-execute SQL that may have expired query results	POST	/api/2.0/genie/spaces/{space_id}/conversations/{conversation_id}/messages/{message_id}/execute-query

